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# Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review



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|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-054833 |
| Article Type: | Original research |
| Date Submitted by the Author: | 24-Jun-2021 |
| Complete List of Authors: | Nguyen, Yen; Johns Hopkins University School of Nursing<br>McNabb, Katherine C.; Johns Hopkins University School of Nursing<br>Farley, Jason E.; Johns Hopkins University School of Nursing<br>Warren, Nicole; Johns Hopkins University School of Nursing |
| Keywords: | Tuberculosis < INFECTIOUS DISEASES, Public health < INFECTIOUS DISEASES, HIV & AIDS < INFECTIOUS DISEASES, PUBLIC HEALTH, REPRODUCTIVE MEDICINE |
| | |



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# Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review

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**Word count (excluding title page, abstract, figures, references and tables)**: 2912

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**ABSTRACT**

**Objective**: To summarize the current evidence of TB-associated pregnancy outcomes, evaluate the state of the science of family planning during TB treatment, and provide recommendations on how to move forward to improve care and outcomes during TB disease.

**Introduction:** Tuberculosis (TB) has long been associated with poor pregnancy outcomes, particularly in countries with a high rate of TB and human immunodeficiency virus (HIV) co-infection. Although family planning is a key strategy to reduce maternal mortality and adverse pregnancy outcomes, it is unclear to what extent this strategy has been addressed in the context of HIV/TB care.

**Methods**: We conducted a systematic review of peer-reviewed literature, from September 2009 to September 2020, on global TB-associated pregnancy outcomes and family planning using the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA).

**Results**: After exclusions, 46 studies were included in this review. To date, observational studies offer the totality of evidence. Most studies reported pregnancy complications for both mothers, fetuses or newborns, as well as maternal and fetal/newborn mortality. Few studies discussed the value of offering family planning to prevent adverse pregnancy outcomes. Only one study examined the effect of a provider training on contraceptive use. Given the evidence on pregnancy outcomes and family planning, recommendations related to integration of family planning into TB and DR-TB programs for this population were provided.

**Conclusion**: Despite well-established adverse pregnancy outcomes, little attention has been paid to family planning as a means to prevent poor pregnancy outcomes for women with TB. Integrating family planning services within a treatment program is essential to improve TB-associated maternal-child health outcomes.

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**Keywords**:

1. Tuberculosis
2. drug-resistant tuberculosis
3. family planning
4. HIV
5. Integrated services

**Strengths and limitations of this study**

1. This is the first known review to evaluate the literature for family planning strategies in TB programs.
2. It is the first review to provide evidence-based recommendations at the practice, program, research and policy levels to guide the integration of family planning into TB and DR-TB services for women of reproductive age.
3. This review was limited by the inclusion of only peer-reviewed, English journal articles published in the last decade. This likely exclude important findings in other languages and important programmatic data at a country level.
4. The variability of study designs, definitions, and methods across this body of literature made comparison of outcomes difficult. Despite the variability, the observational evidence consistently demonstrates adverse maternal and fetal/newborn outcomes across the globe.

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## BACKGROUND

In 2018, about 3.2 million women worldwide contracted tuberculosis (TB) and almost 500,000 died from the disease.(1) Although TB affects more men than women, it disproportionally places women at a higher risk of morbidity and mortality.(2) TB is a non-obstetric cause of maternal mortality, and is especially problematic for women of reproductive age in settings with high rates of TB/HIV co-infection.(2) TB among mothers is associated with a six-fold increase in perinatal deaths and a two-fold risk of premature birth and low birth weight (LBW).(1) Risk of maternal and fetal morbidity and mortality increases by 400% when human immunodeficiency virus (HIV) coinfection is introduced.(1) Drug-resistant TB (DR-TB) occurs in about 10% of TB patients, and can be detrimental to health and survival since treatment regimens are toxic and prolonged (lasting 9 to 18 months), and unsuccessful in 44% of cases.(3,4)

The potential teratogenic effects of TB and DR-TB treatment may add further risks to these already high risk pregnancies: most of the first- and second-line TB drugs are categorized per the United States (US) Food and Drug Administration (FDA) as pregnancy Class C or D.(4) First-line treatment of drug-susceptible TB includes isoniazid, rifampin, pyrazinamide, and ethambutol, all of which are categorized as pregnancy Class C per the US FDA.(4–8) It is nevertheless recommended pregnant women receive drug susceptible TB treatment as early in their pregnancy as possible because the benefits of treatment outweigh the potential adverse outcomes.(4) Most of the second-line TB drugs are also categorized as pregnancy Class C or D. It is up to the providers' discretion to provide and individualize drug regimens for treatment of DR-TB, who may present with advanced disease and therefore are more at risk of adverse outcomes. Given the risks posed to women of reproductive age, quality TB treatment should address comprehensive family planning services to ensure women can make informed choices regarding pregnancy during this high-risk period. Such services would include counseling about the risks associated with becoming pregnant while undergoing DR-TB treatment, providing safe and preferred methods of contraception, and referring women to appropriate care if they intend to conceive or become pregnancy during TB treatment.

Globally, there are no standardized guidelines regarding pregnancy and family planning in the context of drug-resistant TB (DR-TB) management. Country-specific TB and/or DR-TB guidelines recommend avoiding of specific anti-tuberculous drugs, (9) family planning

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counseling, and/or pregnancy testing and contraceptive use.(10–12) However, is unclear whether there are existing studies which evaluated family planning strategies in TB care.

This review examines published research on pregnancy outcomes and family planning strategies for women undergoing treatment for TB. The aims are: 1) to summarize current evidence for TB-associated pregnancy outcomes, 2) to evaluate the state of the science of family planning during TB treatment, and 3) based on the current evidence and WHO family planning guidelines, to offer recommendations to improve care and outcomes.

## METHODS

### Review Design

We conducted a systematic review of peer-reviewed literature on global TB-associated pregnancy outcomes using the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).(13) Given the heterogeneity of study designs and statistical analysis approaches among included studies, we critically reviewed and synthesized findings instead of conducting a meta-analysis.

### Eligibility Criteria

A systematic approach guided the search for articles addressing both TB disease and pregnancy outcomes. Studies were included if they involved pregnant women with TB, DR-TB, or TB/HIV coinfection, and if study variables included pregnancy outcomes, maternal or fetal/birth outcomes, and TB and/or TB/HIV coinfection outcomes. Studies were also included if they examined family planning services among women initiating TB treatment. Articles were excluded if they were published prior to 2009, were not written in English, did not include human subjects, or were not published in a peer-reviewed journal.

### Search and Identification Process

A Boolean search of the electronic databases PubMed, Embase, CINAHL, Cochrane, Web of Science, and Scopus was conducted on September 6, 2020 using the following medical subject heading (MeSH) terms, keywords and associated synonyms in the titles and abstracts: tuberculosis, family planning services, contraceptives, pregnancy outcome, obstetric outcome,

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fetal morbidity, maternal morbidity, fetal mortality, maternal mortality. The Appendix provides a full strategy for the search.

### Data Extraction and Quality Assessment

Citations were imported into Covidence® and duplicates removed. Two reviewers independently reviewed titles and abstracts. Differences in inclusion of articles were resolved through consensus. Once the articles for inclusion were finalized, data extraction and quality assessment were performed. The Joanna Briggs Institute quality appraisal tools were used to assess the quality of each included citation. (14) Quality rating assessment was conducted independently by two raters to identify strengths and weaknesses and guide the interpretation of study findings. Any discrepancies were discussed and reconciled by consensus.

### Ethics

Institutional review board approval was not required.

### Patient and public involvement

Patients and public were not involved in this review.

## RESULTS

### Selection of studies

The initial search of all five databases yielded 1823 articles after 1527 duplicates were removed. The full text of 86 articles were thoroughly evaluated to assess eligibility, 40 of which were excluded based on the pre-identified exclusion criteria.

### Study characteristics

Out of 46 articles included in this review, 45 articles published TB-associated maternal and fetal/neonatal outcomes and 1 article discussed family planning among women with DR-TB (Figure 1). Of these 46 studies, 26 were conducted in low- and middle-income countries, with the majority from India (n=10) and South Africa (n=4). From high income countries, the majority (n=8) were from the US and the United Kingdom (n=4). All studies included observational designs with case reports (n=20), followed by case series (n=9) and retrospective cohort studies

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(n=9). Case control studies (n=4), prospective cohort studies (n=2), secondary data analysis (n=1), and a service delivery improvement project rounded out the literature. Supplemental Table provides details of pregnancy outcomes and other relevant findings for each study included in this review (Supplemental file).

**Current evidence for TB-associated pregnancy outcomes**

Fetal/newborn outcomes

Overall, reported fetal and newborn outcomes were poor. LBW (weight of less than 2,500 grams at birth) was a common fetal outcome for TB infected women with or without HIV coinfection across multiple studies. (15–19) Preterm labor, which adds to the risk of LBW, was reported in two studies for women with TB infection. (20,21) One large retrospective cohort study (n=2064) found increased congenital anomalies (aOR=1.8, 95% C.I 1.24-2.62) among newborns of TB pregnant women.(22) It is unclear if this is attributable to teratogenic effects of TB drugs or other individual variations. Three case reports of congenital TB discussed babies who were born with LBW, two of which mentioned babies born prematurely and who remained hospitalized for at least 30 days.(19,23,24) Other reported poor fetal outcomes included stillbirth, spontaneous abortions, TB infection, and fetal growth restriction.(16,18,20,25–27) In contrast, reports of full-term birth, normal vaginal delivery, and normal neonatal weight were primarily seen with case reports.(28–32) Although multiple case reports mentioned good fetal and newborn outcomes, the majority of findings suggest newborns of mothers with TB tend to have poorer outcomes.

Maternal outcomes

Four large retrospective cohort studies from the US (n=4053, n=2064, n=57393459, n=1450) reported that anemia and postpartum hemorrhage were more common among women with TB infection as compared to those without. (22,33–35) These studies also noted that TB occurred at a higher rates among Hispanic and Black mothers. Fernandez et al. also reported higher rates of TB and HIV coinfection among Black mothers in their study. (33) Dennis et al. proposed that advanced TB disease, commonly seen in ethnic/racial minorities, may be a contributing factor for increased pregnancy complications compared to mothers without TB.(28) None of the studies stratified analysis of pregnancy complications by race or ethnicity.

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Other pregnancy complications that were more common among pregnant women with TB were chorioamnionitis, postpartum anemia, pneumonia, acute respiratory syndrome, and mechanical ventilation.(22,34,36) Additional maternal outcomes included deaths,(25,26,35,37) blood loss, and unfavorable maternal TB outcomes.(21,25,26,34) One case report described a woman with suspected genitourinary TB prior to pregnancy who suffered from severe pre-eclampsia and postpartum complication of pyelonephritis.(23) The majority of women in case reports were successful treated for TB by the end of the treatment.

HIV coinfection outcomes and ART treatment

In studies that included participants with HIV coinfection, increased maternal mortality and fetal mortality were reported.(26,38–40) Other pregnancy outcomes included anemia, pre-eclampsia, LBW, and fetal mortality.(39) Across four studies that analyzed HIV coinfection, reported median age of pregnant women was 29-30 years, with CD4 count at baseline ranging from 97- 353 cells/mm$^3$ (Interquartile Range 11-511).

Our data suggest antiretroviral therapy (ART) is not consistently provided to women with TB and of those, few are virally suppressed. ART was initiated at the same time as TB treatment for 19 women (57%, n=32).(18) In another study, 29 (55%, n=53) women were reported as being on ART at time of delivery.(32) The highest number of HIV coinfected women on ART when they were diagnosed with TB was 55 (68.8%, n=80), with only about 30.8% having an undetectable viral load (<200 copies/mL).(33) Collectively, these studies show that just over half of HIV coinfected women initiating TB treatment are on some form of ART and much fewer are virally suppressed, which may explain why HIV coinfection contributes additional risk for poor outcomes among women and their babies.

DR-TB outcomes

Five studies evaluate pregnancy outcomes among women with DR-TB. A single case series examined the use of a standardized regimen for DR-TB infected pregnant women (n=5), with a reported cure of TB and good maternal outcomes.(32) In a retrospective cohort study looking at medical records of women with DR-TB between 2013 and 2017, the authors found that second-line drugs, bedaquiline and levofloxacin, were predictors of LBW (45% versus 26%, p=0.034), and that women with HIV coinfection had a higher risk of poor pregnancy

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outcomes.(40) The remaining studies, a case report(23) and a case series,(16) provide conflicting results on maternal and fetal outcomes: the case study reported good outcome for mother and baby while the case series noted maternal and fetal/newborn complications and deaths in their descriptive analysis. The validity and generalizability of these studies are limited by their study design and sample size.

The state of the science of family planning during TB treatment

The literature search yielded one study which evaluated family planning uptake at initiation of TB treatment among women with DR-TB.(41) This South African study aimed to increase contraceptive use by securing Depo-Provera stock, training healthcare providers to counsel and deliver Depo-Provera injections at initiation of DR-TB treatment, and referring women to a tertiary hospital if they intended to conceive or became pregnant during treatment. The authors demonstrated an increase in the number of women initiating Depo-Provera at the time of DR-TB treatment. The authors suggested that these interventions are effective in delaying pregnancies during TB treatment.

Though not specifically examining contraceptive use, six studies that evaluated pregnancy outcomes commented on family planning as crucial to the improvement of health outcomes for women of reproductive age during TB treatment after finding high numbers of pregnancies during the study period.(16,18,20,30,31,42) One study noted 36 out of 38 pregnancies were “unplanned” among a DR-TB cohort of women, and another reported fetal exposure to two potentially teratogenic TB drugs (ethionamide and levofloxacin) during the first trimester.(16,42) One study recommended risk counseling for women initiating TB treatment.(17) All of these studies, regardless of design or location, recognized one important theme: integrated family planning services, including risk counseling and provision of effective contraceptives, are an important preventative strategy to reduce adverse maternal and fetal/newborn outcomes.

Given the evidence of outcomes, care and clinical management of TB infected pregnant women described here, as well as limited data on integrated family planning into TB care, it is imperative for national stakeholders to recommend clear strategies to integrate family planning services into TB care. Based on the findings of this review, our team developed a list of recommendations to guide the integration of family planning into TB and DR-TB services for

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women of reproductive age, incorporated WHO family planning recommendations where relevant (Table 1).

In addition to our recommendations for practice, program, and research, local and national level policy action is needed to ensure family planning services are integrated into TB and DR-TB care settings. Our policy-specific recommendations reflect the evidence from this review and best practices from the field of family planning. The United States Agency for International Development and Family Health International (FHI) 360 created guidance for family planning and HIV integration, which may be used as a blueprint and adapted to TB and DR-TB programs. (43,44) In brief, policymakers should monitor family planning data such as contraceptive prevalence rate, unmet need, and demand for family planning among women with TB and DR-TB in order to appropriately direct funding and resources. Allocation of resources should prioritize efforts to integrate family planning services into TB and DR-TB programs and include monitoring and evaluation plan to inform program and service delivery improvement, as well as scale-up. This could be done by measuring additional indicators specific to family planning and TB integration, including service delivery integration, contraceptive method availability, uptake, informed choice, and training and human resources. (43) Healthcare provider training and ensuring supply of contraceptive methods during TB or DR-TB treatment will also be critical. Finally, subsequent updates to WHO TB and DR-TB guidelines should provide recommendations for family planning as part of quality care for women of reproductive age undergoing treatment. Of note, such a recommendation was offered in the companion handbook to the WHO 2014 DR-TB guideline but absent from the 2020 update. (45)

TABLE 1. Recommendations for Integrated TB/DR-TB and Family Planning Services

| **For clinicians** |
|---|
| 1. Women of reproductive age should be tested for pregnancy at initiation of TB or DR-TB treatment as part of quality TB care. (20,30,31,42) |
| 2. Women who are found to be pregnant at treatment initiation should be referred to antenatal care with appropriate follow ups. Providers should not recommend termination of pregnancy on the sole basis of TB diagnosis. (16,35) |
| 3. Women who are not pregnant should be informed of potential risks to maternal and fetal/infant health if they become pregnancy during treatment and counseled to delay pregnancy until after completion of treatment. (17) |
| 4. Family planning counseling should be done at baseline, with information provided to women to include options for contraceptive methods depending on the TB or DR-TB regimen they may be on (such as rifampicin inclusive regimen), side effects they may experience on contraceptive of choice, and how to switch method or discontinue method. Women should be offered a method is the most effective, such as a long-acting reversible contraceptive |

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(LARC), that does not interfere with treatment and is well-suited for her lifestyle and belief. (30,31,51,52)[1]

5. Women who agree to be on a LARC should be educated on the correct use of method and when to follow up. (52)
6. Women who do not agree to be on a contraceptive should be followed up routinely throughout treatment for changes in relationship status, desire for contraceptive, pregnancy testing, and immediate linkage to antenatal care if pregnancy status changes during TB treatment. (52)

**For TB programs**

1. TB treatment site should keep stock of various contraceptive methods including male and female condoms, oral contraceptives, implants, injectables, intrauterine devices to minimize referring women to outside clinics for family planning. (41,52)
2. TB providers should be trained at least annually on family planning counseling, family planning knowledge, and proper techniques to administer contraceptive methods so they may provide these services to women who need them. (41,52)
3. Recommendations for family planning should be standardized and integrated into all TB and DR-TB treatment sites as part of high-quality TB services. (41,48)

**For researchers**

1. Pregnant women should be considered for clinical studies, particular those looking at the efficacy of shorter regimens compared to longer regimens, which expose women to DR-TB drugs for a longer period of time. (47–49)

2. Research, particularly pragmatic and intervention research, should address family planning needs in TB care and implementation of integrated services. (49)

1. This citation refers to the companion handbook to the WHO 2014 DR-TB guidelines, which were updated in 2020. However, in the updated 2020 guidelines family planning was not mentioned.

## DISCUSSION

To our knowledge, this is the first review to summarize research evidence for TB-associated pregnancy outcomes and evaluate the literature for family planning strategies in TB programs. Despite the findings of poor maternal and fetal outcomes, there is a paucity of family planning research for women with TB. Such a gap in literature may be the rationale for the absence of family planning recommendations in the WHO TB and updated WHO DR-TB guidelines. Despite the call by WHO for inclusion of pregnant women in TB and DR-TB clinical trials in facilitate our understanding of how tuberculosis treatment may impact maternal-child health outcomes, there is yet to be a trial that includes pregnant women. (46–49)

Integration of maternal, fetal and infant health care in the setting of TB disease is essential. Knowing that TB disproportionally affects women of reproductive age in low resource settings, (1) there is unquestionably high value in implementing family planning measures as part of TB care. This paucity of research has not gone unnoticed by other scientists. Schnippel et al. (48) wrote a call to action paper suggesting the need for greater integration of family planning

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services in DR-TB facilities in South Africa, however as to date, only one study that specifically evaluated family planning uptake among women initiating DR-TB treatment.(41) This work provided preliminary data on family planning use and the potential benefit of a provider training and contraceptive supply on increasing uptake of Depo-Provera. However, it did not expand on other aspects of family planning services like addressing factors associated with contraceptive use, pregnancy testing, other family planning options beside the injectable Depo-Provera, or discussion of counseling of pregnancy risks during treatment, all of which could have an impact on a woman's decision to use contraception.

It may be safer and beneficial for pregnant women to have the option to be considered for DR-TB research in order to inform future practice.(49) The World Health Organization (WHO) suggests it would be helpful to compare effectiveness of shorter DR-TB regimens in pregnant women.(50) Currently, the 2020 WHO's guidelines recommend placing pregnant women on longer DR-TB regimen.(51) This is primarily due to the exclusion of pregnant women from a large randomized controlled trial testing the efficacy of shorter DR-TB regimen (i.e. STREAM trial). With a longer treatment regimen, there is a greater exposure time in which women could become pregnant, and longer time pregnant women are being exposed to DR-TB drugs. As of now, there is still not sufficient evidence to support a standardized DR-TB regimen for pregnant women. (46) Research guiding regimen selection in pregnant women is urgently needed.

**Strength and limitations of the studies**

All studies in this review were observational, with the majority being case reports, case series, and retrospective cohort studies. Aside from four studies, most had small sample size and primarily reported exploratory and descriptive data. Despite the weakness of the study designs, the observational evidence consistently demonstrates adverse maternal and fetal/newborn outcomes across the globe. This review was limited by the inclusion of only peer-reviewed, English journal articles published in the last decade. This likely exclude important findings in other languages and important programmatic data at a country level. Further, the variability of study design, definitions, and methods across this body of literature made comparison of outcomes difficult. We sought to include all relevant studies regardless of methodological designs to further our understanding of the scientific gaps that exist around the family planning needs and maternal-child outcomes of women with TB.

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## CONCLUSION

TB in pregnancy is as a major risk factor of maternal and fetal morbidity and mortality, especially in low and middle-income countries where resources are limited. Pregnancy outcomes are worsened in the setting of TB/DR-TB, with or without HIV co-infection, and lack of health resources and access to specialized care. Offering family planning services at TB/DR-TB treatment initiation and education on risk of poor pregnancy outcomes should be considered as part of routine care in women of childbearing age.

**Acknowledgments**

This manuscript was supported by The Johns Hopkins School of Nursing, Johns Hopkins School of Nursing REACH Initiative, Jonas Philanthropies, Coverdell Peace Corps Fellows Scholarship, and R01 AI104488 (JF, National Institute of Allergy and Infectious Diseases).

**Contributors**: the idea for this publication and preparation of the first draft was done by YN. YN and KM conducted the systematic search and assessed quality of articles. YN, KM, JF, and NW provided significant contributions to the draft. All authors reviewed the draft manuscript and approved the final draft for publication.

**Funding**: This work received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

**Competing interest**: none declared.

**Data availability statement:** all data relevant to the study are included in this article and supplemental files.

**Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram: identification and selection of relevant articles.**

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## REFERENCES


1. Tuberculosis in women [Internet]. World Health Organization. 2019. Available from: https://www.who.int/tb/publications/tb_women_factsheet.pdf?ua=1
2. Grange J, Adhikari M, Ahmed Y, Mwaba P, Dheda K, Hoelscher M, et al. Tuberculosis in association with HIV/AIDS emerges as a major nonobstetric cause of maternal mortality in Sub-Saharan Africa. Int J Gynecol Obstet. 2010;108(3):181–3.
3. World Health Organization. Global Tuberculosis Report 2019. Geneva, Switzerland; 2019.
4. Mathad JS, Gupta A. Tuberculosis in Pregnant and Postpartum Women: Epidemiology, Management, and Research Gaps. Clin Infect Dis. 2012;55(11):1532–49.
5. Rifampicin. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 15, 2020. Available from www.micromedexsolutions.com. Subscription required to view.
6. Ethambutol. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 4, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
7. Pyrazinamide. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 4, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
8. Isoniazid. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated July 20, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
9. Aboltins C, Crown L, Curtis N, Darby J, Denholm J, Flynn M, et al. Management ,


For peer review only

Control and Prevention of Tuberculosis: Guidelines for health care providers. South Melbourne: Digital House; 2015.

10. Pakistan Chest Society. Diagnosis & Management of Drug-susceptible Tuberculosis: A National Clinical Guideline. 2017.
11. Central TB Division. National Framework for A Gender-Responsive Approach to TB in India. Nirman Bhawan, New Delhi; 2019.
12. National Tuberculosis & Leprosy Control Programme. Guidelines on the Use of the Shorter Regimen and New Drugs in the Clinical and Programmatic Management of Drug Resistant Tuberculosis and Co-Infections. 2017.
13. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ. 2009;339:b2535.
14. Tufanaru C, Munn Z, Aromataris E, Campbell J, Hopp L. Chapter 3: Systematic Reviews of Effectiveness. In: JBI Manual for Evidence Synthesis. JBI; 2020.
15. Asuquo B, Vellore AD, Walters G, Manney S, Mignini L, Kunst H. A case–control study of the risk of adverse perinatal outcomes due to tuberculosis during pregnancy. J Obstet Gynaecol (Lahore). 2012;32(7):635–8.
16. Palacios E, Dallman R, Muñoz M, Hurtado R, Chalco K, Guerra D, et al. Drug-Resistant Tuberculosis and Pregnancy: Treatment Outcomes of 38 Cases in Lima, Peru. Clin Infect Dis. 2009;48(10):1413–9.
17. Lin H-C, Lin H-C, Chen S-F. Increased risk of low birthweight and small for gestational age infants among women with tuberculosis. BJOG An Int J Obstet Gynaecol. 2010;117(5):585–90.
18. Ali A-AA, Abdallah TM, Rayis DA, Adam I. Maternal and perinatal outcomes of

For peer review only

pregnancies associated with tuberculosis in eastern Sudan. Int J Gynecol Obstet. 2011;114(3):286–7.

19. Chang C-W, Wu PW, Yeh C-H, Wong K-S, Wang C-J, Chang C-C. Congenital tuberculosis: case report and review of the literature. Paediatr Int Child Health. 2017; 38(216–219):1–4.

20. Chopra S, Siwatch S, Aggarwal N, Sikka P, Suri V. Pregnancy outcomes in women with tuberculosis: a 10-year experience from an Indian tertiary care hospital. Trop Doct. 2017; 47(2):104–9.

21. Yadav V, Sharma JB, Kachhawa G, Kulshrestha V, Mahey R, Kumari R, et al. Obstetrical and perinatal outcome in pregnant women with extrapulmonary tuberculosis. Indian J Tuberc. 2019; 66(1):158–62.

22. El-Messidi A, Czuzoj-Shulman N, Spence AR, Abenhaim HA. Medical and obstetric outcomes among pregnant women with tuberculosis: a population-based study of 7.8 million births. Am J Obstet Gynecol. 2016; 215(6):797.e1-797.e6.

23. Adhikari EH, Duryea EL, Rac MWF, Sheffield JS. Genitourinary Tuberculosis: A Rare Cause of Obstructive Uropathy in Pregnancy. Case Rep Obstet Gynecol. 2014; 2014(985682):1–4.

24. Inal HA, Ozturk Inal ZH. A Successful Live Twin Birth by In vitro Fertilization in an Untreated Patient with Silent Genital Tuberculosis. Electron J Gen Med. 2015; 12(2):160–3.

25. Suresh S, Sharath BN, Anita S, Lalitha R, Prasad TJ, Rewari BB. TB-HIV co-infection among pregnant women in Karnataka, South India: A case series. J Infect Public Health. 2016; 9(4):465–70.

For peer review only

26. Toro PL, Schneider KL, Carter RJ, Abrams EJ, El-Sadr WM, Howard AA. Maternal and Infant Outcomes With Concurrent Treatment of Tuberculosis and HIV Infection in Pregnant Women. JAIDS J Acquir Immune Defic Syndr. 2011; 56(2):e63–6.

27. Amin S, Stone D, Anderlind C. A 32-Year-Old Woman With Miscarriage, Headache, Hepatitis, and Pulmonary Disease. Chest. 2019; 155(4):e101–5.

28. Alaoui FZF, Rachad M, Chaara H, Bouguern H, Melhouf MA. Peritoneal tuberculosis in pregnancy: a case report. Pan Afr Med J. 2012;12:65.

29. Masukume G, Sengurayi E, Moyo P, Feliu J, Gandanhamo D, Ndebele W, et al. Massive hemoptysis and complete unilateral lung collapse in pregnancy due to pulmonary tuberculosis with good maternal and fetal outcome: a case report. BMC Res Notes. 2013; 6(1):335.

30. Raouf S, Sharma S, Sunanda G V., Jafettasj J, Papaioannou S. Disseminated extra pulmonary tuberculosis in an immune competent pregnant woman. J Obstet Gynaecol (Lahore). 2009; 29(2):148–50.

31. Shital P, Mirza M. Laryngeal & Lower lung field tuberculosis in pregnancy: A case report. Electron J Gen Med. 2018; 15(2):1–5.

32. Tabarsi P, Moradi A, Baghaei P, Marjani M, Shamaei M, Mansouri N, et al. Standardised second-line treatment of multidrug-resistant tuberculosis during pregnancy [Short communication]. Int J Tuberc Lung Dis. 2011; 15(4):547–50.

33. Fernandez D, Salami I, Davis J, Mbah F, Kazeem A, Ash A, et al. HIV-TB Coinfection among 57 Million Pregnant Women, Obstetric Complications, Alcohol Use, Drug Abuse, and Depression. J Pregnancy. 2018; 2018:1–8.

34. Dennis EM, Hao Y, Tamambang M, Roshan TN, Gatlin KJ, Bghigh H, et al. Tuberculosis

For peer review only

during pregnancy in the United States: Racial/ethnic disparities in pregnancy complications and in-hospital death. Jhaveri R, editor. PLoS One. 2018; 13(3):e0194836.

35. Ahmadzia HK, Khorrami N, Carter JA, Stone J, Amdur RL. Impact of human immunodeficiency virus, malaria, and tuberculosis on adverse pregnancy outcomes in the United States. J Perinatol. 2020; 40(2):240–7.

36. Rickman HM, Cohn S, Lala SG, Waja Z, Salazar-Austin N, Hoffmann J, et al. Subclinical tuberculosis and adverse infant outcomes in pregnant women with HIV. Int J Tuberc Lung Dis. 2020; 24(7):681–5.

37. Barbosa OA, Teles FM, Maia ACC, Pessoa GMF, de Alencar IM, de Sousa ET, et al. Disseminated hematogenous tuberculosis in puerperium—case report. Oxford Med Case Reports. 2019; 2019(11):479–81.

38. Bekker A, Schaaf HS, Draper HR, Kriel M, Hesseling AC. Tuberculosis Disease during Pregnancy and Treatment Outcomes in HIV-Infected and Uninfected Women at a Referral Hospital in Cape Town. Wilkinson KA, editor. PLoS One. 2016; 11(11):e0164249.

39. Salazar-Austin N, Hoffmann J, Cohn S, Mashabela F, Waja Z, Lala S, et al. Poor Obstetric and Infant Outcomes in Human Immunodeficiency Virus-Infected Pregnant Women With Tuberculosis in South Africa: The Tshepiso Study. Clin Infect Dis. 2018; 66(6):921–9.

40. Loveday M, Hughes J, Sunkari B, Master I, Hlangu S, Reddy T, et al. Maternal and Infant Outcomes Among Pregnant Women Treated for Multidrug/Rifampicin-Resistant Tuberculosis in South Africa. Clin Infect Dis. 2021; 72(7):1158–68.

41. Cornish EF, Hudson J, Sayers R, Loveday M. Improving access to contraception through integration of family planning services into a multidrug-resistant tuberculosis treatment programme. BMJ Sex Reprod Heal. 2020; 46(2):152–5.

For peer review only

42. Laniado-Laborín R, Carrera-López K, Hernández-Pérez A. Unexpected Pregnancy during Treatment of Multidrug-resistant Tuberculosis. Turkish Thorac J. 2018; 19(4):226–7.

43. Freyder M, Craig L, Kaji A. Monitoring the Integration of Family Planning and HIV Services: A Manual to Support the Use of Indicators to Measure Progress toward PEPFAR's 90–90-90 Targets and Protect Women's Reproductive Rights. Chapel Hill, North Carolina; 2016.

44. FHI 360, World Health Organization, Uniter States Agency for International Development. Strategic Considerations for Strengthening the Linkages between Family Planning and HIV/AIDS Policies, Programs, and Services. Geneva, Switzerland; 2009.

45. World Health Organization. WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-resistant tuberculosis treatment. Geneva, Switzerland; 2020.

46. Nahid P, Mase SR, Migliori GB, Sotgiu G, Bothamley GH, Brozek JL, et al. Treatment of Drug-Resistant Tuberculosis. An Official ATS/CDC/ERS/IDSA Clinical Practice Guideline. Am J Respir Crit Care Med. 2019; 200(10):e93–142.

47. Gupta A, Mathad JS, Abdel-Rahman SM, Albano JD, Botgros R, Brown V, et al. Toward Earlier Inclusion of Pregnant and Postpartum Women in Tuberculosis Drug Trials: Consensus Statements From an International Expert Panel. Clin Infect Dis. 2016; 62(6):761–9.

48. Schnippel K, Ndjeka N, Conradie F, Berhanu R, Claasen Z, Banoo S, et al. A call to action: Addressing the reproductive health needs of women with drug-resistant tuberculosis. South African Med J. 2016; 106(4):333.

49. McKenna L, Frick M, Lee C, Namutamba D, Smit L, Theunissen M, et al. A Community Perspective on the Inclusion of Pregnant Women in Tuberculosis Drug Trials. Clin Infect

For peer review only

Dis. 2017; 65(8):1383–7.

50. World Health Organization. WHO consolidated guidelines on drug-resistant tuberculosis treatment. Geneva, Switzerland; 2019.

51. World Health Organization. Companion handbook to the WHO guidelines for the programmatic management of drug-resistant tuberculosis. Geneva, Switzerland: World Health Organization; 2014.

52. World Health Organization. Quality of care in the provision of sexual and reproductive health services. Geneva, Switzerland; 2011.

53. Ahuja V, Gombar S, Kumar N, Goyal N, Gupta K. Pregnancy with bilateral tubercular pleural effusion: challenges. Trop Doct. 2014; 44(2):116–8.

54. Fois A, Capobianco G, Crivelli P, Meloni G, Pirina P, Cherchi P, et al. Hemoptoe by tuberculosis in near term pregnant woman: A case report. Clin Exp Obstet Gynecol. 2018; 45(1):115–7.

55. Islam B, Islam N, Mehkri H, AlQaasimi M. Abdominal tuberculosis and spontaneous miscarriage. BMJ Case Rep. 2017; 2017:220022.

56. Kaushal S, Dora SK, Thakur S. Spinal tuberculosis with paraplegia in pregnancy: A case report with management of spinal TB in pregnancy. J Nepal Med Assoc. 2015; 53(198):123–5.

57. Lu P, Li F, Wang Y, Xu C, Yu J, Zhang Y, et al. Uncommon Abdominopelvic Tuberculosis in Pregnancy. J Gynecol Surg. 2013; 29(6):306–8.

58. Loughenbury P, Pavlou G, Dunsmuir R. Surgical Treatment of Tuberculous Paraparesis in the Third Trimester: A Report of Two Cases and Review of the Literature. Gynecol Obstet Invest. 2009; 68(3):213–6.

For peer review only

59. Muin DA, Wagner K, Burian R, Ghaem Maghami N, Lapaire O. Brainstem Tuberculoma in Pregnancy. Case Rep Obstet Gynecol. 2015; 2015:1–6.

60. Sharma S, Ahad A, Gupta N, Sharma V. Primary gingival tuberculosis in pregnancy: A rare combination. J Indian Soc Periodontol. 2018; 22(2):178.

61. Stuart R, Lewis A, Ramsden C, Doherty R. Congenital tuberculosis after in-vitro fertilisation. Med J Aust. 2009; 191(1):41–2.

62. Vogel M, Schroten H, Kahl P, Müller A. High Latency of Tuberculosis Manifestation in a Premature Extremely Low Birth Weight Infant with Favorable Outcome. Neonatology. 2014; 105(2):91–4.

63. Badve SA, Ghate SD, Badve MS, Rustagi T, Macchiwala T, Parekh AN, et al. Tuberculosis of spine with neurological deficit in advanced pregnancy: a report of three cases. Spine J. 2011; 11(1):e9–16.

64. Bishara H, Goldstein N, Hakim M, Vinitsky O, Shechter-Amram D, Weiler-Ravell D. Tuberculosis during Pregnancy in Northern Israel, 2002-2012: Epidemiology and Clinical Practices. Isr Med Assoc J. 2015; 17(6):346–50.

65. Knight M, Kurinczuk J, Nelson-Piercy C, Spark P, Brocklehurst P. Tuberculosis in pregnancy in the UK. BJOG An Int J Obstet Gynaecol. 2009; 116(4):584–8.

66. Rathod AK, Singh V, Patil P, Singh H. Is decompressive surgery the only treatment option? A case series of patients with spinal tuberculosis in advanced pregnancy. Eur Spine J. 2017; 26(12):3170–7.

67. Ye R, Wang C, Zhao L, Wu X, Gao Y, Liu H. Characteristics of miliary tuberculosis in pregnant women after in vitro fertilisation and embryo transfer. Int J Tuberc Lung Dis. 2019; 23(2):136–9.

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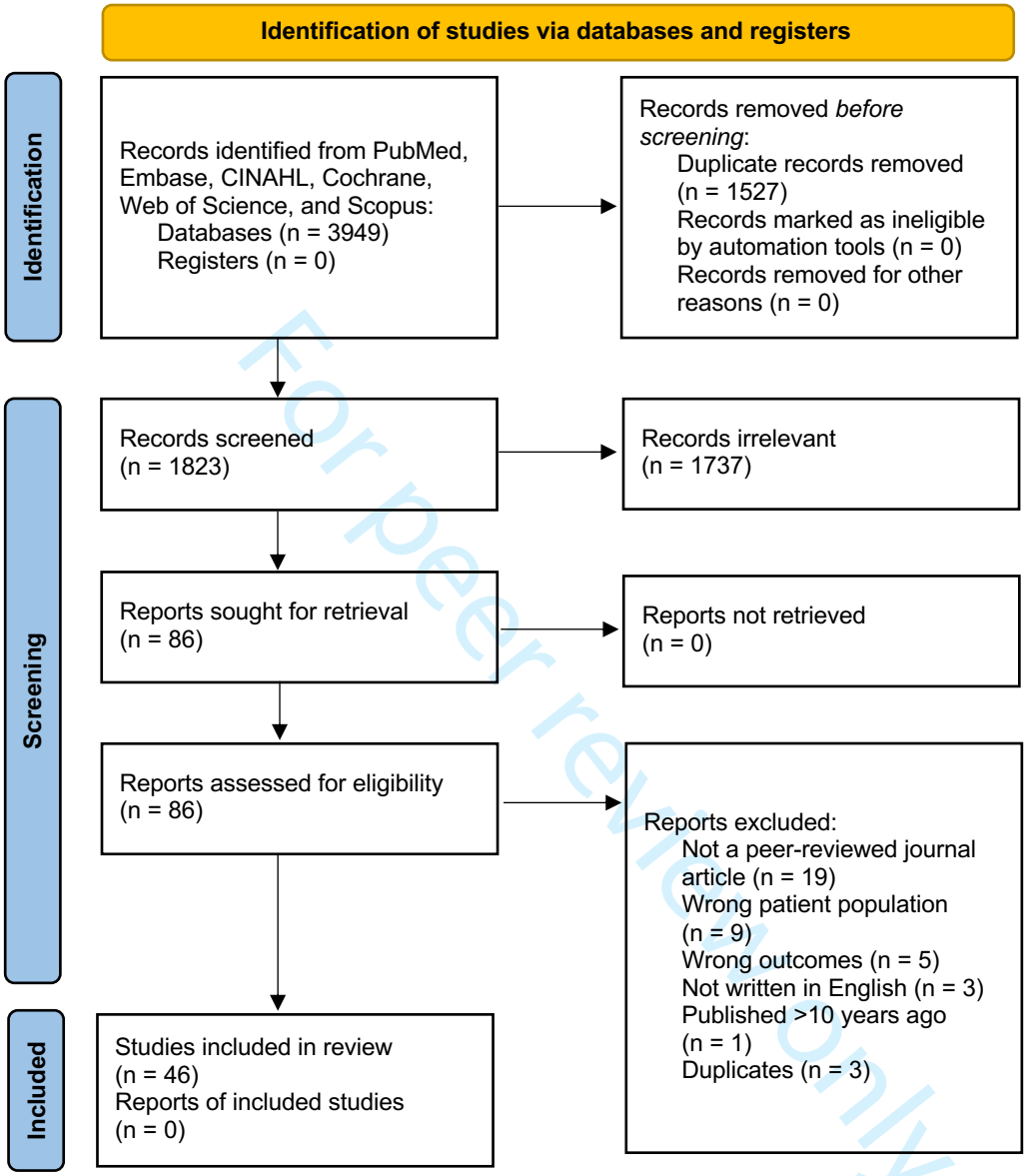
68. Banerjee A, Prateek S, Malik S, Dhingra D. Genital tuberculosis in adolescent girls from low socioeconomic status with acute ectopic pregnancy presenting at a tertiary care hospital in urban Northern India: are we missing an opportunity to treat? Arch Gynecol Obstet. 2012; 286(6):1477–82.

69. LaCourse SM, Greene SA, Dawson-Hahn EE, Hawes SE. Risk of Adverse Infant Outcomes Associated with Maternal Tuberculosis in a Low Burden Setting: A Population-Based Retrospective Cohort Study. Infect Dis Obstet Gynecol. 2016; 2016:1–8.

70. Rendell NL, Batjargal N, Jadambaa N, Dobler CC. Risk of tuberculosis during pregnancy in Mongolia, a high incidence setting with low HIV prevalence. Int J Tuberc Lung Dis. 2016; 20(12):1615–20.

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Figure 1. PRISMA flow diagram: identification and selection of relevant articles




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SUPPLEMENTAL TABLE. Summary of Pregnancy Outcomes and Other Relevant Findings

| Design | First Author (Year), Reference | Setting | Sample | PTB, EPTB, congenital | HIV Co-infection | Maternal outcomes | Fetal/Newborn outcomes | Other relevant findings |
|---|---|---|---|---|---|---|---|---|
| **Case Report** | | | | | | | | |
| Case Report | Adhikari, 2014 (23) | US | 1 (30 yo) | DR EPTB (genitourinary) | no | cured of TB; postpartum complicated by pyelonephritis; fibrosis and chronic inflammation of ureteral segments | induction of labor due to severe pre-eclampsia | NA |
| Case Report | Ahuja, 2014 (53) | India | 1 (27 yo) | PTB | no | weaned off ventilator and discharged home to continue ATT | fetal demise | delayed TB diagnosis; resolved pulmonary embolism; resolved anemia |
| Case Report | Alaoui, 2012 (28) | Morocco | 1 (37 yo 23 weeks gestation) | EPTB (peritoneal) | no | cured of TB | normal birth weight, vaginal delivery 37 weeks | NA |
| Case Report | Amin, 2019 (27) | US | 1 (32 yo | PTB, EPTB (disseminated) | no | premature rupture of membrane | stillbirths of twins | NA |
| Case Report | Barbosa, 2019 (37) | Brazil | 1 (22 yo 22 weeks gestation) | PTB, EPTB (disseminated) | no | death | therapeutic abortion due to suspicion of chorioamnionitis | NA |
| Case Report | Chang, 2018 (19) | Taiwan | 1 (14 day old infant, delivered at 35 weeks) | congenital | no | not discussed | low birth weight, congenital TB | NA |
| Case Report | Fois, 2017 (54) | Italy | 1 (28 yo 36 weeks gestation) | PTB | no | cured of TB | cesarean delivery at 36 weeks, outcome unknown | NA |
| Case Report | Inal, 2015 (24) | Turkey | 1 (34 yo, twins at 27 and 30 weeks gestation) | EPTB (genital) | no | cured of TB | cesarian delivery for both twins at 30 week gestation; both premature and LBW | NA |

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|---|---|---|---|---|---|---|---|---|
| Case Report | Islam, 2017 (55) | Saudi Arabia | 1 (23 yo 23 weeks gestation) | EPTB (abdominal) | no | cured of TB; dilatation and curretage caused uterine perforation and sepsis | spontaneous abortion | resolved sepsis, bilateral pleural effusion; TB treatment was interrupted during transfer of care due to miscommunication |
| Case Report | Kaushal, 2015 (56) | India | 1 (27 yo 35 weeks gestation) | EPTB (spinal) | no | NA | fetal distress, emergency c-section at 37 weeks; ATT prophylaxis x3 months | improvement in motor power over weeks |
| Case Report | Laniado-Laborin, 2017 (42) | Mexico | 1 (15 yo 8 weeks gestation) | DR PTB | no | cured of TB; normal vaginal delivery | NA | fetal exposure to levofloxacin and ethionamide; provider offered therapeutic abortion |
| Case Report | Liu, 2013 (57) | China | 1 (38 yo 28 weeks gestation) | TB (abdominopelvic) | no | cured of TB; abdominal hysterectomy, bilateral salpingectomy, and omentectomy | cesarean delivery at 29 weeks, outcome unknown | NA |
| Case Report | Loughenbury, 2009 (58) | UK | 2 (33 weeks and 33 weeks) | EPTB (spinal) | no | cured of TB; cesarian delivery for both moms | moms given steroids to aid fetal lung maturation; LBW reported for one newborn | NA |
| Case Report | Masukume, 2013 (29) | Zimbabwe | 1 (33 yo 18 weeks gestation) | PTB | no | cured of TB; vaginal delivery at 41 weeks | normal birth weight | delayed TB diagnosis |
| Case Report | Muin, 2015 (59) | Switzerland | 1 (19 yo, 28 weeks gestation) | EPTB (brainstem) | no | cured of TB; failed induction at 40 weeks, underwent c-section | steroids to aid fetal lung maturation | improvement neurological symptoms |

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|---|---|---|---|---|---|---|---|---|
| Case Report | Raouf, 2009 (30) | UK | 1 (23 yo) | EPTB (brain and eye) | no | cured of TB; full term induction vaginal birth; complications resolved with scarring of eye | normal birth weight | Patient stopped ATT by herself 3 weeks before delivery because she wanted to breastfeed |
| Case Report | Sharma, 2018 (60) | India | 1 (25 yo old 4th month of pregnancy) | EPTB (gingiva) | no | cured of TB; normal vaginal delivery | NA | NA |
| Case Report | Shital, 2018 (31) | India | 1 (25 yo 24 weeks gestation) | PTB, EPTB (laryngeal) | no | cure of TB; normal vaginal delivery | normal birth weight | NA |
| Case Report | Stuart, 2009 (61) | Australia | 1 (newborn delivered at 28 weeks) | Congenital | no | TB outcome not reported; no other maternal outcome reported | premature birth, emergency c-section, LBW; ventilator use due to respiratory distress; neonatal jaundice, right-sided Erb's palsy | neutropenia due to ethambutol in newborn; all newborn's complications resolved, and newborn was discharged from hospital |
| Case Report | Vogel, 2014 (62) | Germany | 2 (mother with tuberculous endometritis, newborn with TB diagnosis) | Congenital | no | TB outcome not reported; tuberculous endometritis and deciduitis - unknown outcome | premature birth, extremely LBW, pneumonia, pulmonary hypertension, ventilator use x2 due to cardiac decompensation x2 | all newborn's complications resolved, and newborn was discharged from hospital at 8 months old |
| **Case Series** | | | | | | | | |
| Case Series | Badve, 2011 (63) | India | 3 (third trimester pregnancies, median age=30) | EPTB (spinal) | no | sequela (kyphosis) of spinal TB in 1 woman unresolved | 1 spontaneous abortion; 1 newborn suffers neonatal sepsis in premature care facility; 1 newborn normal birth weight | kyphosis, spinal cord compression |

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|---|---|---|---|---|---|---|---|---|
| Case Series | Bishara, 2015 (64) | Israel | 6 (35-41 weeks gestation, median age=22) | both (n=4 PTB, n=2 EPTB) | no | all 6 cured of TB | 1 intrauterine growth restriction; 2 out of 6 pneumonia post-delivery (moms had EPTB), 2 newborns treated for latent TB | TB diagnostic and treatment delay in all 6 women |
| Case Series | Knight, 2009 (65) | UK | 33 (median age=30) | both (n=16 PTB, n=16 EPTB) | yes (n=5) | 1 elected abortion; 1 death | 2 TB associated early births; 1 fetal demise due to severe maternal TB and intraventricular hemorrhage; 1 developed chronic lung diseae; 1 born with congenital anomaly | 5/6 women were of ethnic minority; delay of TB diagnosis and treatment (median 22 days from onset of symptoms to diagnosis) |
| Case Series | Palacios, 2009 (16) | Peru | 38 (mean age=24.4 SD=5.8) | DR PTB (n=37), EPTB (n=1) | yes (n=3) | 23 (n=38) cured of TB; 2 episodes of vaginal bleed, 1 placenta previa, 1 premature membrane rupture; 4 clinical TB deterioration, 8 deaths | 5 spontaneous abortions; 1 stillborn; of 32 infants born alive: three with LBW <2500g, one fetal distress; two latent TB treated, 1 active TB treated at mo 9 | treatment was suspended with confirmed pregnancy (n=14); second-line drugs may be continued despite teratogenic risk; most had advanced disease at baseline (n=36) |
| Case Series | Rathod, 2017 (66) | India | 5 (third trimester pregnancies, median age=24) | EPTB (spinal) | no | 3 normal vaginal deliveries | 1 spontaneous abortion; 1 fetal demise due to adverse reaction to anesthesia | 3 with resolution of TB, no sequelae; 1with mild back pain; 1 with mild neck pain; 1 patient with recovery of paraplegia; 1 with destruction of spinal vertebra and kyphosis; 1 |

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| | | | | | | | | with 15 degrees kyphosis |
| Case Series | Suresh, 2015 (25) | India | 17 (median age=24) | both | yes | 5 deaths, 3 died before delivery, 4 elected abortions | 1 death | high maternal mortality in low-resource high HIV burden countries |
| Case Series | Tabarsi, 2011 (32) | Iran | 5 (median age=22) | DR PTB | NA | cured of TB; normal vaginal delivery | normal weight | offer of therapeutic abortion; treatment was suspended with confirmed pregnancy and resumed at patient's decision |
| Case Series | Toro, 2011 (26) | sub-Saharan Africa | 33 (median age=29) | Not reported | yes | 1 death, 1 elected abortion | rate of stillbirths more than twice that compared to women receiving ART alone during pregnancy (15.2%, n=5); 1 stillbirth was HIV infected | NA |
| Case Series | Ye, 2019 (67) | China | 6 (mean age 32.17± 3.87 years) | EPTB | no | cured of TB | 3 spontaneous abortions, 3 elected abortions | delayed TB diagnosis (10-32 days from onset of symptoms to diagnosis) |
| **Case-control** | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|
| Case-control | Asuquo (2012) (15) | UK | 24 (pregnant TB) + 72 (pregnant control, no TB) | PTB (n=13), EPTB (n=10), both (n=1) | yes (n=1) | cured of TB not reported | LBW (p=0.03) explained by pre-term delivery (OR 179.9, p=<0.001) | complications include hepatitis, cholestasis, and vomiting |
| Case-control | Banerjee, 2012 (68) | India | 37 (n=17 cases, n=20 controls) | EPTB | no | more adolescent girls with ectopic pregnancy had genital TB (p=0.03) | NA | NA |
| Case-control | Chopra, 2017 (20) | India | 50 women matched with TB control | both | NA | anemia (23% vs. 4% control); two died due to TB disease progression | 3x more likely to have fetal growth retardation (OR 3.2, 95% CI 1.3-7.9), preterm labor x5 more likely (OR 5.9 95% CI 2.5-13.9) | NA |
| Case-control | Lin, 2010 (17) | Taiwan | 761 women matched with 3805 on age and year of delivery | not reported | no | NA | small for gestational age (aOR 1.22, 95% CI 1.00-1.49); low birth weight (aOR 1.35, 95% CI 1.01-1.81) | NA |
| **Cohort Study** | | | | | | | | |
| Retrospective Cohort Study | Ahmadzia, 2020 (35) | US | 1450 pregnant women with TB | PTB, EPTB | NA | TB group more likely to experience pre-term labor (13.0% versus 6.8%, p=0.004), to have UTIs ( 7.0% versus 3.3%, p=0.02), and undergo cesarian delivery (30.9% versus 21.5%, p=0.007) | fetal distress more likely in fetuses with mothers infected with TB compared to control group (5% versus 2.1%, p=0.02) | NA |
| Retrospective Cohort Study | Ali, 2011 (18) | Sudan | 42 | 35 PTB, 7 EPTB | yes (5) | anemia, 2 deaths | preterm birth, LBW | NA |
| Prospective Cohort Study | Bekker, 2016 (38) | SA | 74 | both | yes (53) | 5 deaths in HIV-coinfected; LBW infant associated with unfavorable maternal TB treatment outcome (p=0.009) | 6 infants died, 4 stillbirths, all to HIV positive women; 49 prematurity, 44 LBW | NA |

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|---|---|---|---|---|---|---|---|---|
| Retrospective Cohort Study | Dennis, 2018 (34) | US | 57,393,459 | not reported | yes | rate of pregnancy complications (severe pre-eclampsia, eclampsia, placenta previa/accreta/abruption, antepartum/postpartum hemorrhage, sepsis, anemia) 80% higher among TB infected mothers compared to non-TB infected mothers | NA | non-Hispanic black mothers have an aOR 2.77 (95% C.I. 2.35, 3.25) of non-TB related death compared to white mothers; TB infected mothers have similar risk level across all ethnicities (white, black, Hispanic); over 1/3 of all TB cases were diagnosed among mothers of Hispanic origin |
| Retrospective Cohort Study | El-Messidi, 2016 (22) | US | 2064 | not reported | NA | increased aOR of chorioamnionitis (OR=1.35, 95% C.I., 1.04-1.74), postpartum anemia ((OR=1.51; 95% C.I., 1.22-1.87), PNA, ARDS, mechanical vent, maternal mortality increased among pregnant women with TB (OR=6.27, 95% C.I. 2.01-19.58) | congenital anomalies (aOR=1.8, 95% C.I 1.24-2.62) | Medicaid the most common payer for TB-related hospitalizations; more pregnant women with TB from lower income quartiles |

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|---|---|---|---|---|---|---|---|---|
| Retrospective Cohort Study | Fernandez, 2018 (33) | US | 4053 | not reported | yes (110) | rate of postpartum hemorrhage highest in TB only group (45.5 per 1,000 hospitalizations); compared to HIV & TB negative mothers, HIV monoinfected or TB monoinfected mothers have significantly higher odds of pregnancy complications (HIV monoinfected, aOR=1.4, 95% C.I. 1.32-1.47; TB monoinfected, aOR=1.91, 95% C.I. 1.64-2.23) | NA | Rate of TB among HIV-positive mothers was 21 times higher compared to rate of TB in HIV-negative mothers |
| Retrospective Cohort Study | LaCourse, 2016 (69) | US | 134 matched by year delivery and TB status | not reported | NA | completed treatment or cured of TB (67%), treatment failure (3%), death (3%) | LBW (aRR=3.74, 95% C.I. 1.40-10.00); SGA (uRR=1.95, 95% C.I. 1.11-3.41) | NA |
| Retrospective Cohort Study | Loveday, 2020 (40) | SA | 108 | DR PTB | yes (88) | completed treatment or cured of TB (67%), treatment failure (3%), death (3%) | fetal and neonatal deaths (9%), treated for DR PTB in the first year of life (2%); LBW for those exposed to bedaquilline (45% vs 24%, p=0.034) | low baseline hemoglobin a predictor of poor TB treatment outcome (uHR 0.67, p= 0.006); women living with HIV had a higher risk of a poor pregnancy outcomes (uHR 3.35; p=0.030); bedaquilline and levofloxacin are predictors of LBW |
| Retrospective Cohort Study | Rendell, 2016 (70) | Mongolia | 107 | PTB and EPTB | no | 10/85 elected abortion after TB diagnosis | NA | 2/82 TB treatment failure, 4/82 lost to follow-up; higher incidence of TB in pregnancy |

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| | | | | | | | | compared to general population (IRR 1.31, CI 1.08-1.59) |
| Prospective Cohort Study | Salazar-Austin, 2007 (39) | SA | 80 (pregnant women with TB) matched with 155 controls (no TB) | both | yes | anemia more in cases than control; pre-eclampsia 8-fold more common among case patients than controls (5.2% vs. 0.7%, p=0.03); higher rates of maternal hospitalization (25% vs. 11%, p=0.005) | infants born to HIV/TB moms twice as likely to have LBW (20.8% vs. 10.7%, p=0.04), prolonged hospitalizations at birth (51% vs. 16%, p=<0.001) than born to HIV only; infants born to HIV/TB moms had 10- fold increased infant mortality risk (p=<0.001) | NA |
| Retrospective Cohort Study | Yadav, 2019 (21) | India | 30 | EPTB | no | significant increased incidence of oligoamnios and preterm rupture of membrane/preterm delivery compared to non-TB patients (13.3% versus 1.6%, p=0.001); death secondary to shock and end-organ failure from miliary TB; 7 women underwent emergent cesarian delivery for fetal distress | 1 stillborn due to severe miliary TB in mother; LBW in newborns of moms with TB compared to non-TB moms (2324.26±379.5 grams versus 2712.3±635.7, p=0.001); more incidence of small for gestation age fetuses in moms with EPTB compared to non-TB moms. (33.3% versus 7.8%, p=0.001) | NA |

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**Other**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Secondary Data Analysis | Rickman, 2020 (36) | US | 7 | not reported | yes | 43% of women with TB experienced complications compared to 35% of control (p=1.0, Fisher's exact) | 43% infants experienced complications including TB, neonatal sepsis, and infant hospitalization compared to 11% of control (p=0.045, Fisher's exact) | in a cohort of women living with HIV, 7 subclinical cases were identified and discussed in this study |

Abbreviations: MTCT-Plus, maternal to child transmission-Plus; PTB, pulmonary TB; EPTB, extrapulmonary TB; DR PTB, drug resistant pulmonary TB; DR EPTB, drug resistant extrapulmonary TB; LBW, low birth weight <2500g; SD, standard deviation; SGA, small for gestational age; PNA, pneumonia; ARDS, acute respiratory distress syndrome; NA, not applicable; aOR, adjusted odds ratio; aRR, adjusted risk ratio; uHR, unadjusted hazard ratio; NA, not applicable

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**Appendix. Detailed Search of Systematic Review.**

**Pubmed (565 citations)**
("Tuberculosis"[Mesh] OR tuberculos* [tiab] OR "tb" [tiab])
AND
("Pregnancy Outcome"[Mesh] OR ((pregnan* [tiab] OR obstetric [tiab] OR maternal [tiab] OR fetal [tiab]) AND (outcome* [tiab] OR morbidit* [tiab] OR mortalit* [tiab])))

(84 citations)
("family planning services" [MESH] "OR family planning" [tiab] OR "contraception" [tiab] OR "contracept*" [tiab]) AND ("Tuberculosis"[Mesh] OR tuberculos* [tiab] OR "tb" [tiab])

**Embase (660 citations)**
'tuberculosis'/exp OR tuberculos*:ti,ab OR 'tb':ti,ab
AND
'pregnancy outcome'/exp OR (((pregnan* OR obstetric* OR maternal* OR fetal*) NEAR/5 (outcome* OR morbidity OR mortality)):ti,ab)

**(249)**
'tuberculosis'/exp OR tuberculos*:ti,ab OR 'tb':ti,ab
AND
'family planning'/exp OR ((family planning services):ti,ab OR (contraception OR contracept*):ti,ab)

**CINAHL (308 citations)**
(MH "Tuberculosis+") OR ( tuberculos* OR "tb" )
AND
(MH "Pregnancy Outcomes") OR ( ((pregnan* OR obstetric* OR maternal OR fetal) N5 (outcome* OR morbidity OR mortality)) )

**(60)**
(MH "Tuberculosis+") OR ( tuberculos* OR "tb" )
AND
(MH "family planning") OR ((family planning services) OR (contraception OR contracept*))

**COCHRANE (136 citations - Reviews and Trials Only)**
#1 MeSH descriptor: [Tuberculosis] explode all trees 2103
#2 tuberculos* or tb 10858
#3 #1 or #2 10865
#4 MeSH descriptor: [Pregnancy Outcome] explode all trees 3446
#5 ((pregnan* or obstetric* or maternal or birth or fetal) near/5 outcome*) OR ((fetal or maternal) near/5 morbidity) OR ((fetal or maternal) near/5 mortality) 17349
#6 #4 or #5 17645
#7 #3 and #6 129

For peer review only

**(159 citations – Review and Trials Only)**

| | | |
|---|---|---|
| #8 | MeSH descriptor: [Family Planning Services] explode all trees | 226 |
| #9 | contraception or contracept* | 14614 |
| #10 | #8 or #9 | 14669 |
| #11 | #3 and #10 | 159 |

**WEB OF SCIENCE (308 citations)**

TOPIC: (tuberculos* OR tb)
AND
TOPIC: ((pregnan* OR obstetric* OR maternal OR fetal) NEAR/5 (outcome* OR morbidity OR mortality))
Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC.

TOPIC: (tuberculos* OR tb)
AND
TOPIC: (family planning OR family planning services OR contraception OR contracept*)
Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC.

**SCOPUS (851 citations)**

( TITLE-ABS-KEY ( tuberculos* OR tb ) AND ( TITLE-ABS-KEY ( ( pregnan* W/5 outcome* ) OR ( obstetric* W/5 outcome* ) OR ( maternal W/5 outcome* ) OR ( maternal W/5 morbidity* ) OR ( maternal W/5 mortality ) OR ( fetal W/5 outcome* ) OR (fetal W/5 morbidity) OR ( fetal W/5 mortality ) ) ) )

**(362)**

( TITLE-ABS-KEY ( tuberculos* OR tb ) AND (family planning OR family planning services OR contraception OR contracept*)

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | Title page |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | 4 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | 5 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | 5 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | 5 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | 5-6, Appendix |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | 5-6 |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | 5-6 |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | 5 |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | 5 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | 6 |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | N/a |
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | 5 |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | 5 |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | N/a |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | 5 |
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | N/a |
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | N/a |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | N/a |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| Certainty assessment | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | 6 |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | 6, Figure 1 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | 6 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | Supplemental material |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | N/a |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Supplemental material, as appropriate |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | 6 |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | N/a |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | N/a |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | N/a |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | N/a |
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | 12 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | 11-12 |
| | 23b | Discuss any limitations of the evidence included in the review. | 12 |
| | 23c | Discuss any limitations of the review processes used. | 12 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | 12, table 1 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | N/a |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | N/a |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | N/a |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | 13 |
| Competing interests | 26 | Declare any competing interests of review authors. | 13 |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | 13 |

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*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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# Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review



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|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-054833.R1 |
| Article Type: | Original research |
| Date Submitted by the Author: | 06-Jan-2022 |
| Complete List of Authors: | Nguyen, Yen; Johns Hopkins University School of Nursing<br>McNabb, Katherine C.; Johns Hopkins University School of Nursing<br>Farley, Jason E.; Johns Hopkins University School of Nursing<br>Warren, Nicole; Johns Hopkins University School of Nursing |
| Primary Subject Heading: | Public health |
| Secondary Subject Heading: | Reproductive medicine |
| Keywords: | Tuberculosis < INFECTIOUS DISEASES, Public health < INFECTIOUS DISEASES, HIV & AIDS < INFECTIOUS DISEASES, PUBLIC HEALTH, REPRODUCTIVE MEDICINE |
| | |



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# Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review

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**Word count (excluding title page, abstract, figures, references and tables)**: 4449

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**ABSTRACT**

**Title: Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: a systematic review**

**Objectives: 1**. Summarize and evaluate the current evidence of TB-associated pregnancy outcomes; 2. Evaluate the state of the science of family planning during TB treatment; 3. Provide recommendations to move forward to improve care and outcomes during TB disease.

**Methods:** A systematic review of peer-reviewed literature was conducted across PubMed, Embase, CINAHL, Cochrane, Web of Science, and Scopus from September 2009 to November 2021 on global TB-associated pregnancy outcomes and family planning strategies for women undergoing treatment for TB and DR-TB. Data were extracted using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) . Quality assessment was conducted using the Joanna-Briggs Institute Critical Appraisal Tools and the level of evidence was reported using the Johns Hopkins Evidence-Based Practice (JHEBP) guidelines.

**Results**: 69 studies were included in this review. Case reports, case series, case controls, cohort studies, secondary data analyses, and a service delivery improvement project conducted in 26 countries made up the totality of the evidence. Most studies reported pregnancy complications for mothers including anemia, postpartum hemorrhage, and deaths. For fetuses or newborns, low birth weight, premature birth, and spontaneous or induced abortions are commonly reported. Few studies discussed the value of offering family planning to prevent adverse pregnancy outcomes. Only one study examined the effect of a provider training on contraceptive use with reported increased contraceptive use.

**Discussion:** Integrating family planning services within a treatment program is essential to reduce adverse TB-associated maternal-child outcomes. Despite well-established adverse pregnancy outcomes, little attention has been paid to family planning to prevent poor pregnancy outcomes for women with TB/DR-TB. Recommendations for clinicians, TB programs, and researchers are provided and reflect evidence presented in the review.

**Funding:** This work received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

**Registration:** This systematic review is not registered.

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**Keywords**:

1. Tuberculosis
2. drug-resistant tuberculosis
3. family planning
4. HIV
5. Integrated services

**Strengths and limitations of this study**

1. A systematic search of six major electronic databases was conducted and results reported per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
2. Inclusion of case reports and case series allows for synthesis of important clinical data that is often not reported in systematic reviews.
3. Review was limited by the inclusion of only peer-reviewed, English journal articles published in the last 11 years, which likely excluded important findings in other languages and important programmatic data at a country level.
4. The variability of study designs, outcome definitions, and methods across this body of literature made comparison of outcomes difficult and therefore we were unable to conduct a meta-analysis.

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## BACKGROUND

In 2018, about 3.2 million women worldwide contracted tuberculosis (TB) and almost 500,000 died from the disease.(1) Although TB affects more men than women, it disproportionally places women at a higher risk of morbidity and mortality.(2) TB is a non-obstetric cause of maternal mortality and is especially dangerous for women of reproductive age in settings with high rates of TB or drug-resistant TB (DR-TB), and HIV co-infection.(2) TB among mothers is associated with a six-fold increase in perinatal deaths and a two-fold risk of premature birth and low birth weight (LBW).(1) Risk of maternal and fetal morbidity and mortality increases by 400% when human immunodeficiency virus (HIV) coinfection is introduced.(1)

The potential teratogenic effects of TB and DR-TB treatment may add further risks to these already high-risk pregnancies: most of the first- and second-line TB drugs are categorized per the United States (US) Food and Drug Administration (FDA) as pregnancy Class C or D.(3) First-line treatment of drug-susceptible TB includes isoniazid, rifampin, pyrazinamide, and ethambutol, all of which are categorized as pregnancy Class C per the US FDA.(3–7) It is nevertheless recommended pregnant women receive drug susceptible TB treatment as early in their pregnancy as possible because the benefits of treatment outweigh the potential adverse outcomes.(3) DR-TB occurs in about 10% of TB patients and can be detrimental to health and survival since treatment regimens are toxic and prolonged (lasting 9 to 18 months), and unsuccessful in 44% of cases.(3,8) Most of the second-line TB drugs are also categorized as pregnancy Class C or D. It is up to the providers' discretion to provide and individualize drug regimens for treatment of DR-TB, who may present with advanced disease and therefore are more at risk of adverse outcomes.

Comprehensive family planning services have the potential to reduce women's risk of adverse pregnancy outcomes in the context of TB and DR-TB. Such services would include counseling about the risks associated with becoming pregnant while undergoing TB or DR-TB treatment, providing safe and preferred methods of contraception, and referring women to appropriate care if they intend to conceive or become pregnancy during TB treatment.(9) Family planning counseling by a TB or DR-TB provider at initiation of TB or DR-TB treatment is especially important given that there are known drug-drug interactions between first line TB

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drugs, rifampicin, antiretroviral drugs (e.g. protease inhibitors) and contraceptives such as combined contraceptive pills.(10)

Despite reported poor outcomes among pregnant women with TB, DR-TB, and HIV coinfection, they have not been systematically summarized with family planning strategies to reduce risks explored. In addition, there are no international guidelines regarding pregnancy and family planning in the context of drug-resistant TB (DR-TB) management. Country-specific TB and/or DR-TB guidelines recommend family planning counseling, and/or pregnancy testing and contraceptive use(11–13), or avoidance of specific anti-tuberculous drugs.(14) However, they do not describe how to integrate these services into TB care. Without a more comprehensive understanding of the risks to and prevention strategies for childbearing women with TB, DR-TB, and/or HIV coinfection, it will be difficult to improve services or guide additional research.

This systematic review examines published research on pregnancy outcomes and family planning strategies for women undergoing treatment for TB and DR-TB. The objectives are: 1) to summarize and evaluate current evidence for TB-associated pregnancy outcomes, 2) to evaluate the state of the science of family planning during TB treatment, and 3) to provide recommendations to move forward to improve care and outcomes.

## METHODS

### Review Design

We conducted a systematic review of peer-reviewed literature on global TB-associated pregnancy outcomes using the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).(15) Given the heterogeneity of study designs, outcome definitions, and statistical analysis approaches among included studies, we critically reviewed and synthesized findings instead of conducting a meta-analysis.

### Eligibility Criteria

A systematic approach guided the search for articles addressing TB-related pregnancy outcomes. Studies were included if they assessed patient population of pregnant women with active TB, DR-TB, or TB/HIV coinfection, and primary outcome variables included pregnancy outcomes, maternal or fetal/birth outcomes, and TB or TB/HIV coinfection outcomes. Studies were also included if they examined outcomes of family planning services among women

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initiating TB treatment (counseling of pregnancy risks, pregnancy testing, contraceptive uptake). Pregnancy outcomes were any outcome a study defined as maternal outcome or any maternal complication: pre-eclampsia, eclampsia, anemia, hemorrhage, placental abruption/previa, premature rupture of membrane, unplanned cesarian section or induction, death, or TB, DR-TB, or TB-HIV coinfection outcomes as defined below. Studies were also included if they assessed fetal complications or fetal/birth outcomes, which may include premature birth defined as birth prior to 37 weeks, low birth weight (LBW) defined as weight below 2500 grams, stillbirth, spontaneous abortions/fetal demise or induced abortions, and congenital TB infection in newborns up to 6 months postpartum. TB, DR-TB, or TB/HIV coinfection outcomes include cure/treatment completion, treatment failure, lost to follow up, or fetal or maternal death.

Articles were excluded if they were not peer reviewed articles (e.g., commentaries, communication briefs, editorials, conference abstracts or posters), literature reviews, published prior to 2009, were not written in English, and did not include human subjects. We excluded studies older than 2009 to ensure only the most up-to-date evidence was synthesized. We also excluded studies that did not have a primary aim to analyze and discuss pregnancy outcomes as defined above.

**Search and Identification Process**

A Boolean search of the electronic databases PubMed, Embase, CINAHL, Cochrane, Web of Science, and Scopus was conducted through November 5, 2021 using the following medical subject heading (MeSH) terms, keywords and associated synonyms in the titles and abstracts: tuberculosis, family planning services, contraceptives, pregnancy outcome, obstetric outcome, fetal morbidity, maternal morbidity, fetal mortality, maternal mortality. The Appendix provides a full strategy for the search.

**Data Extraction and Quality Assessment**

Citations were imported into Covidence® and duplicates removed. Two reviewers independently reviewed titles and abstracts. Differences in inclusion of articles were resolved through consensus. Once the articles for inclusion were finalized, data extraction and quality assessment were performed. The level of evidence of each study was assessed using the Johns Hopkins Evidence-Based Practice (JHEBP) guidelines.(16) The Joanna Briggs Institute (JBI)

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quality appraisal tools were used to assess the quality of each included citation; studies that fulfilled most to all items of the respective quality appraisal tool indicate they are of high quality and minimal bias.(17) Quality rating assessment was conducted independently by two raters to identify strengths and weaknesses and guide the interpretation of study findings. Any discrepancies were discussed and reconciled by consensus. The critical appraisal of included studies is tabulated in the Supplemental Table 1 for review (Supplemental Table 1. Critical Appraisals of the Included Studies).

**Ethics**

Institutional review board approval was not required.

**Patient and public involvement**

Patients and the public were not involved in this review.

## RESULTS

### Selection of studies

The initial search of all five databases yielded 2967 articles after 1362 duplicates were removed. The full text of 176 articles were thoroughly evaluated to assess eligibility, 107 of which were excluded based on the pre-identified exclusion criteria (Figure 1. PRISMA flow diagram: identification and selection of relevant articles).

### Study characteristics

Of the 69 articles included in this review, 68 articles published TB-associated maternal and fetal/neonatal outcomes and 1 article discussed family planning among women with DR-TB. Of these 69 studies, 51 were conducted in low- and middle-income countries, with the majority from India (n=12), China (n=10), and South Africa (n=7). Among the high-income countries, the majority (n=8) were from the US and the United Kingdom (n=3).

### Level of Evidence and Quality appraisal

All studies included were of level III and V evidence per JHEBP guidelines. Level III evidence included non-experimental, observational designs with the majority being retrospective

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cohort studies (n=15), followed by case control studies (n=3) and prospective cohort studies (n=3), and secondary data analysis (n=2). Level V evidence included case reports (n=32), followed by case series (n=13) and a service delivery improvement project. There was no experimental (Level I), mixed-methods (Level II) or qualitative study identified in our search.

The majority of included studies met most or all items of the quality assessment based on the JBI appraisal tool (n=60) with 9 fulfilling less than or half of the items of the quality assessment. Because this review aims to provide comprehensive evidence on pregnancy outcomes among women with TB and DR-TB and family planning strategies among this population, we included all studies that met the inclusion criteria regardless of quality or level of evidence. Supplemental Table 1 provides details of the quality assessment and level of evidence of appraised studies. Supplemental Table 2 provides details of pregnancy outcomes and other relevant findings for each study included in this review, including study designs, outcome measures, and statistical results (Supplemental Table 2. Summary of Pregnancy Outcomes and Other Relevant Findings).

**Current evidence for TB-associated pregnancy outcomes**

Fetal/newborn outcomes

Overall, reported fetal and newborn outcomes were poor. LBW was a common fetal outcome for TB infected women with or without HIV coinfection across multiple studies (case reports n=7; case series n=2; cohort study n=8).(18-37) Preterm labor, which adds to the risk of LBW, was reported in multiple studies (case report n=5; case series n=1; case controls n=1; cohort studies n=4) among women with TB infection.(18,21,23,25,26,30,36,38-42) A case control study found that preterm labor was five times more likely among women with TB (OR 5.9 95% CI 2.5-13.9).(41) One case series and four cohort studies had similar findings: there were higher numbers of preterm births among women with TB versus those who did not.(21,30,36,40,42) A large retrospective cohort study (n=24149664) found that pregnant women with TB were more likely to experience preterm labor (13.0% versus 6.8%, p=0.004)(37), and this trend of higher preterm labor rates persisted with other smaller case series and cohort studies.(21,30,40,42)

One large retrospective cohort study (n=2064) found increased congenital anomalies (aOR=1.8, 95% C.I 1.24-2.62) among newborns of pregnant women with TB.(43) It is unclear if

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this is attributable to teratogenic effects of TB drugs or other individual variations. A total of 8 studies evaluated congenital TB cases. Six case reports of congenital TB discussed babies who were born with LBW, two of which mentioned babies born prematurely and who remained hospitalized for at least 30 days. (22,26,27,44-45) TB outcomes for newborns with congenital TB include: resolution or improvement of TB at follow up, loss to follow up, died without TB treatment, or unknown.(22,26,28,40,44-46) Other reported poor fetal outcomes included stillbirth (case reports n=1, case series n=2; cohort studies n=3) (30,47,49), spontaneous abortions/fetal demise (case reports n=3, case series n=6 cohort study n=1) (19,29,48,50-55), or induced abortions (case reports n=4, series n=1, cohort study n=2, other n=1) (42,48,55-60) and fetal growth restriction/retardation (case series n=1, case control n=1).(41,61) One case series reported stillbirth, spontaneous abortions/fetal demise and induced abortions (48), and one cohort study reported spontaneous and induced abortions.(62) In contrast, reports of full-term birth, normal vaginal delivery, and normal neonatal weight were primarily seen with case reports.(28–32) (63-69) Although multiple case reports mentioned good fetal and newborn outcomes, the majority of findings suggest newborns of mothers with TB tend to have poorer outcomes.

Maternal outcomes

Three large retrospective cohort studies from the US reported that anemia and postpartum hemorrhage were more common among women with TB infection as compared to those without TB (n=4053 rate of composite pregnancy complications including anemia and postpartum hemorrhage 80% higher among women with TB; n=2064 anemia OR 1.51, 95% CI 1.22-1.87; n=4053 rate of postpartum hemorrhage highest in TB only group, 45.5 per 1,000 hospitalizations compared to TB and HIV negative group, 25.8 per 1000, similar to anemia, 216.6 compared to 102.7 per 1000 hospitalizations).(43,70,71) Two prospective cohort studies found similar findings regarding pregnancy complications including anemia and pre-eclampsia (n=80, more anemia in women with TB versus no TB, 27.5% versus 11.0%, p=0.001, pre-eclampsia 8-fold more common among women with TB versus no TB, 5.2% vs. 0.7%, p=0.03; n=26, 57.7% with adverse events related to TB treatment).(35,62) Anemia was similarly seen in a case control study (n=50, anemia 23% vs. 4% control, p-value not reported) and a retrospective cohort study (n=42 among 15252 pregnant women, anemia was seen in 83.3% versus 61.9%, p=0.02).(41,21) In the US, Fernandez et al. (70) reported highest rates of TB-HIV coinfection (49.9%) among

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Black mothers compared to Hispanic (16.7%) and non-Hispanic white mothers (% data suppressed) in their study. Dennis et al. proposed that advanced TB disease, commonly seen in ethnic/racial minorities, may be a contributing factor for increased pregnancy complications compared to mothers without TB.(71) None of the studies stratified analysis of pregnancy complications by race or ethnicity.

A retrospective cohort study (n=2064) reported other pregnancy complications more commonly seen among pregnant women with TB including chorioamnionitis (OR=1.35, 95% C.I., 1.04-1.74), pneumonia (OR 8.42, 95% C.I., 5.77-12.29), acute respiratory syndrome (OR, 2.85, 95% CI, 1.35-6.10), and mechanical ventilation (OR 3.33, 95% C.I. 1.66-6.68).(43) Case reports also found similar pregnancy complications.(56,58) Multiple case reports and case series described delayed TB diagnosis and complicated management of TB and pregnancy.(19,24,28,29,39,44,45,47,48,50,52,54,55,57-59,61,64,66,68,72-76) Some resulted in recommendation and subsequent termination of pregnancy(56,57,59,77,78), induced abortion(48,62,79), or unplanned cesarian section or induction.(24,25,27,29,36,38,39,53,72,74) For example, one case report described a woman with suspected genitourinary TB prior to pregnancy who suffered from severe pre-eclampsia and postpartum complication of pyelonephritis with unplanned induction.(72)

In regard to TB outcomes, the majority of women in cases reported treatment completion, cure, or improvement of clinical symptoms (23-25,38,39,45,50-52,58,59,63-69,72-75,78,80). Other TB-related maternal outcomes included deaths (19,21,28,30,32,33,36,41,42,47-49,56,81,82), lost to follow up (41,83), TB treatment failure (57), or unknown (18,22,44,26). Across 16 case series (19,29,48,53,61,76,77,81,83), a case control study (41), and cohort studies (31,33,42,62,82,84) that provided descriptive data of TB outcomes, there were a total of 402 cases. Of those, 259 (64%) had an outcome of treatment completed/cured, 49 (12%) were lost to follow up, 29 (7%) died, 5 (1%) failed treatment, 4 (1%) deteriorated in clinical condition, 5 (1%) with unknown outcome, and 46 were on treatment at time of data collection. Among studies that analyzed association of TB and pregnancy outcomes, one study found that women who gave birth to newborns with LBW were 3.83 times more likely to have poor TB outcome such as death, treatment failure, or loss to follow up compared to those delivering newborns with normal weight (95% CI 1.40–10.53, p = 0.009) (30). El-Messidi and colleagues (43) found that compared with pregnant women without TB, maternal mortality was higher among those with

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TB (OR=6.27, 95% C.I. 2.01-19.58). Similarly, compared to HIV and TB negative mothers, those with HIV or TB have significantly higher odds of pregnancy complications (HIV, aOR=1.4, 95% C.I. 1.32-1.47; TB, aOR=1.91, 95% C.I. 1.64-2.23). (70) Although the majority of case reports described favorable TB outcomes for women, they also described difficult management of disease, or significant adverse sequelae of TB. Furthermore, larger studies found a large proportion of women with adverse TB outcomes associated with TB disease.

HIV coinfection outcomes and ART treatment

Across 9 studies that analyzed data among women with TB-HIV coinfection (1 case series, sub-Saharan Africa; 3 prospective cohort studies, SA; 4 retrospective cohort studies, US and SA; 1 secondary data analysis, US), 4 reported median age of pregnant women between 29-30.1 years (Interquartile Range [IQR] 25-35), and 6 reported CD4 count at baseline ranging from 11-565 cells/mm$^3$ (IQR 11-565). (30,33,35,41,42,49,62,70,85) These studies found increased maternal mortality and fetal mortality. Bekker et al. (30) described higher number of deaths among women with TB-HIV coinfection compared to those with TB only (9% versus 0%, p=0.313) and 10 fetal deaths (n=55 among mothers with HIV), suggesting higher mortality among this population compared to those with TB only. Similarly, another study found 91% (10/11) of preterm labor were among those with TB-HIV coinfection compared to those with TB only (42). One study found no statistical significance in association between TB-HIV coinfection and maternal or fetal outcome (62).

Other pregnancy outcomes included anemia (27% versus 11%, p=0.001), pre-eclampsia (5.2% versus 0.7%, p=0.03), LBW (2950g versus 3060 g, p=0.04), and fetal mortality (68 deaths versus 7 per 1000 live births, p<0.001) among women with TB-HIV coinfection compared to women with HIV and no TB.(35) One study found that HIV-coinfection increased risk of unfavorable pregnancy outcome of preterm labor and/or LBW (unadjusted hazard ratio 3.35, p=0.030) (33). Another found that rates of pregnancy were the highest among women with TB-HIV coinfection (331.28 per 1,000 hospitalizations) compared to those with HIV alone (272.2 per 1,000 hospitalizations) (70).

Our data suggest antiretroviral therapy (ART) is not consistently provided to women with TB and HIV coinfection and of those, few are virally suppressed. ART was initiated at the same time as TB treatment for 19 women (57%, n=32).(49) In another study, 29 (55%, n=53) women

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were reported as being on ART at time of delivery.(30) The highest number of HIV coinfected women on ART at time of TB diagnosis was 55 (68.8%, n=80); only about 30.8% had an undetectable viral load (<200 copies/mL) compared to those without TB (47.7%, p<0.001).(35) Collectively, these studies show that just over half of HIV coinfected women initiating TB treatment are on some form of ART and much fewer are virally suppressed, which may explain why HIV coinfection contributes additional risk for poor outcomes among women and their babies. Unfortunately, no study analyzed pregnancy outcome among pregnant women with TB-HIV coinfection who were on ART versus not on ART, which may limit our understanding of how HIV treatment may impact pregnancy outcomes among this population.

DR-TB outcomes

Ten studies evaluate pregnancy outcomes among women with DR-TB. (19,33,42,57,62,72,77,78,82,84) A single case series examined the use of a standardized regimen for DR-TB infected pregnant women (n=5), with a reported cure of TB and good maternal outcomes.(77) In a retrospective cohort study looking at medical records of women with DR-TB between 2013 and 2017, the authors found that second-line drugs, bedaquiline and levofloxacin, were predictors of LBW (45% versus 26%, p=0.034), and that women with HIV coinfection had a higher risk of poor pregnancy outcomes.(33) Four studies reported successful treatment outcome (complete or cure) for the majority of their samples (83.3%, 61%, 65.4% and 88%) and lost to follow up (11.1%, 28.6%, 26.9% and 12%) (42,62,82,84). The remaining studies, three case reports (72,78,57) and a case series (19), provide conflicting results on maternal and fetal outcomes: two case studies reported good outcome for mother and baby, 1 case study reported treatment failure with induced abortion, while the case series noted maternal and fetal/newborn complications and deaths in their descriptive analysis. The validity and generalizability of these studies are limited by their study design and sample size.

The state of the science of family planning during TB treatment

The literature search yielded one study which evaluated family planning uptake at initiation of TB treatment among women with DR-TB.(60) This South African study aimed to increase contraceptive use by securing Depo-Provera stock, training healthcare providers to counsel and deliver Depo-Provera injections at initiation of DR-TB treatment and referring

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women to a tertiary hospital if they intended to conceive or became pregnant during treatment. The authors demonstrated an increase in the number of women initiating Depo-Provera at the time of DR-TB treatment. The authors suggested that these interventions are effective in delaying pregnancies during TB treatment.

Though not specifically examining contraceptive use, six studies that evaluated pregnancy outcomes commented on family planning as crucial to the improvement of health outcomes for women of reproductive age during TB treatment after finding high numbers of pregnancies during the study period.(19,20,41,69,75,78) One study noted 36 out of 38 pregnancies were "unplanned" among a DR-TB cohort of women, and another reported fetal exposure to two potentially teratogenic TB drugs (ethionamide and levofloxacin) during the first trimester.(19,78) One study recommended risk counseling for women initiating TB treatment.(20) All of these studies, regardless of design or location, recognized one important theme: integrated family planning services, including risk counseling and provision of effective contraceptives, are an important preventative strategy to reduce adverse maternal and fetal/newborn outcomes.

## DISCUSSION

To our knowledge, this is the first review to evaluate evidence for TB-associated pregnancy outcomes and the literature for family planning strategies in TB programs. We found that there were consistent evidence across level III and V studies of poor TB-associated pregnancy outcomes including fetal outcomes (LBW, premature birth, spontaneous or induced abortions) and maternal outcomes (anemia, pre-eclampsia, and deaths). Despite the findings of poor maternal and fetal outcomes, there is a paucity of family planning research for women with TB and DR-TB. Such a gap in literature may be the rationale for the absence of family planning recommendations in the WHO TB and updated WHO DR-TB guidelines.(86)

Integration of maternal, fetal and infant health care in the setting of TB disease is essential. Knowing that TB disproportionally affects women of reproductive age in low resource settings, (1) there is unquestionably high value in implementing family planning measures as part of TB care. This paucity of research has not gone unnoticed by other scientists. Schnippel et al. (87) wrote a call-to-action paper suggesting the need for greater integration of family planning services in DR-TB facilities in South Africa, however as to date, only one study specifically

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evaluated family planning uptake among women initiating DR-TB treatment.(60) This work provided preliminary data on family planning use and the potential benefit of a provider training and contraceptive supply on increasing uptake of Depo-Provera. However, it did not expand on other aspects of family planning services like addressing factors associated with contraceptive use, pregnancy testing, other family planning options beside the injectable Depo-Provera, or discussion of counseling of pregnancy risks during treatment, all of which could have an impact on a woman's decision to use contraception.

Given the evidence of outcomes, care and clinical management of pregnant women with TB or DR-TB described in this review, as well as limited data on integrated family planning into TB care, it is imperative for national stakeholders to recommend clear strategies to integrate family planning services into TB care. Our team developed a list of recommendations to guide the integration of family planning into TB and DR-TB services for women of reproductive age, incorporating the evidence from this systematic review, WHO family planning recommendations as well as the United States Agency for International Development and Family Health International (FHI) 360 guidelines where relevant (Table 1. Recommendations for Integrated TB/DR-TB and Family Planning Services). (88,89,90)

Due to the limited studies on family planning strategies among women with TB or DR-TB, as well as low level of evidence of included studies (level III and V), it is important for future research to conduct more rigorous studies with higher level of evidence, such as experimental or mixed methods studies to explore family planning needs and strategies at the individual, programmatic, and policy levels. Furthermore, it may be safer and beneficial for pregnant women to have the option to be considered for TB and DR-TB research in order to facilitate our understanding of how TB and DR-TB treatment may impact maternal child health outcomes and inform future practice.(87,91-93) Specifically, the WHO suggests it would be helpful to compare effectiveness of shorter DR-TB regimens in pregnant women.(94) As of now, the 2020 WHO's guidelines recommend placing pregnant women on longer DR-TB regimen.(94) This is primarily due to the exclusion of pregnant women from a large randomized controlled trial testing the efficacy of shorter DR-TB regimen (e.g. STREAM trial). (95) With a longer treatment regimen, there is a greater exposure time in which women could become pregnant, and longer time pregnant women are being exposed to DR-TB drugs. As of now, there

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is still not sufficient evidence to support a standardized DR-TB regimen for pregnant women. (91) Research guiding regimen selection in pregnant women is urgently needed in the long-term.

In addition to our recommendations for practice, program, and research, local and national level policy action is needed to ensure family planning services are integrated into TB and DR-TB care settings. Our policy-specific recommendations reflect the evidence from this review and best practices from the field of family planning. FHI 360 created guidance for family planning and HIV integration, which may be used as a blueprint and adapted to TB and DR-TB programs. (89,90) In brief, policymakers should monitor family planning data such as contraceptive prevalence rate, unmet need, and demand for family planning among women with TB and DR-TB to appropriately direct funding and resources. Allocation of resources should prioritize efforts to integrate family planning services into TB and DR-TB programs and include monitoring and evaluation plan to inform program and service delivery improvement, as well as scale-up. This could be done by measuring additional indicators specific to family planning and TB integration, including service delivery integration, contraceptive method availability, uptake, informed choice, and training and human resources. (89) Healthcare provider training and ensuring supply of contraceptive methods during TB or DR-TB treatment will also be critical. Finally, subsequent updates to WHO TB and DR-TB guidelines should provide recommendations for family planning as part of quality care for women of reproductive age undergoing treatment. Of note, such a recommendation was offered in the companion handbook to the WHO 2014 DR-TB guideline but absent from the 2020 update. (86,96)

TABLE 1. Recommendations for Integrated TB/DR-TB and Family Planning Services

| **For clinicians**[1] |
|---|
| 1. Women of reproductive age should be tested for pregnancy at initiation of TB or DR-TB treatment as part of quality TB care. (41,75,69,78) |
| 2. Women who are found to be pregnant at treatment initiation should be referred to antenatal care with appropriate follow ups and more intensive monitoring. (19,20) |
| 3. Women who are not pregnant should be informed of potential risks to maternal and fetal/infant health if they become pregnancy during treatment and counseled to delay pregnancy until after completion of treatment. (73) |
| 4. Family planning counseling should be done at baseline, with information provided to women to include options for contraceptive methods depending on the TB or DR-TB regimen they may be on (such as rifampicin inclusive regimen), side effects they may experience on contraceptive of choice, and how to switch method or discontinue method. Women should be offered a method is the most effective, such as a long-acting reversible contraceptive (LARC), that does not interfere with either TB/DR-TB or HIV treatment and is well-suited for her lifestyle and belief. (69,75,88,96)[2] |
| 5. Women who agree to be on a LARC should be educated on the correct use of method and when to follow up. (88) |

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6. Women who do not agree to be on a contraceptive should be followed up routinely throughout treatment for changes in relationship status, desire for contraceptive, pregnancy testing, and immediate linkage to antenatal care if pregnancy status changes during TB treatment. (88)

**For TB programs**

1. TB treatment site should keep stock of various contraceptive methods including male and female condoms, oral contraceptives, implants, injectables, intrauterine devices to minimize referring women to outside clinics for family planning. (60,88)
2. TB providers should be trained at least annually on family planning counseling, family planning knowledge, and proper techniques to administer contraceptive methods so they may provide these services to women who need them. (60,88)
3. Recommendations for family planning should be standardized and integrated into all TB and DR-TB treatment sites as part of high-quality TB services. (60,87)

**For researchers**

1. Pregnant women should be considered for clinical studies, particular those looking at the efficacy of shorter regimens compared to longer regimens, which expose women to DR-TB drugs for a longer period of time. (97,92,93)
2. Research, particularly pragmatic and intervention research, should address family planning needs in TB care and the implementation of integrated services. (93)
3. Given a lack of high level of evidence of studies, future research on pregnancy outcomes and family planning among women with TB or DR-TB should prioritize more rigorous study designs where possible, for example, level 1 experimental designs, level 2 mixed methods design, or level 3 prospective cohort studies with larger sample size.

1. Clinicians refer to providers who have direct patient contact including doctors, nurses, and pharmacists
2. This citation refers to the companion handbook to the WHO 2014 DR-TB guidelines, which were updated in 2020. However, in the updated 2020 guidelines family planning was not mentioned.

## Strength and limitations of the studies

There were several limitations to our systematic review. This review was limited by the inclusion of only peer-reviewed, English journal articles published in the 11 years. This likely excludes important findings in other languages and important programmatic data at a country level. Most studies in this review were of lower level of evidence, with the majority being case reports, case series, and retrospective cohort studies. The inclusion of lower level of evidence such as case reports allow for a comprehensive view of available clinical data across the globe. Aside from five cohort studies which took place in the US and China, most had small sample sizes and primarily reported exploratory and descriptive data. This may reduce the ability to generalize results beyond these studies to places that had higher burden of TB, DR-TB, and/or HIV coinfection with less access to health resources. Only one study evaluated family planning strategies as part of a quality improvement project. It would be important for future research to consider more rigorous study designs given the low level of evidence of current studies (e.g. experimental or mixed methods studies). This is especially important to consider for studies that evaluate family planning strategies for this high-risk population of women. It is also important

For peer review only

that future research prioritize studies with larger sample size in high TB, DR-TB, and/or HIV coinfection burden settings to evaluate pregnancy outcomes and family planning strategies since the majority of the larger cohort studies took place in US, a high-income country with low burden of disease.

Another limitation of this review is the variability of study design, definitions, and methods, which made comparison of outcomes difficult. Definitions of pregnancy outcomes were often heterogenous across multiple studies; some studies looked at TB, DR-TB, HIV coinfection outcomes as we defined in our review, some looked at fetal/newborn outcomes such as fetal distress, fetal demise versus stillbirth and intrauterine growth restriction or growth retardation rather than LBW or preterm birth without establishing clear definitions. Some failed to discuss TB/DR-TB outcomes of mothers or newborns. There were studies that did not define preterm birth or LBW, and there were studies that explored different pregnancy complications as a composite outcome, all of which made it difficult to cohesively conduct a meta-analysis. We sought to include all relevant studies regardless of methodological designs or quality to further our understanding of the scientific gaps that exist around maternal-child outcomes of women with TB or DR-TB and their family planning needs in both the clinical and research settings. Despite the weakness of the study designs and inclusion of studies of all quality types, the observational evidence consistently demonstrates adverse maternal and fetal/newborn outcomes across the globe in the last decade.

## CONCLUSION

TB in pregnancy is as a major risk factor of maternal and fetal morbidity and mortality, especially in low and middle-income countries where resources are limited. Pregnancy outcomes are worsened in the setting of TB/DR-TB, with or without HIV co-infection, and lack of health resources and access to specialized care. Offering family planning services at TB/DR-TB treatment initiation and education on risk of poor pregnancy outcomes should be considered as part of routine care in women of childbearing age.

**Acknowledgments**

This manuscript was supported by The Johns Hopkins School of Nursing, Johns Hopkins School of Nursing REACH Initiative, Jonas Philanthropies, Coverdell Peace Corps Fellows Scholarship,

For peer review only

and R01 AI104488 (JF, National Institute of Allergy and Infectious Diseases).

**Contributors**: the idea for this publication and preparation of the first draft was done by YN. YN and KM conducted the systematic search and assessed quality of articles. YN, KM, JF, and NW provided significant contributions to the draft and all its revisions. All authors reviewed the draft manuscript and approved the final draft for publication.

**Funding**: This work received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

**Competing interest**: none declared.

**Data availability statement:** all data relevant to the study are included in this article, appendix and supplemental files.

**Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram: identification and selection of relevant articles.**

For peer review only

## REFERENCES


1. Tuberculosis in women [Internet]. World Health Organization. 2019. Available from: https://www.who.int/tb/publications/tb_women_factsheet.pdf?ua=1
2. Grange J, Adhikari M, Ahmed Y, Mwaba P, Dheda K, Hoelscher M, et al. Tuberculosis in association with HIV/AIDS emerges as a major nonobstetric cause of maternal mortality in Sub-Saharan Africa. Int J Gynecol Obstet. 2010;108(3):181–3.
3. Mathad JS, Gupta A. Tuberculosis in Pregnant and Postpartum Women: Epidemiology, Management, and Research Gaps. Clin Infect Dis. 2012;55(11):1532–49.
4. Rifampicin. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 15, 2020. Available from www.micromedexsolutions.com. Subscription required to view.
5. Ethambutol. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 4, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
6. Pyrazinamide. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 4, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
7. Isoniazid. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated July 20, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
8. World Health Organization. Global Tuberculosis Report 2019. Geneva, Switzerland; 2019.


For peer review only

9. Hancock NL, Stuart GS, Tang JH, Chibwesha CJ, Stringer JS, Chi BH. Renewing focus on family planning service quality globally. Contraception and Reproductive Medicine. 2016 Dec;1(1):1-6.

10. World Health Organization. Reproductive Health. Medical eligibility criteria for contraceptive use. Geneva, Switzerland; 2015.

11. Pakistan Chest Society. Diagnosis & Management of Drug-susceptible Tuberculosis: A National Clinical Guideline. 2017.

12. Central TB Division. National Framework for A Gender-Responsive Approach to TB in India. Nirman Bhawan, New Delhi; 2019.

13. National Tuberculosis & Leprosy Control Programme. Guidelines on the Use of the Shorter Regimen and New Drugs in the Clinical and Programmatic Management of Drug Resistant Tuberculosis and Co-Infections. 2017.

14. Aboltins C, Crown L, Curtis N, Darby J, Denholm J, Flynn M, et al. Management , Control and Prevention of Tuberculosis: Guidelines for health care providers. South Melbourne: Digital House; 2015.

15. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ. 2009;339:b2535.

16. Dearholt SL, Dang, D. Johns Hopkins nursing evidence-based practice: model and guidelines (3rd ed.). Indianapolis, IN: Sigma Theta Tau International; 2017.

17. Tufanaru C, Munn Z, Aromataris E, Campbell J, Hopp L. Chapter 3: Systematic Reviews of Effectiveness. In: JBI Manual for Evidence Synthesis. JBI; 2020.

For peer review only

18. Asuquo B, Vellore AD, Walters G, Manney S, Mignini L, Kunst H. A case–control study of the risk of adverse perinatal outcomes due to tuberculosis during pregnancy. J Obstet Gynaecol (Lahore). 2012;32(7):635–8.

19. Palacios E, Dallman R, Muñoz M, Hurtado R, Chalco K, Guerra D, et al. Drug-Resistant Tuberculosis and Pregnancy: Treatment Outcomes of 38 Cases in Lima, Peru. Clin Infect Dis. 2009;48(10):1413–9.

20. Lin H-C, Lin H-C, Chen S-F. Increased risk of low birthweight and small for gestational age infants among women with tuberculosis. BJOG An Int J Obstet Gynaecol. 2010;117(5):585–90.

21. Ali A-AA, Abdallah TM, Rayis DA, Adam I. Maternal and perinatal outcomes of pregnancies associated with tuberculosis in eastern Sudan. Int J Gynecol Obstet. 2011;114(3):286–7.

22. Chang C-W, Wu PW, Yeh C-H, Wong K-S, Wang C-J, Chang C-C. Congenital tuberculosis: case report and review of the literature. Paediatr Int Child Health. 2017; 38(216–219):1–4.

23. Inal HA, Ozturk Inal ZH. A Successful Live Twin Birth by In vitro Fertilization in an Untreated Patient with Silent Genital Tuberculosis. Electron J Gen Med. 2015; 12(2):160–3.

24. Kaushal S, Dora SK, Thakur S. Spinal tuberculosis with paraplegia in pregnancy: A case report with management of spinal TB in pregnancy. J Nepal Med Assoc. 2015; 53(198):123–5.

For peer review only

25. Loughenbury P, Pavlou G, Dunsmuir R. Surgical Treatment of Tuberculous Paraparesis in the Third Trimester: A Report of Two Cases and Review of the Literature. Gynecol Obstet Invest. 2009; 68(3):213–6.

26. Sagar T, Gupta K, Rani M, Kaur IR. Disseminated tuberculosis in a newborn infant. Journal of Family Medicine and Primary Care. 2016 Jul;5(3):695.

27. Stuart R, Lewis A, Ramsden C, Doherty R. Congenital tuberculosis after in-vitro fertilisation. Med J Aust. 2009; 191(1):41–2.

28. Yeh JJ, Lin SC, Lin WC. Congenital tuberculosis in a neonate: A case report and literature review. Frontiers in Pediatrics. 2019 Jun 21;7:255.

29. Badve SA, Ghate SD, Badve MS, Rustagi T, Macchiwala T, Parekh AN, et al. Tuberculosis of spine with neurological deficit in advanced pregnancy: a report of three cases. Spine J. 2011; 11(1):e9–16.

30. Bekker A, Schaaf HS, Draper HR, Kriel M, Hesseling AC. Tuberculosis Disease during Pregnancy and Treatment Outcomes in HIV-Infected and Uninfected Women at a Referral Hospital in Cape Town. Wilkinson KA, editor. PLoS One. 2016; 11(11):e0164249.

31. Ali RF, Siddiqi DA, Malik AA, Shah MT, Khan AJ, Hussain H, Chandir S. Integrated tuberculosis screening during antenatal visits to improve Tuberculosis diagnosis and care: Results from a pilot project in Pakistan. International Journal of Infectious Diseases. 2021 Jun 1.

32. LaCourse SM, Greene SA, Dawson-Hahn EE, Hawes SE. Risk of Adverse Infant Outcomes Associated with Maternal Tuberculosis in a Low Burden Setting: A Population-Based Retrospective Cohort Study. Infect Dis Obstet Gynecol. 2016; 2016:1–8.

For peer review only

33. Loveday M, Hughes J, Sunkari B, Master I, Hlangu S, Reddy T, et al. Maternal and Infant Outcomes Among Pregnant Women Treated for Multidrug/Rifampicin-Resistant Tuberculosis in South Africa. Clin Infect Dis. 2021; 72(7):1158–68.

34. Sade S, Wainstock T, Sheiner E, Pariente G. Perinatal Outcome and Long-Term Infectious Morbidity of Offspring Born to Women with Known Tuberculosis. Journal of Clinical Medicine. 2020 Sep;9(9):2768.

35. Salazar-Austin N, Hoffmann J, Cohn S, Mashabela F, Waja Z, Lala S, et al. Poor Obstetric and Infant Outcomes in Human Immunodeficiency Virus-Infected Pregnant Women With Tuberculosis in South Africa: The Tshepiso Study. Clin Infect Dis. 2018; 66(6):921–9.

36. Yadav V, Sharma JB, Kachhawa G, Kulshrestha V, Mahey R, Kumari R, et al. Obstetrical and perinatal outcome in pregnant women with extrapulmonary tuberculosis. Indian J Tuberc. 2019; 66(1):158–62.

37. Ahmadzia HK, Khorrami N, Carter JA, Stone J, Amdur RL. Impact of human immunodeficiency virus, malaria, and tuberculosis on adverse pregnancy outcomes in the United States. J Perinatol. 2020; 40(2):240–7.

38. Fois A, Capobianco G, Crivelli P, Meloni G, Pirina P, Cherchi P, et al. Hemoptoe by tuberculosis in near term pregnant woman: A case report. Clin Exp Obstet Gynecol. 2018; 45(1):115–7.

39. Lu P, Li F, Wang Y, Xu C, Yu J, Zhang Y, et al. Uncommon Abdominopelvic Tuberculosis in Pregnancy. J Gynecol Surg. 2013; 29(6):306–8.

40. Du J, Dong S, Jia S, Zhang Q, Hei M. Clinical characteristics and post-discharge follow-up analyses of 10 infants with congenital tuberculosis: A retrospective observational study. Pediatric Investigation. 2021 Jun;5(2):86-93.

For peer review only

41. Chopra S, Siwatch S, Aggarwal N, Sikka P, Suri V. Pregnancy outcomes in women with tuberculosis: a 10-year experience from an Indian tertiary care hospital. Trop Doct. 2017; 47(2):104–9.

42. Mokhele I, Jinga N, Berhanu R, Dlamini T, Long L, Evans D. Treatment and Pregnancy Outcomes of Pregnant Women Exposed to Second-line Anti-tuberculosis Drugs in South Africa.

43. El-Messidi A, Czuzoj-Shulman N, Spence AR, Abenhaim HA. Medical and obstetric outcomes among pregnant women with tuberculosis: a population-based study of 7.8 million births. Am J Obstet Gynecol. 2016; 215(6):797.e1-797.e6.

44. Klimko A, Brandt A, Brustan MI. A Case of Miliary Perinatal Tuberculosis in a Preterm Newborn Infant Presenting as Peritonitis. Cureus. 2020 May;12(5).

45. Lee JS, Lim CH, Kim E, Lim H, Lee Y, Choung JT, Yoo Y. Congenital miliary tuberculosis in an 18-day-old boy. Korean Journal of Pediatrics. 2016 Nov;59(Suppl 1):S64.

46. Peng W, Yang J, Liu E. Analysis of 170 cases of congenital TB reported in the literature between 1946 and 2009. Pediatric Pulmonology. 2011 Dec;46(12):1215-24.

47. Rajput D, Gupta A, Roshan R, Kumar A. Stillbirth as the primary manifestation of disseminated tuberculosis in a young immunocompetent mother with multiple perforations of the ileum. BMJ Case Reports CP. 2021 Feb 1;14(2):e239386.

48. Li Q, Song Y, Chen H, Xie L, Gao M, Ma L, Huang Y. Retrospective Analysis of 28 Cases of tuberculosis in pregnant Women in china. Scientific Reports. 2019 Oct 25;9(1):1-8.

For peer review only

49. Toro PL, Schneider KL, Carter RJ, Abrams EJ, El-Sadr WM, Howard AA. Maternal and Infant Outcomes With Concurrent Treatment of Tuberculosis and HIV Infection in Pregnant Women. JAIDS J Acquir Immune Defic Syndr. 2011; 56(2):e63–6.

50. Islam B, Islam N, Mehkri H, AlQaasimi M. Abdominal tuberculosis and spontaneous miscarriage. BMJ Case Rep. 2017; 2017:220022.

51. Jacquemyn Y, Van Casteren C, Luijks M, Colpaert C. Disseminated tuberculosis in pregnancy unknown to doctors in Western Europe case presentation:'part of the routine study in infertility'. Case Reports. 2012 Sep 13;2012:bcr2012006227.

52. Ahuja V, Gombar S, Kumar N, Goyal N, Gupta K. Pregnancy with bilateral tubercular pleural effusion: challenges. Trop Doct. 2014; 44(2):116–8.

53. Gai X, Chi H, Cao W, Zeng L, Chen L, Zhang W, Song D, Wang Y, Liu P, Li R, Sun Y. Acute miliary tuberculosis in pregnancy after in vitro fertilization and embryo transfer: a report of seven cases. BMC Infectious Diseases. 2021 Dec;21(1):1-9.

54. Rathod AK, Singh V, Patil P, Singh H. Is decompressive surgery the only treatment option? A case series of patients with spinal tuberculosis in advanced pregnancy. Eur Spine J. 2017; 26(12):3170–7.

55. Ye R, Wang C, Zhao L, Wu X, Gao Y, Liu H. Characteristics of miliary tuberculosis in pregnant women after in vitro fertilisation and embryo transfer. The International Journal of Tuberculosis and Lung Disease. 2019 Feb 1;23(2):136-9.

56. Barbosa OA, Teles FM, Maia ACC, Pessoa GMF, de Alencar IM, de Sousa ET, et al. Disseminated hematogenous tuberculosis in puerperium—case report. Oxford Med Case Reports. 2019; 2019(11):479–81.

For peer review only

57. Raznatovska OM, Fedorec AV, Shalmina MO, Grekova TA. Clinical features of pregnancy in multidrug-resistant tuberculosis and type 1 diabetes mellitus comorbidities (a case report). Медичні перспективи. 2019;24(3).

58. Shi YF, Shi XH, Zhang Y, Chen JX, Lai WX, Luo JM, Ba JH, Wang YH, Chen JN, Wu BQ. Disseminated Tuberculosis Associated Hemophagocytic Lymphohistiocytosis in a Pregnant Woman With Evans syndrome: A Case Report and Literature Review. Frontiers in Immunology. 2021 Jun 10;12:2302.

59. Zhu Z, Zhang M, Li Y. Anti-tuberculosis drug-induced acute liver failure requiring transplantation in the second trimester of pregnancy: a case report. BMC Pegnancy and Childbirth. 2021 Dec;21(1):1-6.

60. Cornish EF, Hudson J, Sayers R, Loveday M. Improving access to contraception through integration of family planning services into a multidrug-resistant tuberculosis treatment programme. BMJ Sex Reprod Heal. 2020; 46(2):152–5.

61. Bishara H, Goldstein N, Hakim M, Vinitsky O, Shechter-Amram D, Weiler-Ravell D. Tuberculosis during Pregnancy in Northern Israel, 2002-2012: Epidemiology and Clinical Practices. Isr Med Assoc J. 2015; 17(6):346–50.

62. Walt MV, Masuku S, Botha S, Nkwenika T, Keddy KH. Retrospective record review of pregnant women treated for rifampicin-resistant tuberculosis in South Africa. PloS one. 2020 Sep 24;15(9):e0239018.

63. Alaoui FZF, Rachad M, Chaara H, Bouguern H, Melhouf MA. Peritoneal tuberculosis in pregnancy: a case report. Pan Afr Med J. 2012;12:65.

For peer review only

64. Brouwer A, Degrieck N, Rasschaert M, Lockefeer F, Huizing M, Tjalma W. Tuberculous mastitis presenting as a lump: a mimicking disease in a pregnant woman case report and review of literature. Acta Clinica Belgica. 2014 Oct 1;69(5):389-94.

65. Erenel H, Alpay V, Alakbarova U, Mut A, Atilgan H, Demiroz AS, Ocal P. Intraperitoneal tuberculosis abscess in pregnancy presenting as an adnexal mass: a case report. Journal of Obstetrics and Gynaecology. 2021 Feb 17;41(2):305-7.

66. John M, Chhikara A, John DM, Khawar N, Brown B, Narula P. Diagnosis of tuberculosis in an asymptomatic child, sibling, and symptomatic pregnant mother in new York City by tuberculin skin testing and the importance of screening high-risk urban populations for tuberculosis. The American Journal of Case Reports. 2018;19:1004.

67. Lahbabi M, Brini J, Massaoudi K. Tuberculous peritonitis in pregnancy: a case report. Journal of Medical Case Reports. 2014 Dec;8(1):1-4.

68. Masukume G, Sengurayi E, Moyo P, Feliu J, Gandanhamo D, Ndebele W, et al. Massive hemoptysis and complete unilateral lung collapse in pregnancy due to pulmonary tuberculosis with good maternal and fetal outcome: a case report. BMC Res Notes. 2013; 6(1):335.

69. Shital P, Mirza M. Laryngeal & Lower lung field tuberculosis in pregnancy: A case report. Electron J Gen Med. 2018; 15(2):1–5.

70. Fernandez D, Salami I, Davis J, Mbah F, Kazeem A, Ash A, et al. HIV-TB Coinfection among 57 Million Pregnant Women, Obstetric Complications, Alcohol Use, Drug Abuse, and Depression. J Pregnancy. 2018; 2018:1–8.

For peer review only

71. Dennis EM, Hao Y, Tamambang M, Roshan TN, Gatlin KJ, Bghigh H, et al. Tuberculosis during pregnancy in the United States: Racial/ethnic disparities in pregnancy complications and in-hospital death. Jhaveri R, editor. PLoS One. 2018; 13(3):e0194836.

72. Adhikari EH, Duryea EL, Rac MWF, Sheffield JS. Genitourinary Tuberculosis: A Rare Cause of Obstructive Uropathy in Pregnancy. Case Rep Obstet Gynecol. 2014; 2014(985682):1–4.

73. Lin S, Guan W, LaZhou C, Shi Y. Left lung hypoplasia with a right tuberculous pleural effusion after childbirth: A case report. Medicine. 2018 May;97(21).

74. Muin DA, Wagner K, Burian R, Ghaem Maghami N, Lapaire O. Brainstem Tuberculoma in Pregnancy. Case Rep Obstet Gynecol. 2015; 2015:1–6.

75. Raouf S, Sharma S, Sunanda G V., Jafettasj J, Papaioannou S. Disseminated extra pulmonary tuberculosis in an immune competent pregnant woman. J Obstet Gynaecol (Lahore). 2009; 29(2):148–50.

76. Modi P, Khanna R, Reddy N, Patankar A, Patel S, Nair G, Gopal S, Uppe A. COVID-19 and tuberculosis co-infection in pregnancy–A case series and review. Journal of Mother and Child. 2021 Oct 29.

77. Tabarsi P, Moradi A, Baghaei P, Marjani M, Shamaei M, Mansouri N, et al. Standardised second-line treatment of multidrug-resistant tuberculosis during pregnancy [Short communication]. Int J Tuberc Lung Dis. 2011; 15(4):547–50.

78. Laniado-Laborín R, Carrera-López K, Hernández-Pérez A. Unexpected Pregnancy during Treatment of Multidrug-resistant Tuberculosis. Turkish Thorac J. 2018; 19(4):226–7.

For peer review only

79. Rendell NL, Batjargal N, Jadambaa N, Dobler CC. Risk of tuberculosis during pregnancy in Mongolia, a high incidence setting with low HIV prevalence. Int J Tuberc Lung Dis. 2016; 20(12):1615–20.

80. Sharma S, Ahad A, Gupta N, Sharma V. Primary gingival tuberculosis in pregnancy: A rare combination. J Indian Soc Periodontol. 2018; 22(2):178.

81. Suresh S, Sharath BN, Anita S, Lalitha R, Prasad TJ, Rewari BB. TB-HIV co-infection among pregnant women in Karnataka, South India: A case series. J Infect Public Health. 2016; 9(4):465–70.

82. Baluku JB, Bongomin F. Treatment outcomes of pregnant women with drug-resistant tuberculosis in Uganda: A retrospective review of 18 cases. International Journal of Infectious Diseases. 2021 Apr 1;105:230-3.

83. Acquah R, Mohr-Holland E, Daniels J, Furin J, Loveday M, Mudaly V, Reuter A. Outcomes of Children Born to Pregnant Women With Drug-resistant Tuberculosis Treated With Novel Drugs in Khayelitsha, South Africa: A Report of Five Patients. The Pediatric Infectious Disease Journal. 2021 May;40(5):e191.

84. van de Water BJ, Brooks MB, Huang CC, Trevisi L, Lecca L, Contreras C, Galea J, Calderon R, Yataco R, Murray M, Becerra MC. Tuberculosis clinical presentation and treatment outcomes in pregnancy: a prospective cohort study. BMC Infectious Diseases. 2020 Dec;20(1):1-8.

85. Rickman HM, Cohn S, Lala SG, Waja Z, Salazar-Austin N, Hoffmann J, et al. Subclinical tuberculosis and adverse infant outcomes in pregnant women with HIV. Int J Tuberc Lung Dis. 2020; 24(7):681–5.

For peer review only

86. World Health Organization. WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-resistant tuberculosis treatment. Geneva, Switzerland; 2020.

87. Schnippel K, Ndjeka N, Conradie F, Berhanu R, Claasen Z, Banoo S, et al. A call to action: Addressing the reproductive health needs of women with drug-resistant tuberculosis. South African Med J. 2016; 106(4):333.

88. World Health Organization. Quality of care in the provision of sexual and reproductive health services. Geneva, Switzerland; 2011.

89. Freyder M, Craig L, Kaji A. Monitoring the Integration of Family Planning and HIV Services: A Manual to Support the Use of Indicators to Measure Progress toward PEPFAR's 90–90-90 Targets and Protect Women's Reproductive Rights. Chapel Hill, North Carolina; 2016.

90. FHI 360, World Health Organization, Uniter States Agency for International Development. Strategic Considerations for Strengthening the Linkages between Family Planning and HIV/AIDS Policies, Programs, and Services. Geneva, Switzerland; 2009.

91. Nahid P, Mase SR, Migliori GB, Sotgiu G, Bothamley GH, Brozek JL, et al. Treatment of Drug-Resistant Tuberculosis. An Official ATS/CDC/ERS/IDSA Clinical Practice Guideline. Am J Respir Crit Care Med. 2019; 200(10):e93–142.

92. Gupta A, Mathad JS, Abdel-Rahman SM, Albano JD, Botgros R, Brown V, et al. Toward Earlier Inclusion of Pregnant and Postpartum Women in Tuberculosis Drug Trials: Consensus Statements From an International Expert Panel. Clin Infect Dis. 2016; 62(6):761–9.

For peer review only

93. McKenna L, Frick M, Lee C, Namutamba D, Smit L, Theunissen M, et al. A Community Perspective on the Inclusion of Pregnant Women in Tuberculosis Drug Trials. Clin Infect Dis. 2017; 65(8):1383–7.

94. World Health Organization. WHO consolidated guidelines on drug-resistant tuberculosis treatment. Geneva, Switzerland; 2019.

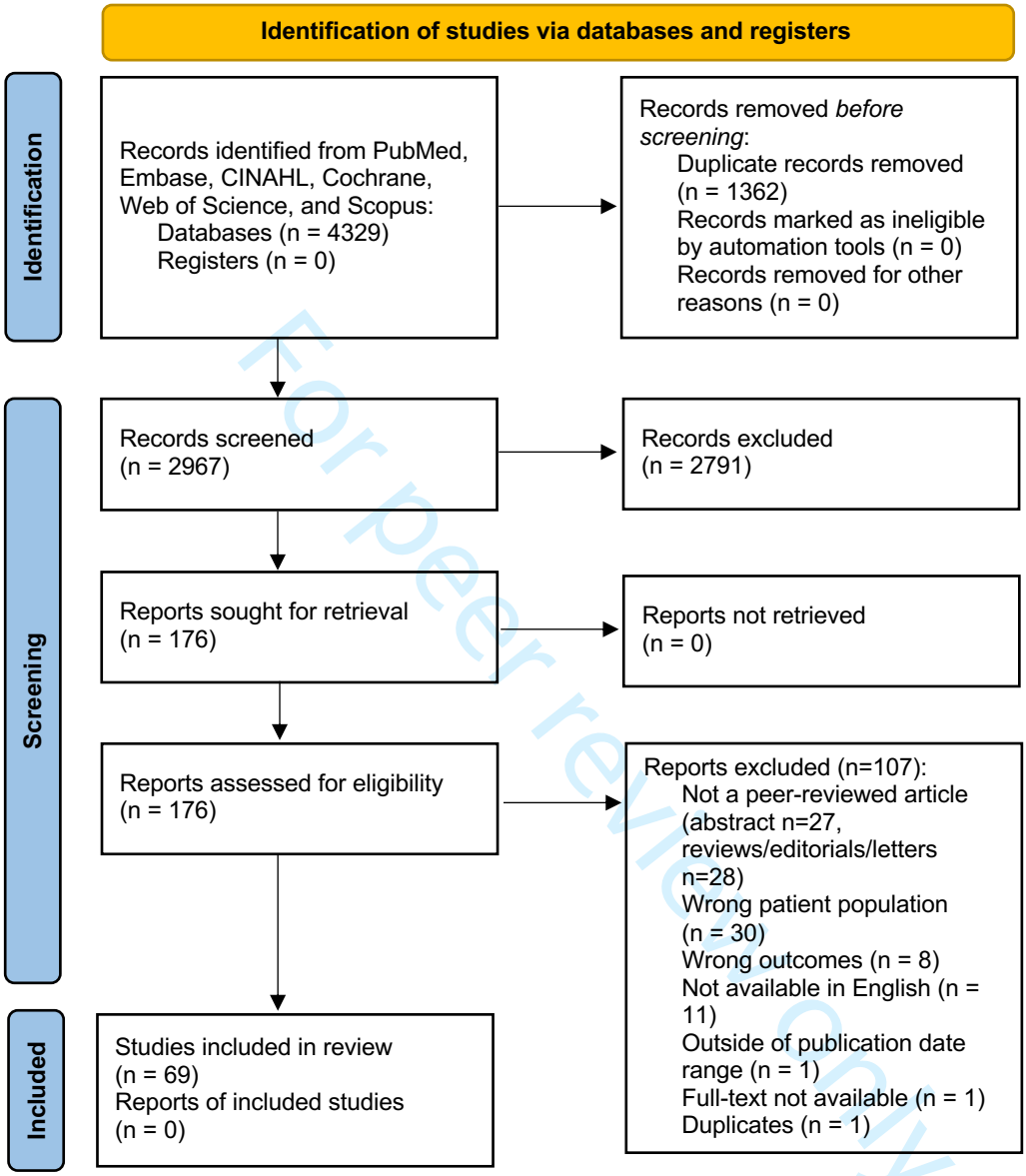
95. Nunn AJ, Rusen ID, Van Deun A, Torrea G, Phillips PP, Chiang CY, Squire SB, Madan J, Meredith SK. Evaluation of a standardized treatment regimen of anti-tuberculosis drugs for patients with multi-drug-resistant tuberculosis (STREAM): study protocol for a randomized controlled trial. Trials. 2014 Dec;15(1):1-0.

96. World Health Organization. Companion handbook to the WHO guidelines for the programmatic management of drug-resistant tuberculosis. Geneva, Switzerland: World Health Organization; 2014.

97. Sun Q, Zhang H, Zhang Y, Peng Z, Lu J, Ma X. Increased Risk of Stillbirth among Women whose Partner Has Tuberculosis. BioMed research international. 2021 Sep 15;2021.

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Figure 1. PRISMA flow diagram: identification and selection of relevant articles




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Supplemental Table 1. Critical Appraisal of the included studies

| Design | Authors | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Case Reports (Level of evidence: 5)** | | Demographic characterstics | Patient's history | Clinical presentation | Diagnostics and assessment | Intervention or treatment procedures | Post-intervention clinical condition | Adverse events identified? | Provision of takeaway lessions | | | Number of criteria met (quality) n/8 |
| Case Report | Adhikari (2014)[72] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Ahuja (2014)[52] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Alaoui (2012)[63] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Barbosa (2019)[56] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Brouwer (2014)[64] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Chang (2018)[22] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Erenel (2019)[65] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Fois (2017)[38] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Inal (2015)[23] | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 7 |
| Case Report | Islam (2017)[50] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Jacquemyn (2021)[51] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | John (2018)[66] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Kaushal (2015)[24] | x | ✓ | ✓ | x | ✓ | x | ✓ | ✓ | | | 5 |
| Case Report | Kimko (2020)[44] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Lahbabi (2014)[67] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Laniado-Laborin (2017)[78] | x | ✓ | ✓ | x | ✓ | ✓ | x | ✓ | | | 5 |
| Case Report | Lee (2015)[45] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Lin (2018)[73] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Lu (2013)[39] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Loughenbury (2009)[25] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Masukume (2013)[68] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Muin (2015)[74] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Rajput (2021)[47] | ✓ | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | | | 6 |
| Case Report | Raznatovska (2021)[57] | ✓ | x | ✓ | ✓ | x | x | ✓ | ✓ | | | 5 |
| Case Report | Sagar (2016)[26] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Sharma (2018)[80] | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 7 |
| Case Report | Shi (2021)[58] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Shital (2018)[69] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Stuart (2009)[27] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Yeh (2019)[28] | ✓ | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | | | 7 |
| Case Report | Zhu (2021)[59] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| **Case Series (Level of evidence: 5)** | | Clear criteria for inclusion? | Condition measured in a standard, reliable way for all participants? | Valid methods for identification of condition included? | Consecutive inclusion of participants? | Complete inclusion of participants? | Clear reporting of demographics? | Clear reporting of clinical information? | Outcomes or follow up results clearly reported? | Clear reporting of presenting sites/clinics demographic information? | Statistical analysis appropriate? | Number of criteria met (quality) n/10 |
| Case Series | Acquah (2021)[83] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Case Series | Badve (2011)[29] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | x | x | 6 |
| Case Series | Bishara (2015)[61] | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Case Series | Du (2021)[40] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Li (2019)[48] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Modi (2021)[76] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Palacios (2009)[19] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Case Series | Peng (2011)[46] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Rathod (2017)[54] | ✓ | ✓ | x | x | x | ✓ | ✓ | ✓ | x | x | 5 |
| Case Series | Suresh (2015)[81] | ✓ | ✓ | x | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | 8 |
| Case Series | Tabarsi (2011)[77] | ✓ | ✓ | x | x | x | ✓ | ✓ | ✓ | ✓ | ✓ | 7 |
| Case Series | Toro (2011)[49] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | 9 |

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| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Case Series | Ye (2019)[55] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | | 9 |
| **Case-control (Level of evidence: 3)** | | Groups comparable? | Cases and controls appropriately matched? | Same criteria for identification of cases and controls? | Exposure measured in a standard, reliable, and valid way? | Exposure measured in the same way for cases and controls? | Confounding factors stated? | Strategies to deal with confounding factors stated? | Outcomes assessed in a standard, reliable, and valid way for cases and controls? | Exposure period of interest long enough to be meaningful? | Appropriate statistical analysis? | | Number of criteria met (quality) n/10 |
| Case-control | Asuquo (2012)[18] | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | ✓ | x | x | | 7 |
| Case-control | Chopra (2017)[41] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | ✓ | | 9 |
| Case-control | Lin (2010)[20] | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | 9 |
| **Cohort Study (Level of evidence: 3)** | | Groups similar and recruited from the same population? | Exposures measured similarly to exposed and unexposed groups? | Exposure measured in a valid and reliable way? | Confounding factors identified? | Strategies to deal with confounding factors stated? | Groups free of outcome(s) at start of study? | Outcomes measured in valid and reliable way? | Follow up time sufficiently long enough for outcome(s) to occur? | Follow up complete? Reasons for loss to follow up described? | Were strategies to address incomplete follow up used? | Appropriate statistical analysis? | Number of criteria met (quality) n/11 |
| Retrospective Cohort Study | Ahmadzia (2020)[37] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | Ali (2011)[21] | ✓ | | | x | x | ✓ | ✓ | ✓ | | | x | 4 |
| Retrospective Cohort Study | Ali (2021)[31] | | | ✓ | | | ✓ | ✓ | ✓ | ✓ | | ✓ | 6 |
| Retrospective Cohort Study | Baluku (2021)[82] | | | ✓ | x | x | ✓ | ✓ | ✓ | | | ✓ | 5 |
| Prospective Cohort Study | Bekker (2016)[30] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | x | ✓ | 9 |
| Retrospective Cohort Study | Dennis (2018)[71] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | x | ✓ | 9 |
| Retrospective Cohort Study | El-Messidi (2016)[43] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | Fernandez (2018)[70] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | LaCourse (2016)[32] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | Loveday (2020)[33] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | 10 |
| Retrospective Cohort Study | Mokhele (2021)[42] | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | | ✓ | 9 |
| Retrospective Cohort Study | Rendell (2016)[79] | x | x | ✓ | x | x | | ✓ | x | | | ✓ | 3 |
| Prospective Cohort Study | Salazar-Austin (2007)[35] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | | ✓ | 8 |
| Retrospective Cohort Study | Sade (2020)[34] | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Retrospective Cohort Study | Sun (2021)[97] | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Retrospective Cohort Study | van der Walt (2020)[62] | | | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 6 |
| Retrospective Cohort Study | Yadav (2019)[36] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | | ✓ | 8 |
| **Other (Level of evidence: 3)** | | Groups similar and recruited from the same population? | Exposures measured similarly to exposed and unexposed groups? | Exposure measured in a valid and reliable way? | Confounding factors identified? | Strategies to deal with confounding factors stated? | Groups free of outcome(s) at start of study? | Outcomes measured in valid and reliable way? | Follow up time sufficiently long enough for outcome(s) to occur? | Follow up complete? Reasons for loss to follow up described? | Were strategies to address incomplete follow up used? | Appropriate statistical analysis? | Number of criteria met (quality) n/11 |
| Secondary Data Analysis | Rickman (2020)[85] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Secondary Data Analysis | van de Water (2020)[84] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 8 |
| **Project Improvement (Level of evidence: 5)** | | Cause and effect? | Participants in each comparison group similar? | Participants received similar treatment? | Control group? | Multiple measurements of outcome pre and post intervention? | Follow up complete? | Outcomes in comparison groups measured the same way? | Outcomes measured in a reliable way? | Appropriate statistical analysis used? | | | Number of criteria met (quality) n/9 |
| Project Improvement | Cornish (2020)[60] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | | | 7 |

***Blank cells denote not applicable

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Supplemental Table 2. Summary of Pregnancy Outcomes and Other Relevant Findings

| Design | First Author (Year), Reference | Setting | Sample | PTB, EPTB, congenital | HIV Co-infection | Maternal outcomes | Fetal/Newborn outcomes | Other relevant findings |
|---|---|---|---|---|---|---|---|---|
| **Case Report** | | | | | | | | |
| Case Report | Adhikari (2014)[72] | US | 1 (30 yo) | DR EPTB (genitourinary) | no | cured of TB | induction of labor due to severe pre-eclampsia at term | postpartum complicated by pyelonephritis; fibrosis and chronic inflammation of ureteral segments |
| Case Report | Ahuja (2014)[52] | India | 1 (27 yo) | PTB | no | discharged home to continue ATT | fetal demise | delayed TB diagnosis; status post ventilator with resolved pulmonary embolism; resolved anemia |
| Case Report | Alaoui (2012)[63] | Morocco | 1 (37 yo 23 weeks gestation) | EPTB (peritoneal) | no | cured of TB | normal birth weight, delivered at term: vaginal delivery 37 weeks | NA |
| Case Report | Barbosa (2019)[56] | Brazil | 1 (22 yo 22 weeks gestation) | PTB, EPTB (disseminated) | no | death | induced abortion due to suspicion of chorioamnionitis | NA |
| Case Report | Brouwer (2014)[64] | Belgium | 1 (33 yo 4 weeks gestation) | EPTB (left breast) | no | completion of TB treatment | delivered at term | misdiagnosis of TB and delayed treatment x13 weeks from presentation |
| Case Report | Chang (2017)[22] | Taiwan | 1 (14 day old infant, delivered at 35 weeks) | congenital | no | not discussed | LBW; completed TB treatment, gradual resolution of miliary TB | NA |
| Case Report | Erenel (2021)[65] | Turkey | 1 (24 yo 15 weeks gestation) | EPTB (intraperitoneal) | no | Completed treatment | delivered at term, normal weight, negative for TB; BCG vaccine given; INH prophylaxis given x6 months | NA |
| Case Report | Fois (2017)[38] | Italy | 1 (28 yo 36 weeks gestation) | PTB | no | cured of TB | preterm cesarean delivery at 36 weeks, outcome unknown | NA |
| Case Report | Inal (2015)[23] | Turkey | 1 (34 yo, twins at 27 and 30 weeks gestation) | EPTB (genital) | no | cured of TB | preterm cesarian delivery for both twins at 30 week gestation; LBW | NA |
| Case Report | Islam (2017)[50] | Saudi Arabia | 1 (23 yo 23 weeks gestation) | EPTB (abdominal) | no | cured of TB | spontaneous abortion | dilatation and curretage caused uterine perforation and sepsis; resolved sepsis, bilateral pleural effusion; TB treatment was interrupted during transfer of care due to miscommunication |
| Case Report | Jacquemyn (2012)[51] | Belgium | 1 (28 yo 13 weeks gestation) | PTB, EPTB (disseminated) | no | cured of TB | spontaneous abortion | acid-fast bacilli on examination of placenta, but not fetus |
| Case Report | John (2018)[66] | US | 1 (37 yo 19 weeks gestation) | PTB | no | recovered fully | healthy infant at birth, treated with isoniazid prophylaxis, negative T-spot test at age 3-month and isoniazid was discontinued | mother had delayed diagnosis of TB; cough, fever, night sweats, 10-lb weight loss in the last 3 months - T-spot test + (30mm induration) and referred to pulmonologist - placed on azithromycin for presumptive PNA; admitted to inpatient TB treatment when 2-yo daughter was screened + for TB |
| Case Report | Kaushal (2015)[24] | India | 1 (27 yo 35 weeks gestation) | EPTB (spinal) | no | continued TB treatment after discharge; doing well at 3 month post-discharge | fetal distress, emergency cesarian section at 37 weeks; ATT prophylaxis x3 months; no congenital TB at 3 month post-discharge | improvement in motor power over weeks |
| Case Report | Klimko (2020)[44] | Romania | 1 (27 days old infant delivered at 33 weeks) | congenital | no | mom diagnosed with latent TB and isolated from infant; TB treatment outcome not discussed | improved TB; premature birth, LBW (2100g) | presents with hepatosplenomegaly, respiratory distress, hypotonic, irritable; misdiagnosis of late-onset culture-negative neonatal sepsis - patient's condition worsened prompting TB screening of mother, paracenthesis performed + for TB and started on treatment |

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|---|---|---|---|---|---|---|---|---|
| Case Report | Lahbabi (2014)[67] | Morocco | 1 (31 yo 22 weeks gestation) | EPTB (peritoneal) | no | cured of TB | delivered at term, normal weight, no TB | NA |
| Case Report | Laniado-Laborin (2017)[78] | Mexico | 1 (15 yo 8 weeks gestation) | DR PTB | no | cured of TB; normal vaginal delivery | NA | fetal exposure to levofloxacin and ethionamide; provider offered induced abortion |
| Case Report | Lee (2015)[45] | South Korea | 1 (18 days old infant) | congenital | no | No symptoms at 8-month follow up | improvement of lung infiltration and no symptoms after 8 months | delayed diagnosis of TB in mother; baby's initial presentation of respiratory distress and fever, initial misdiagnosis of PNA, then started on TB treatment on day #2 of hospitalization |
| Case Report | Lin (2018)[73] | China | 1 (31 yo, gestation age unknown) | PTB | no | cured of TB | spontaneous vaginal delivery; TB outcome on neonate not provided | 1 month delay of TB diagnosis after presenting with symptoms |
| Case Report | Lu (2013)[39] | China | 1 (38 yo 28 weeks gestation) | TB (abdominopelvic) | no | cured of TB | preterm cesarean delivery at 29 weeks, outcome unknown | abdominal hysterectomy, bilateral salpingectomy, and omentectomy |
| Case Report | Loughenbury (2009)[25] | UK | 2 (33 weeks and 33 weeks) | EPTB (spinal) | no | cured of TB | preterm cesarian delivery for both at 33 and 34 weeks; LBW reported (1980g and 2100g) | moms given steroids to aid fetal lung maturation |
| Case Report | Masukume (2013)[68] | Zimbabwe | 1 (33 yo 18 weeks gestation) | PTB | no | cured of TB; vaginal delivery at 41 weeks | normal birth weight | delayed TB diagnosis |
| Case Report | Muin (2015)[74] | Switzerland | 1 (19 yo, 28 weeks gestation) | EPTB (brainstem) | no | cured of TB; failed induction at 40 weeks, underwent cesarian section | not discussed | improvement neurological symptoms; steroids to aid fetal lung maturation; recommendation for treatment via a multidisciplinary approach inclusive of neonatal care |
| Case Report | Rajput (2021)[47] | India | 1 (21 yo female postpartum) | EPTB (intestinal) | no | died from TB due to sepsis secondary to 20 ileal perforations from TB and delayed presentation to the hospital; asymptomatic TB during pregnancy with symptom manifestation postpartum | full term stillbirth | delayed TB diagnosis, was not started on TB treatment prior to death |
| Case Report | Raouf (2009)[75] | UK | 1 (23 yo) | EPTB (brain and eye) | no | cured of TB; complications resolved with scarring of eye | full term induction vaginal birth; normal birth weight | Patient stopped ATT by herself 3 weeks before delivery because she wanted to breastfeed |
| Case Report | Raznatovska (2019)[57] | Ukraine | 1 (38 yo became pregnant during TB treatment) | DR PTB | no | treatment failure and extensive resistant TB by month 7 of treatment; recommended induced abortion based on clinical indications of: non-conversion of culture after 3 months of DR-TB treatment, nephropathy and intoxication syndrome secondary to diabetes, adenomyosis and bloody vaginal discharge | induced abortion due to advanced TB disease | Initial diagnosis found resistance to rifampicin, unclear what treatment patient was on; on month 1, sputum found resistance to further isoniazid and pyrazinamide, unclear what therapy was placed on her; on month 7, was further found to be resistant to ofloxacin, kanamycin, and capreomycin, unclear what therapy was placed on her at this time |
| Case Report | Sagar (2016)[26] | India | 1 (20 day old baby) | Congenital | no | unknown TB outcome; diagnosed with PTB at 8 month pregnancy and started on TB treatment 8 days prior to baby's presentation of fever, decreased appetite, and rapid breathing | preterm birth at 24 week gestation; LBW 1800g at 20 days old; X-ray after 12 month treatment normal | treated x12 months on isoniazid, rifampicin, pyrazinamide, and amikacin |
| Case Report | Sharma (2018)[80] | India | 1 (25 yo old 4th month of pregnancy) | EPTB (gingiva) | no | cured of TB; normal vaginal delivery | NA | NA |
| Case Report | Shi (2021)[58] | China | 1 (26 yo 15 week gestation) | PTB, EPTB (disseminated) | no | improved with combined therapeutic therapy; follow up at 2 month doing well on isoniazid and rifampicin; Patient with history of Evans syndrome on daily corticosteroid. On day 3 hospitalization, condition deteriorated; final diagnoses included disseminated TB-associated hemophagocytic lymphohistiocytosis, severe PNA, respiratory failure, severe acute respiratory distress syndrome | fetal death day 5 of hospitalization, caseating tuberculous granulomas found in placenta | was initiated on isoniazid, rifampicin, ethambutol, pyrazinamide, and moxifloxacin, then on "four drug…regimen" with methylprednisolone and immunoglobin and plasma exchange with improvement and d/c day 19 |

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|---|---|---|---|---|---|---|---|---|
| Case Report | Shital (2018)[69] | India | 1 (25 yo 24 weeks gestation) | PTB, EPTB (laryngeal) | no | cure of TB; normal vaginal delivery | normal birth weight | NA |
| Case Report | Stuart (2009)[27] | Australia | 1 (newborn delivered at 28 weeks) | Congenital | no | TB outcome not reported; no other maternal outcome reported | premature birth, emergency cesarian section, LBW; ventilator use due to respiratory distress; neonatal jaundice, right-sided Erb palsy | neutropenia due to ethambutol in newborn; all newborn's complications resolved and newborn was discharged from hospital |
| Case Report | Yeh (2019)[28] | Taiwan | 1 (8 day old neonate) | Congenital | no | initiated on TB treatment but died 3 days later | delivered at 37 weeks with LBW (2380g); presented with fever x2 days; diagnosed with TB day 20, initiated on TB treatment at day 24; no TB outcome reported | delayed TB diagnosis; 33 yo mother with dry cough 1 week postpartum, admitted to ICU for altered mental status 24 day postpartum, tested for TB due to suspected TB in neonate |
| Case Report | Zhu (2021)[59] | China | 1 (26 yo 11 weeks gestation) | EPTB (plural space) | no | TB treatment completion without relapse or liver damage at month 8 follow up; underwent liver transplant at 17 week gestation; uterine curettage day 27 due to incomplete abortion | induced termination of pregnancy by induction at 21 week gestation (30 day postop liver transplant of mother) due to ventricle widening in the fetus | TB treatment changed 3 times: isoniazid, rifampicin, and pyrazinamide at initiation; then amikacin, levofloxacin, and meropenem after liver transplant; then linezolid, levofloxacin, and pyridoxine at postop day 20 |
| **Case Series** | | | | | | | | |
| Case Series | Acquah (2021)[83] | South Africa | 5 (18-29 years old) | DR PTB | yes (n=3) | 3 cured, 2 lost to follow up | delivered at term, no congenital abnormalities, none developed TB or HIV | all women were exposed to regimens containing bedaquiline, delamanid, and linezolid |
| Case Series | Badve (2011)[29] | India | 3 (third trimester pregnancies, median age=30) | EPTB (spinal) | no | 1 woman with residual spastic paraparesis, otherwise complete neurological recovery in the other 2 women; unclear TB outcomes | 1 spontaneous abortion; urgent 2 cesarian section with 1 LBW (2400g); 1 newborn suffers neonatal sepsis in premature care facility; 1 newborn normal birth weight | kyphosis, spinal cord compression |
| Case Series | Bishara (2015)[61] | Israel | 6 (35-41 weeks gestation, median age=22) | both (n=4 PTB, n=2 EPTB) | no | all 6 cured of TB | 1 intrauterine growth restriction; 2 out of 6 PNA post-delivery (moms had EPTB), 2 newborns treated for latent TB | TB diagnostic and treatment delay in all 6 women |
| Case Series | Du (2021)[40] | China | 10 (mean gestational age 36.6 SD 2.2 weeks) | congenital | no | none treated for TB during pregnancy though 1 woman was diagnosed during pregnancy | 5 were preterm infants (<37 weeks); during treatment: 4 failed to thrive, 3 had neurological sequelae including seizures, 1 had seizure requiring shunt and physical therapy; at 2 month follow up: 1 baby lost to follow up, five fully recovered, 3 babies had hepatotoxicity secondary to TB treatments; at 6 months of age, 3/9 babies had normal growth and development | NA |
| Case Series | Gai (2021)[53] | China | 7 (ages 28-34) | PTB, EPTB (miliary TB, with 2 TB meningitis cases) | not reported | achieved pregnancy after in-vitro fertilization; 4 spontaneous abortion, 3 induced abortions; all recovered after TB treatment | NA | clinical pregnancy and live birth rates significantly lower in infertile patients with untreated prior PTB than in those without |
| Case Series | Li (2019)[48] | China | 28 pregnant women with TB (IQR 27-33) | PTB, EPTB | no | 10/28 were initially misdiagnosed as other respiratory illnesses; 27 cured of TB, 1 death | 4 spontaneous abortions, 9 induced abortions; 4 neonates were healthy; 1 small for gestational age, 1 stillbirth, 2 congenital TB, 1 died at 4 months old | 35.7% of cases of misdiagnosis potentially due to atypical presentation leading to treatment delays and adverse pregnancy outcomes |

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|---|---|---|---|---|---|---|---|---|
| Case Series | Modi (2021)[76] | Inida | 2 pregnant women with TB-COVID-19 coinfection (third trimester pregnancy) | PTB | no | patient 1 treated with quadruple therapy for TB and remdesivir and oxygen in ICU x10 days and was discharged; patient 2 underwent c-section at 38 weeks and managed in ICU with O2 and remdesivir and quadruple therapy for TB | term delivery with normal weight baby for patient 2; no discussion regarding patient 1's fetus | NA |
| Case Series | Palacios (2009)[19] | Peru | 38 (mean age=24.4 SD=5.8) | DR PTB (n=37), EPTB (n=1) | yes (n=3) | 23 (n=38, 61%) cured of TB; 4 clinical TB deterioration, 8 deaths; 2 episodes of vaginal bleed, 1 placenta previa, 1 premature membrane rupture | 5 spontaneous abortions; 1 stillborn; of 32 infants born alive: three with LBW <2500g, one fetal distress; two latent TB treated, 1 active TB treated at mo 9 | treatment was suspended with confirmed pregnancy (n=14); second-line drugs may be continued despite teratogenic risk; most had advanced disease at baseline (n=36) |
| Case Series | Peng (2011)[46] | China | 6 neonates (5-42 days old) | congenital | NA | 3 mothers had miliary TB, 1 had genital TB, 3 unknown TB; outcomes of mothers not discussed/unknown | 4 full term, 2 premature babies; all 6 babies demonstrated cough and hepatosplenomegaly, 4 with rales, 3 fever, respiratory distress, and poor feeding; two with abdominal distention, 1 jaundice, 1 with ascites; 4 improved with TB treatment, 2 died without TB treatment initiation | study also conducted a literature review of congenital TB - 95% of cases had mothers with active TB during pregnancy; laboratory results may cause misdiagnosis of other bacterial infections vs TB |
| Case Series | Rathod (2017)[54] | India | 5 (third trimester pregnancies, median age=24) | EPTB (spinal) | no | 3 normal vaginal deliveries | 1 spontaneous abortion; 1 fetal demise due to adverse reaction to anesthesia | 3 with resolution of TB, no sequelae; 1with mild back pain; 1 with mild neck pain; 1 patient with recovery of paraplagia; 1 with destruction of spinal vertebra and kyphosis; 1 with 15 degrees kyphosis |
| Case Series | Suresh (2015)[81] | India | 17 (median age=24) | both | yes | 5 deaths, 3 died before delivery, 4 elected abortions | 1 death | high maternal mortality in low-resource high HIV burden countries |
| Case Series | Tabarsi (2011)[77] | Iran | 5 (median age=22) | DR PTB | NA | cured of TB; normal vaginal delivery | normal weight | offer of induced abortion; treatment was suspended with confirmed pregnancy and resumed at patient's decision |
| Case Series | Toro (2011)[49] | sub-Saharan Africa | 33 (median age=29 IQR 27-31) | Not reported | yes | 32 survived, 25 with live births, 1 induced abortion, 5 spontaneous abortions/stillbirths, 1 infant outcome unknown | rate of spontaneous abortion/stillbirths among women with TB-HIV coinfection more than twice that compared to women receiving ART alone during pregnancy (15.2% versus 5.9%); 1 stillbirth was HIV+ | analyzed women with TB-HIV coinfection; median gestational age 33 IQR 27-37, median CD4 97 IQR 48-196 |
| Case Series | Ye (2019)[55] | China | 6 (mean age 32.17± 3.87 years) | EPTB | no | cured of TB | 3 spontaneous abortions, 3 elected abortions | delayed TB diagnosis (10-32 days from onset of symptoms to diagnosis) |
| **Case-control** | | | | | | | | |
| Case-control | Asuquo (2012)[18] | UK | 24 (pregnant TB) + 72 (pregnant control, no TB) | PTB (n=13), EPTB (n=10), both (n=1) | yes (n=1) | cured of TB not reported | LBW (p=0.03) explained by pre-term delivery (OR 179.9, p=<0.001) | complications include hepatitis, cholestasis, and vomiting |
| Case-control | Chopra (2017)[41] | India | 50 women matched with TB control | both | NA | anemia (23% vs. 4% control, p-value not reported); 2/50 died due to TB disease progression, 2/50 lost to follow up | 3x more likely to have fetal growth retardation (OR 3.2, 95% CI 1.3-7.9), preterm labor x5 more likely (OR 5.9 95% CI 2.5-13.9) | NA |

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|---|---|---|---|---|---|---|---|---|
| Case-control | Lin (2010)[20] | Taiwan | 761 women matched with 3805 on age and year of delivery | not reported | no | NA | SGA (aOR 1.22, 95% CI 1.00-1.49); LBW (aOR 1.35, 95% CI 1.01-1.81) | NA |
| **Cohort Study** | | | | | | | | |
| Retrospective Cohort Study | Ahmadzia (2020)[37] | US | 1450 TB cases out of 24,149,664 deliveries | PTB, EPTB | NA | TB group more likely to experience pre-term labor (90, 13% versus 162435, 6.8%, p=0.004), to have UTIs ( 7.0% versus 3.3%, p=0.02), and undergo cesarian delivery (30.9% versus 21.5%, p=0.007) compared to those without TB | fetal distress more likely in fetuses with mothers infected with TB compared to control group (5% versus 2.1%, p=0.02) | NA |
| Retrospective Cohort Study | Ali (2011)[21] | Sudan | 42 (mean ae 30.7, SD 3.7) out of 15252 pregnant women | 35 PTB, 7 EPTB | yes (n=5) | anemia (83.3% versus 61.9%, p=0.02) compared to those without TB, 2 deaths in women with TB-HIV coinfection | LBW for newborns to women with tuberculosis (28.5% vs 14.3%, p=0.03); higher numbers of preterm births (28.5% versus 19.0%, p=0.04) compared to those without TB | NA |
| Retrospective Cohort Study | Ali (2021)[31] | Pakistan | 27 pregnant women with TB out of 2,896 screened | 26 PTB, 1 DR PTB | NA | Compared to those without TB, women with TB had higher prevalence of reported weight loss (59.2% vs 23.0%, P <0.01); 25 of women with TB successfully linked for treatment; 24 completed treatment, 1 lost to follow up | 26 live births, 1 intrauterine death at 32 weeks, 8 had LBW (<2500g), 1 had very LBW (<1700g); 16 gastric samples collected and no vertical transmission was found | NA |
| Retrospective Cohort Study | Baluku (2021)[82] | Uganda | 18 (mean age 27.5, SD 5.2 years) | DR PTB | yes (n=8) | Treatment success among 15 (83.3%) patients, 2 (11.1%) were lost to follow up and 1 (5.6%) died; elevated liver enzymes (81.8%, n = 11) and hearing loss (33.3%, n = 15) were the most frequently encountered drug adverse events | NA | 6 were pregnant at initiation of TB treatment |
| Prospective Cohort Study | Bekker (2016)[30] | SA | 74 comparingmaternal TB-HIV and TB | both (14% DR PTB) | yes (n=53, 72% sample) | 5 deaths in HIV-coinfected attributed to TB; women delivering LBW infants were 3.83 times more likely to have unfavourable TB treatment outcome such as death, treatment failure, loss to follow up (95% CI 1.40–10.53, p = 0.009), compared to women delivering infants weighing> 2500 grams. | 6 infants died, 4 stillbirths, all to HIV positive women; 49 premature, 44 LBW, 2 congenital TB; 57 infants given isoniazid prophylaxis, 39 with outcomes at 6 months: 24/39 (62%) completed treatment, 13/39 (33%) did not, and 2/39 (5%) died; of 55 HIV exposed infants, HIV PCR test performed on 45, 3 were + and initiated on ART | median CD4 count 155 IQR 11-565, 29/53 (55%) on ART at time delivery, 10 (19%) no ART, Women with HIV-coinfection had more severe EPTB (15% versus 0%); only 64% had documented ART use |
| Retrospective Cohort Study | Dennis (2018)[71] | US | 4053 TB positive among 57393459 pregnancy and delivery hospitalizations | not reported | yes | rate of pregnancy complications (severe pre-eclampsia, eclampsia, placenta previa/accreta/abruption, antepartum/ postpartum hemorrhage, sepsis, anemia) 80% higher among TB infected mothers compared to non-TB infected mothers (Composite rate 293.2 TB positive versus 166.5 TB negative) | NA | non-Hispanic black mothers have an aOR 2.77 (95% C.I. 2.35, 3.25) of non-TB related death compared to white mothers; TB infected mothers have similar risk level across all ethnicities (white, black, Hispanic); over 1/3 of all TB cases were diagnosed among mothers of Hispanic origin |
| Retrospective Cohort Study | El-Messidi (2016)[43] | US | 2064 | not reported | NA | increased aOR of chorioamniotitis (OR=1.35, 95% C.I., 1.04-1.74), postpartum anemia (OR=1.51; 95% C.I., 1.22-1.87), PNA (OR 8.42, 95% C.I., 5.77-12.29), ARDS (OR, 2.85; 95% CI, 1.35-6.10), mechanical vent (OR 3.33, 95% C.I. 1.66-6.68), maternal mortality increased among pregnant women with TB (OR=6.27, 95% C.I. 2.01-19.58) | congenital anomalies (aOR=1.8, 95% C.I 1.24-2.62) | Medicaid the most common payer for TB-related hospitalizations; more pregnant women with TB from lower income quartiles |

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|---|---|---|---|---|---|---|---|---|
| Retrospective Cohort Study | Fernandez (2018)[70] | US | 4053 TB positive cases among 57393459 pregnancy and delivery hospitalizations | not reported | yes (n=110) | rate of postpartum hemorrhage highest in TB only group (45.5 per 1,000 hospitalizations) compared to TB and HIV negative group (25.8 per 1000), similar to anemia (216.6 compared to 102.7 per 1000 hospitalizations); compared to HIV & TB negative mothers, mothers with HIV or TB have significantly higher odds of pregnancy complications (HIV, aOR=1.4, 95% C.I. 1.32-1.47; TB, aOR=1.91, 95% C.I. 1.64-2.23) | NA | Rate of TB among HIV-positive mothers was 21 times higher compared to rate of TB in HIV-negative mothers; low socioeconomic status and having public or no health insurance were biggest risk factors for HIV or TB monoinfection and HIV-TB coinfection' pregnancy complications highest among HIV-coinfected mothers (331.28 per 1,000 hospitalizations), followed by those with TB mono-infection (292.1 per 1,000 hospitalizations) |
| Retrospective Cohort Study | LaCourse (2016)[32] | US | 134 matched by year delivery and TB status | not reported | NA | NA | LBW (aRR=3.74, 95% C.I. 1.40-10.00); SGA (uRR=1.95, 95% C.I. 1.11-3.41) | NA |
| Retrospective Cohort Study | Loveday (2020)[33] | SA | 108 | DR PTB | yes (n=88, 81%) | completed treatment or cured of TB (67%), treatment failure (3%), death (3%) | fetal and neonatal deaths (9%), treated for DR PTB in the first year of life (2%); LBW for those exposed to bedaquilline (45% vs 24%, p=0.034) | mean age 28, SD ±6.13; baseline cd4 median 353 IQR 165-511<br>low baseline hemoglobin a predictor of poor TB treatment outcome (uHR 0.67, p= 0.006); women living with HIV had a higher risk of a poor pregnancy outcomes (uHR 3.35; p=0.030); bedaquilline and levofloxacin are predictors of LBW |
| Retrospective Cohort Study | Mokhele (2021)[42] | SA | 35 | DR PTB | yes (n=24) | Overall 48.6% had adverse events; 82.4% had 1 and 17.6 two or more; most prevalent was nephrotoxicity (35.3%), anemia (23..5%), hypokalemia (23.5%), ototoxicity and rash (11.8%); HIV-negative patients had fewer adverse events (18.2% vs 62.5%, p<0.05); of 28 with TB outcomes, 17 completed treatment, 8 lost to follow up, 3 died; for 20 women with pregnancy outcomes, 65% of these had an adverse pregnancy outcomes | of 20 known pregnancy outcomes, there were 4 term births, 11 preterm births, 1 miscarriage, 1 neonatal death, and 3 pregnancy terminations; 10/11 preterm births were from mothers with HIV coinfection and 1 neonatal death of mother with HIV coinfection as well | median age 30.1 IQR 28-35, median CD4 145 IQR 51-301<br>Patients with HIV coinfection had more severe immunosuppression (median baseline CD4 145) with more adverse outcomes; authors propose that DR PTB & HIV coinfection during pregnancy are cause of poor birth outcomes, recommending increased TB prevention strategies and TB treatment in antenatal care & HIV programs |
| Retrospective Cohort Study | Rendell (2016)[79] | Mongolia | 107 | PTB and EPTB | no | 10/85 elected abortion after TB diagnosis | NA | 2/82 TB treatment failure, 4/82 lost to follow-up; higher incidence of TB in pregnancy compared to general population (IRR 1.31, CI 1.08-1.59) |
| Retrospective Cohort Study | Sade (2020)[34] | Israel | 46 neonates and mothers with TB compared to 243636 mothers without TB | PTB, latent TB | yes (n=2 mothers) | 3 cases of placental abruption, 1 resulting in perinatal death; placental abruption is significantly associated with mothers with TB compared to those without (23.1% vs. 2.9%, p<0.001); TB was an independent risk factor for placental abruption and very LBW when controlling for maternal age (10.76, 95% CI 3.37–34.36) and gestational age (11.27, 95% CI 3.44-36.96) | higher rates of very LBW (<1500g) newborns in women with latent TB when compared to those with active TB during pregnancy, latent TB, and those without TB (0.0%, 5.6%, and 0.6%, p < 0.001); no significant difference in long-term infectious morbidity between children of mothers with and without TB | NA |

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|---|---|---|---|---|---|---|---|---|
| Prospective Cohort Study | Salazar-Austin (2007)[35] | SA | 80 (pregnant women with TB) matched with 155 controls (no TB) | PTB and DR PTB | yes | anemia more in cases than control (27.5% versus 11.0%, p=0.001); pre-eclampsia 8-fold more common among case patients than controls (5.2% vs. 0.7%, p=0.03); higher rates of maternal hospitalization (25% vs. 11%, p=0.005) | infants born to HIV/TB moms twice as likely to have LBW (20.8% vs. 10.7%, p=0.04), prolonged hospitalizations at birth (51% vs. 16%, p=<0.001) than born to HIV only; infants born to HIV/TB moms had 10- fold increased infant mortality risk (p=<0.001) | median age 29 IQR 26-31, median gestation 30 weeks (26-34), CD4 among cases 242 IQR 137-379) compared to controls 369, IQR 263-476, p<0.001), less cases were virally suppressed than controls (30.8% versus 47.7%, p<0.001); over 90% of both cases and controls were on ART at screening |
| Retrospective Cohort Study | Sun (2021)[97] | China | 6389 cases (pregnant women or partner with TB) matched with 31945 controls (couple without TB) | PTB | unknown | no significant differences in PTB (OR: 0.87, 95% CI 0.66–1.12) between women with and without TB; women whose partner had TB had a higher risk of stillbirth (OR 2.13, 95% CI 1.10–3.86) than women whose partner did not have TB | no significant differences in stillbirth (OR 1.48, 95% CI 0.61–3.09) and LBW (OR 1.04, 95% CI 0.79–1.35) between women with and without TB | also stratified analysis by TB treatment and found similar results of pregnancy outcomes |
| Prospective Cohort Study | van der Walt (2020)[62] | SA | 26 | DR PTB | yes (n=20) | 65.4% successful treatment outcome; no statistical significance between HIV coinfection and pregnancy outcome of mother or fetus; 57.7% had at least one adverse drug event (GI disturbance, neuropsychiatric symptom, rash, thrombocytopenia, arthalgia, dizziness, hearing loss, anemia) | Birth outcome significantly associated with the trimester in which DR TB treatment was initated (60% in 1st trimester vs 90% in 2nd vs 100% in 3rd, p=0.036); 2 abortions/miscarriages, 1 stillbirth, unclear of abortions were induced or spontaneous | mean age 29 years, SD ±5.1; missing information of type of delivery, complications of pregnancy, and birth weight |
| Retrospective Cohort Study | Yadav (2019)[36] | India | 30 | EPTB | no | significant increased incidence of oligoamnios and preterm rupture of membranc/preterm delivery compared to non-TB patients (13.3% versus 1.6%, p=0.001); death secondary to shock and end-organ failure from miliary TB. 7 women underwent emergent cesarian delivery for fetal distress | 1 stillborn due to severe miliary TB in mother; LBW in newborns of moms with TB compared to non-TB moms (2324.26±379.5 grams versus 2712.3±635.7, p=0.001); more incidence of SGA in moms with EPTB compared to non-TB moms. (33.3% versus 7.8%, p=0.001) | NA |
| **Other** | | | | | | | | |
| Service Improvement Project | Cornish (2020)[60] | SA | 56 (ages 13-50) pre-intervention, 58 post-intervention | DR PTB | yes (n=34 pre-intervention, n=23 post-intervention) | 3 pregnancies during treatment before improvement project; after improvement project: 0 pregnancy | 2 termination of pregnancy; 1 uncomplicated c-section at full term before improvement project | training intervention included family planning training of DRTB nurses and stocking the DRTB unit with long-acting injectable contraceptive; proportion of women on injectable contraceptives by the time of DRTB treatment initiation increased more than 3-fold, from 23.9% to 77.4% (p<0.0001) |
| Secondary Data Analysis | Rickman (2020)[85] | US | 235 pregnant women (n=73 with TB, n=162 without TB of which 7 later identified to have TB | not reported | yes | 43% of prevalent cases experienced complications (hypertension, antepartum hemorrhage, rupture of membranes, unplanned cesarian section, sepsis, maternal death, spontaneous abortion, stillbirth, prematurity, LBW, SGA, perinatal death, fetal abnormalities) compared to 35% of control (p=1.0) | 43% infants experienced complications including TB, neonatal sepsis, and infant hospitalization among TB prevalent cases compared to 11% of control (p=0.045) | median age 29 IQR 25-31, CD4 361 IQR 303-510; ART at baseline 2/7 (29%), 1 undetectable VL; in a cohort of women living with HIV, 7 subclinical cases were identified and discussed in this study; subclinical maternal TB may result in elevated risk of adverse infant outcomes |
| Secondary Data Analysis | van de Water (2020)[84] | Peru | 36 pregnant women with TB compared with 1298 non-pregnant with TB | PTB and DR PTB | yes (n=1 in pregnant and n=28 in non-pregnant women | 97.1% pregnant women were HIV-negative; 28 had reported successful TB outcome (cured, completed treatment, treatment ended early by clinical team), 1 was treatment failure; 8 pregnant women had DR PTB - 7 completed treatment/cured, 1 lost to follow up | neonate outcome not reported | median age of pregnant women 24.5 (IQR 21-30.5) |

Abbreviations: MTCT-Plus, maternal to child transmission-Plus; PTB, pulmonary TB; EPTB, extrapulmonary TB; DR PTB, drug resistant pulmonary TB; DR EPTB, drug resistant extrapulmonary TB; LBW, low birth weight <2500g;

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SD, standard deviation; SGA, small for gestational age; PNA, pneumonia; ARDS, acute respiratory distress syndrome; NA, not applicable; aOR, adjusted odds ratio; aRR, adjusted risk ratio; uHR, unadjusted hazard ratio; NA, not applicable

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Appendix. Detailed Search of Systematic Review.

Pubmed (1160 citations)
("Tuberculosis"[Mesh] OR tuberculos* [tiab] OR "tb" [tiab])
AND
("Pregnancy Outcome"[Mesh] OR ((pregnan* [tiab] OR obstetric [tiab] OR maternal [tiab] OR fetal [tiab]) AND (outcome* [tiab] OR morbidit* [tiab] OR mortalit* [tiab] OR "family planning services" [MESH] "OR family planning" [tiab] OR "contraception" [tiab] OR "contracept*" [tiab])))

Embase (1370 citations)
'tuberculosis'/exp OR tuberculos*:ti,ab OR 'tb':ti,ab
AND
'pregnancy outcome'/exp OR (((pregnan* OR obstetric* OR maternal* OR fetal*) NEAR/5 (outcome* OR morbidity OR mortality)):ti,ab) OR 'family planning'/exp OR ((family planning services):ti,ab OR (contraception OR contracept*):ti,ab)

CINAHL (504 citations)
(MH "Tuberculosis+") OR ( tuberculos* OR "tb" )
AND
(MH "Pregnancy Outcomes") OR ( ((pregnan* OR obstetric* OR maternal OR fetal) N5 (outcome* OR morbidity OR mortality)) ) OR (MH "family planning") OR ((family planning services) OR (contraception OR contracept*))

WEB OF SCIENCE (678 citations)
TOPIC: (tuberculos* OR tb)
AND
TOPIC: ((pregnan* OR obstetric* OR maternal OR fetal) NEAR/5 (outcome* OR morbidity OR mortality) OR family planning OR family planning services OR contraception OR contracept*)

COCHRANE ( citations - Reviews and Trials Only)
#1 MeSH descriptor: [Tuberculosis] explode all trees 2116
#2 tuberculos* or tb 12120
#3 #1 or #2 12125
#4 MeSH descriptor: [Pregnancy Outcome] explode all trees 3820
#5 ((pregnan* or obstetric* or maternal or birth or fetal) near/5 outcome*) OR ((fetal or maternal) near/5 morbidity) OR ((fetal or maternal) near/5 mortality) 21371
#6 #4 or #5 21739
#7 MeSH descriptor: [Family Planning Services] explode all trees 264
#8 contraception or contracept* 15515
#9 #7 or #8 15576
#10 #6 or #9 37590
#11 #3 and #10 317

SCOPUS (310 citations)

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TITLE-ABS-KEY(( tuberculos* OR tb ) AND (family planning OR family planning services OR contraception OR contracept* OR ((pregnan* W/5 outcome*) OR (obstetric* W/5 outcome*) OR (maternal W/5 outcome*) OR (maternal W/5 morbidity*) OR (maternal W/5 mortality) OR (fetal W/5 outcome*) OR (fetal W/5 morbidity) OR (fetal W/5 mortality))))

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | Title page |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | 4 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | 5 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | 5-6 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | 6 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | 6, Appendix |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | 6-7 |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | 6-7 |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | 6 |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | 6 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | 6-7 |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | Supplemental Table 2 |
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | 7 |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | 6-7 |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | 8, Supplemental Table 1<br>Supplemental Table 2 |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | 6 |
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | N/a |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | N/a |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | 6-7 |
| Certainty assessment | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | 6-7 |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | 7, Figure 1 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | 7 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | Supplemental Table 2 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | Supplemental Table 1 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Supplemental Table 2, as appropriate |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | 7-8<br>Supplemental Table 1 |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | N/a |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | N/a |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | N/a |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | N/a |
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | 7-8 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | 13-14 |
| | 23b | Discuss any limitations of the evidence included in the review. | 14,16-17 |
| | 23c | Discuss any limitations of the review processes used. | 16-17 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | 14-16, Table 1 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | 2 |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | N/a |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | N/a |

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## PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | 17 |
| Competing interests | 26 | Declare any competing interests of review authors. | 17-18 |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | 17-18 |

*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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# Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review

| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-054833.R2 |
| Article Type: | Original research |
| Date Submitted by the Author: | 21-Feb-2022 |
| Complete List of Authors: | Nguyen, Yen; Johns Hopkins University School of Nursing<br>McNabb, Katherine C.; Johns Hopkins University School of Nursing<br>Farley, Jason E.; Johns Hopkins University School of Nursing<br>Warren, Nicole; Johns Hopkins University School of Nursing |
| Primary Subject Heading: | Public health |
| Secondary Subject Heading: | Reproductive medicine |
| Keywords: | Tuberculosis < INFECTIOUS DISEASES, Public health < INFECTIOUS DISEASES, HIV & AIDS < INFECTIOUS DISEASES, PUBLIC HEALTH, REPRODUCTIVE MEDICINE |
| | |

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# Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review

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**Word count (excluding title page, abstract, figures, references and tables)**: 4449

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**ABSTRACT**

**Title:** Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: a systematic review

**Objectives:** 1. Summarize and evaluate the current evidence of TB-associated pregnancy outcomes; 2. Evaluate the state of the science of family planning during TB treatment; 3. Provide recommendations to move forward to improve care and outcomes during TB disease.

**Design:** Systematic review using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines.

**Data Sources:** PubMed, Embase, CINAHL, Cochrane, Web of Science, and Scopus were searched from September 2009 to November 2021.

**Eligibility Criteria:** Studies were included if they assessed pregnant women with active TB, DR-TB, or TB/HIV coinfection and examined pregnancy, maternal, fetal/birth, and TB or TB/HIV coinfection outcomes. Studies were also included if they examined family planning services among women initiating TB treatment.

**Data extraction and synthesis:** Two independent reviewers extracted data using PRISMA guidelines and conducted quality assessment using the Joanna-Briggs Institute Critical Appraisal Tools. The level of evidence was reported using the Johns Hopkins Evidence-Based Practice guidelines.

**Results:** 69 studies were included in this review. Case reports, case series, case controls, cohort studies, secondary data analyses, and a service delivery improvement project conducted in 26 countries made up the totality of the evidence. Most studies reported pregnancy complications for mothers (anemia, postpartum hemorrhage, deaths) and fetuses or newborns (low birth weight, premature birth, and spontaneous or induced abortions). Few studies discussed the value of offering family planning to prevent adverse pregnancy outcomes. One study examined the effect of a provider training on contraceptive use with reported increased contraceptive use.

**Conclusions:** Integrating family planning services within a TB treatment program is essential to reduce adverse TB-associated maternal-child outcomes. Despite well-established adverse pregnancy outcomes, little attention has been paid to family planning to prevent poor pregnancy outcomes for women with TB/DR-TB. Recommendations for clinicians, TB programs, and researchers are provided and reflect evidence presented in this review.

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**Keywords**:

1. Tuberculosis
2. drug-resistant tuberculosis
3. family planning
4. HIV
5. Integrated services

**Strengths and limitations of this study**

1. A systematic search of six major electronic databases was conducted and results reported per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
2. Inclusion of case reports and case series allows for synthesis of important clinical data that is often not reported in systematic reviews.
3. Review was limited by the inclusion of only peer-reviewed, English journal articles published in the last 11 years, which likely excluded important findings in other languages and important programmatic data at a country level.
4. The variability of study designs, outcome definitions, and methods across this body of literature made comparison of outcomes difficult and therefore we were unable to conduct a meta-analysis.

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## BACKGROUND

In 2018, about 3.2 million women worldwide contracted tuberculosis (TB) and almost 500,000 died from the disease.(1) Although TB affects more men than women, it disproportionally places women at a higher risk of morbidity and mortality.(2) TB is a non-obstetric cause of maternal mortality and is especially dangerous for women of reproductive age in settings with high rates of TB or drug-resistant TB (DR-TB), and HIV co-infection.(2) TB among mothers is associated with a six-fold increase in perinatal deaths and a two-fold risk of premature birth and low birth weight (LBW).(1) Risk of maternal and fetal morbidity and mortality increases by 400% when human immunodeficiency virus (HIV) coinfection is introduced.(1)

The potential teratogenic effects of TB and DR-TB treatment may add further risks to these already high-risk pregnancies: most of the first- and second-line TB drugs are categorized per the United States (US) Food and Drug Administration (FDA) as pregnancy Class C or D.(3) First-line treatment of drug-susceptible TB includes isoniazid, rifampin, pyrazinamide, and ethambutol, all of which are categorized as pregnancy Class C per the US FDA.(3–7) It is nevertheless recommended pregnant women receive drug susceptible TB treatment as early in their pregnancy as possible because the benefits of treatment outweigh the potential adverse outcomes.(3) DR-TB occurs in about 10% of TB patients and can be detrimental to health and survival since treatment regimens are toxic and prolonged (lasting 9 to 18 months), and unsuccessful in 44% of cases.(3,8) Most of the second-line TB drugs are also categorized as pregnancy Class C or D. It is up to the providers' discretion to provide and individualize drug regimens for treatment of DR-TB, who may present with advanced disease and therefore are more at risk of adverse outcomes.

Comprehensive family planning services have the potential to reduce women's risk of adverse pregnancy outcomes in the context of TB and DR-TB. Such services would include counseling about the risks associated with becoming pregnant while undergoing TB or DR-TB treatment, providing safe and preferred methods of contraception, and referring women to appropriate care if they intend to conceive or become pregnancy during TB treatment.(9) Family planning counseling by a TB or DR-TB provider at initiation of TB or DR-TB treatment is especially important given that there are known drug-drug interactions between first line TB

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drugs, rifampicin, antiretroviral drugs (e.g. protease inhibitors) and contraceptives such as combined contraceptive pills.(10)

Despite reported poor outcomes among pregnant women with TB, DR-TB, and HIV coinfection, they have not been systematically summarized with family planning strategies to reduce risks explored. In addition, there are no international guidelines regarding pregnancy and family planning in the context of drug-resistant TB (DR-TB) management. Country-specific TB and/or DR-TB guidelines recommend family planning counseling, and/or pregnancy testing and contraceptive use(11–13), or avoidance of specific anti-tuberculous drugs.(14) However, they do not describe how to integrate these services into TB care. Without a more comprehensive understanding of the risks to and prevention strategies for childbearing women with TB, DR-TB, and/or HIV coinfection, it will be difficult to improve services or guide additional research.

This systematic review examines published research on pregnancy outcomes and family planning strategies for women undergoing treatment for TB and DR-TB. The objectives are: 1) to summarize and evaluate current evidence for TB-associated pregnancy outcomes, 2) to evaluate the state of the science of family planning during TB treatment, and 3) to provide recommendations to move forward to improve care and outcomes.

## METHODS

### Review Design

We conducted a systematic review of peer-reviewed literature on global TB-associated pregnancy outcomes using the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).(15) Given the heterogeneity of study designs, outcome definitions, and statistical analysis approaches among included studies, we critically reviewed and synthesized findings instead of conducting a meta-analysis.

### Eligibility Criteria

A systematic approach guided the search for articles addressing TB-related pregnancy outcomes. Studies were included if they assessed patient population of pregnant women with active TB, DR-TB, or TB/HIV coinfection, and primary outcome variables included pregnancy outcomes, maternal or fetal/birth outcomes, and TB or TB/HIV coinfection outcomes. Studies were also included if they examined outcomes of family planning services among women

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initiating TB treatment (counseling of pregnancy risks, pregnancy testing, contraceptive uptake). Pregnancy outcomes were any outcome a study defined as maternal outcome or any maternal complication: pre-eclampsia, eclampsia, anemia, hemorrhage, placental abruption/previa, premature rupture of membrane, unplanned cesarian section or induction, death, or TB, DR-TB, or TB-HIV coinfection outcomes as defined below. Studies were also included if they assessed fetal complications or fetal/birth outcomes, which may include premature birth defined as birth prior to 37 weeks, low birth weight (LBW) defined as weight below 2500 grams, stillbirth, spontaneous abortions/fetal demise or induced abortions, and congenital TB infection in newborns up to 6 months postpartum. TB, DR-TB, or TB/HIV coinfection outcomes include cure/treatment completion, treatment failure, lost to follow up, or fetal or maternal death.

Articles were excluded if they were not peer reviewed articles (e.g., commentaries, communication briefs, editorials, conference abstracts or posters), literature reviews, published prior to 2009, were not written in English, and did not include human subjects. We excluded studies older than 2009 to ensure only the most up-to-date evidence was synthesized. We also excluded studies that did not have a primary aim to analyze and discuss pregnancy outcomes as defined above.

**Search and Identification Process**

A Boolean search of the electronic databases PubMed, Embase, CINAHL, Cochrane, Web of Science, and Scopus was conducted through November 5, 2021 using the following medical subject heading (MeSH) terms, keywords and associated synonyms in the titles and abstracts: tuberculosis, family planning services, contraceptives, pregnancy outcome, obstetric outcome, fetal morbidity, maternal morbidity, fetal mortality, maternal mortality. The Appendix provides a full strategy for the search.

**Data Extraction and Quality Assessment**

Citations were imported into Covidence® and duplicates removed. Two reviewers independently reviewed titles and abstracts. Differences in inclusion of articles were resolved through consensus. Once the articles for inclusion were finalized, data extraction and quality assessment were performed. The level of evidence of each study was assessed using the Johns Hopkins Evidence-Based Practice (JHEBP) guidelines.(16) The Joanna Briggs Institute (JBI)

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quality appraisal tools were used to assess the quality of each included citation; studies that fulfilled most to all items of the respective quality appraisal tool indicate they are of high quality and minimal bias.(17) Quality rating assessment was conducted independently by two raters to identify strengths and weaknesses and guide the interpretation of study findings. Any discrepancies were discussed and reconciled by consensus. The critical appraisal of included studies is tabulated in the Supplemental Table 1 for review (Supplemental Table 1. Critical Appraisals of the Included Studies).

**Ethics**

Institutional review board approval was not required.

**Patient and public involvement**

Patients and the public were not involved in this review.

## RESULTS

### Selection of studies

The initial search of all five databases yielded 2967 articles after 1362 duplicates were removed. The full text of 176 articles were thoroughly evaluated to assess eligibility, 107 of which were excluded based on the pre-identified exclusion criteria (Figure 1. PRISMA flow diagram: identification and selection of relevant articles).

### Study characteristics

Of the 69 articles included in this review, 68 articles published TB-associated maternal and fetal/neonatal outcomes and 1 article discussed family planning among women with DR-TB. Of these 69 studies, 51 were conducted in low- and middle-income countries, with the majority from India (n=12), China (n=10), and South Africa (n=7). Among the high-income countries, the majority (n=8) were from the US and the United Kingdom (n=3).

### Level of Evidence and Quality appraisal

All studies included were of level III and V evidence per JHEBP guidelines. Level III evidence included non-experimental, observational designs with the majority being retrospective

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cohort studies (n=15), followed by case control studies (n=3) and prospective cohort studies (n=3), and secondary data analysis (n=2). Level V evidence included case reports (n=32), followed by case series (n=13) and a service delivery improvement project. There was no experimental (Level I), mixed-methods (Level II) or qualitative study identified in our search.

The majority of included studies met most or all items of the quality assessment based on the JBI appraisal tool (n=60) with 9 fulfilling less than or half of the items of the quality assessment. Because this review aims to provide comprehensive evidence on pregnancy outcomes among women with TB and DR-TB and family planning strategies among this population, we included all studies that met the inclusion criteria regardless of quality or level of evidence. Supplemental Table 1 provides details of the quality assessment and level of evidence of appraised studies. Supplemental Table 2 provides details of pregnancy outcomes and other relevant findings for each study included in this review, including study designs, outcome measures, and statistical results (Supplemental Table 2. Summary of Pregnancy Outcomes and Other Relevant Findings).

**Current evidence for TB-associated pregnancy outcomes**

Fetal/newborn outcomes

Overall, reported fetal and newborn outcomes were poor. LBW was a common fetal outcome for TB infected women with or without HIV coinfection across multiple studies (case reports n=7; case series n=2; cohort study n=8).(18-37) Preterm labor, which adds to the risk of LBW, was reported in multiple studies (case report n=5; case series n=1; case controls n=1; cohort studies n=4) among women with TB infection.(18,21,23,25,26,30,36,38-42) A case control study found that preterm labor was five times more likely among women with TB (OR 5.9 95% CI 2.5-13.9).(41) One case series and four cohort studies had similar findings: there were higher numbers of preterm births among women with TB versus those who did not.(21,30,36,40,42) A large retrospective cohort study (n=24149664) found that pregnant women with TB were more likely to experience preterm labor (13.0% versus 6.8%, p=0.004)(37), and this trend of higher preterm labor rates persisted with other smaller case series and cohort studies.(21,30,40,42)

One large retrospective cohort study (n=2064) found increased congenital anomalies (aOR=1.8, 95% C.I 1.24-2.62) among newborns of pregnant women with TB.(43) It is unclear if

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this is attributable to teratogenic effects of TB drugs or other individual variations. A total of 8 studies evaluated congenital TB cases. Six case reports of congenital TB discussed babies who were born with LBW, two of which mentioned babies born prematurely and who remained hospitalized for at least 30 days. (22,26,27,44-45) TB outcomes for newborns with congenital TB include: resolution or improvement of TB at follow up, loss to follow up, died without TB treatment, or unknown.(22,26,28,40,44-46) Other reported poor fetal outcomes included stillbirth (case reports n=1, case series n=2; cohort studies n=3) (30,47,49), spontaneous abortions/fetal demise (case reports n=3, case series n=6 cohort study n=1) (19,29,48,50-55), or induced abortions (case reports n=4, series n=1, cohort study n=2, other n=1) (42,48,55-60) and fetal growth restriction/retardation (case series n=1, case control n=1).(41,61) One case series reported stillbirth, spontaneous abortions/fetal demise and induced abortions (48), and one cohort study reported spontaneous and induced abortions.(62) In contrast, reports of full-term birth, normal vaginal delivery, and normal neonatal weight were primarily seen with case reports.(28–32) (63-69) Although multiple case reports mentioned good fetal and newborn outcomes, the majority of findings suggest newborns of mothers with TB tend to have poorer outcomes.

Maternal outcomes

Three large retrospective cohort studies from the US reported that anemia and postpartum hemorrhage were more common among women with TB infection as compared to those without TB (n=4053 rate of composite pregnancy complications including anemia and postpartum hemorrhage 80% higher among women with TB; n=2064 anemia OR 1.51, 95% CI 1.22-1.87; n=4053 rate of postpartum hemorrhage highest in TB only group, 45.5 per 1,000 hospitalizations compared to TB and HIV negative group, 25.8 per 1000, similar to anemia, 216.6 compared to 102.7 per 1000 hospitalizations).(43,70,71) Two prospective cohort studies found similar findings regarding pregnancy complications including anemia and pre-eclampsia (n=80, more anemia in women with TB versus no TB, 27.5% versus 11.0%, p=0.001, pre-eclampsia 8-fold more common among women with TB versus no TB, 5.2% vs. 0.7%, p=0.03; n=26, 57.7% with adverse events related to TB treatment).(35,62) Anemia was similarly seen in a case control study (n=50, anemia 23% vs. 4% control, p-value not reported) and a retrospective cohort study (n=42 among 15252 pregnant women, anemia was seen in 83.3% versus 61.9%, p=0.02).(41,21) In the US, Fernandez et al. (70) reported highest rates of TB-HIV coinfection (49.9%) among

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Black mothers compared to Hispanic (16.7%) and non-Hispanic white mothers (% data suppressed) in their study. Dennis et al. proposed that advanced TB disease, commonly seen in ethnic/racial minorities, may be a contributing factor for increased pregnancy complications compared to mothers without TB.(71) None of the studies stratified analysis of pregnancy complications by race or ethnicity.

A retrospective cohort study (n=2064) reported other pregnancy complications more commonly seen among pregnant women with TB including chorioamnionitis (OR=1.35, 95% C.I., 1.04-1.74), pneumonia (OR 8.42, 95% C.I., 5.77-12.29), acute respiratory syndrome (OR, 2.85, 95% CI, 1.35-6.10), and mechanical ventilation (OR 3.33, 95% C.I. 1.66-6.68).(43) Case reports also found similar pregnancy complications.(56,58) Multiple case reports and case series described delayed TB diagnosis and complicated management of TB and pregnancy.(19,24,28,29,39,44,45,47,48,50,52,54,55,57-59,61,64,66,68,72-76) Some resulted in recommendation and subsequent termination of pregnancy(56,57,59,77,78), induced abortion(48,62,79), or unplanned cesarian section or induction.(24,25,27,29,36,38,39,53,72,74) For example, one case report described a woman with suspected genitourinary TB prior to pregnancy who suffered from severe pre-eclampsia and postpartum complication of pyelonephritis with unplanned induction.(72)

In regard to TB outcomes, the majority of women in cases reported treatment completion, cure, or improvement of clinical symptoms (23-25,38,39,45,50-52,58,59,63-69,72-75,78,80). Other TB-related maternal outcomes included deaths (19,21,28,30,32,33,36,41,42,47-49,56,81,82), lost to follow up (41,83), TB treatment failure (57), or unknown (18,22,44,26). Across 16 case series (19,29,48,53,61,76,77,81,83), a case control study (41), and cohort studies (31,33,42,62,82,84) that provided descriptive data of TB outcomes, there were a total of 402 cases. Of those, 259 (64%) had an outcome of treatment completed/cured, 49 (12%) were lost to follow up, 29 (7%) died, 5 (1%) failed treatment, 4 (1%) deteriorated in clinical condition, 5 (1%) with unknown outcome, and 46 were on treatment at time of data collection. Among studies that analyzed association of TB and pregnancy outcomes, one study found that women who gave birth to newborns with LBW were 3.83 times more likely to have poor TB outcome such as death, treatment failure, or loss to follow up compared to those delivering newborns with normal weight (95% CI 1.40–10.53, p = 0.009) (30). El-Messidi and colleagues (43) found that compared with pregnant women without TB, maternal mortality was higher among those with

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TB (OR=6.27, 95% C.I. 2.01-19.58). Similarly, compared to HIV and TB negative mothers, those with HIV or TB have significantly higher odds of pregnancy complications (HIV, aOR=1.4, 95% C.I. 1.32-1.47; TB, aOR=1.91, 95% C.I. 1.64-2.23). (70) Although the majority of case reports described favorable TB outcomes for women, they also described difficult management of disease, or significant adverse sequelae of TB. Furthermore, larger studies found a large proportion of women with adverse TB outcomes associated with TB disease.

HIV coinfection outcomes and ART treatment

Across 9 studies that analyzed data among women with TB-HIV coinfection (1 case series, sub-Saharan Africa; 3 prospective cohort studies, SA; 4 retrospective cohort studies, US and SA; 1 secondary data analysis, US), 4 reported median age of pregnant women between 29-30.1 years (Interquartile Range [IQR] 25-35), and 6 reported CD4 count at baseline ranging from 11-565 cells/mm$^3$ (IQR 11-565). (30,33,35,41,42,49,62,70,85) These studies found increased maternal mortality and fetal mortality. Bekker et al. (30) described higher number of deaths among women with TB-HIV coinfection compared to those with TB only (9% versus 0%, p=0.313) and 10 fetal deaths (n=55 among mothers with HIV), suggesting higher mortality among this population compared to those with TB only. Similarly, another study found 91% (10/11) of preterm labor were among those with TB-HIV coinfection compared to those with TB only (42). One study found no statistical significance in association between TB-HIV coinfection and maternal or fetal outcome (62).

Other pregnancy outcomes included anemia (27% versus 11%, p=0.001), pre-eclampsia (5.2% versus 0.7%, p=0.03), LBW (2950g versus 3060 g, p=0.04), and fetal mortality (68 deaths versus 7 per 1000 live births, p<0.001) among women with TB-HIV coinfection compared to women with HIV and no TB.(35) One study found that HIV-coinfection increased risk of unfavorable pregnancy outcome of preterm labor and/or LBW (unadjusted hazard ratio 3.35, p=0.030) (33). Another found that rates of pregnancy were the highest among women with TB-HIV coinfection (331.28 per 1,000 hospitalizations) compared to those with HIV alone (272.2 per 1,000 hospitalizations) (70).

Our data suggest antiretroviral therapy (ART) is not consistently provided to women with TB and HIV coinfection and of those, few are virally suppressed. ART was initiated at the same time as TB treatment for 19 women (57%, n=32).(49) In another study, 29 (55%, n=53) women

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were reported as being on ART at time of delivery.(30) The highest number of HIV coinfected women on ART at time of TB diagnosis was 55 (68.8%, n=80); only about 30.8% had an undetectable viral load (<200 copies/mL) compared to those without TB (47.7%, p<0.001).(35) Collectively, these studies show that just over half of HIV coinfected women initiating TB treatment are on some form of ART and much fewer are virally suppressed, which may explain why HIV coinfection contributes additional risk for poor outcomes among women and their babies. Unfortunately, no study analyzed pregnancy outcome among pregnant women with TB-HIV coinfection who were on ART versus not on ART, which may limit our understanding of how HIV treatment may impact pregnancy outcomes among this population.

DR-TB outcomes

Ten studies evaluate pregnancy outcomes among women with DR-TB. (19,33,42,57,62,72,77,78,82,84) A single case series examined the use of a standardized regimen for DR-TB infected pregnant women (n=5), with a reported cure of TB and good maternal outcomes.(77) In a retrospective cohort study looking at medical records of women with DR-TB between 2013 and 2017, the authors found that second-line drugs, bedaquiline and levofloxacin, were predictors of LBW (45% versus 26%, p=0.034), and that women with HIV coinfection had a higher risk of poor pregnancy outcomes.(33) Four studies reported successful treatment outcome (complete or cure) for the majority of their samples (83.3%, 61%, 65.4% and 88%) and lost to follow up (11.1%, 28.6%, 26.9% and 12%) (42,62,82,84). The remaining studies, three case reports (72,78,57) and a case series (19), provide conflicting results on maternal and fetal outcomes: two case studies reported good outcome for mother and baby, 1 case study reported treatment failure with induced abortion, while the case series noted maternal and fetal/newborn complications and deaths in their descriptive analysis. The validity and generalizability of these studies are limited by their study design and sample size.

The state of the science of family planning during TB treatment

The literature search yielded one study which evaluated family planning uptake at initiation of TB treatment among women with DR-TB.(60) This South African study aimed to increase contraceptive use by securing Depo-Provera stock, training healthcare providers to counsel and deliver Depo-Provera injections at initiation of DR-TB treatment and referring

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women to a tertiary hospital if they intended to conceive or became pregnant during treatment. The authors demonstrated an increase in the number of women initiating Depo-Provera at the time of DR-TB treatment. The authors suggested that these interventions are effective in delaying pregnancies during TB treatment.

Though not specifically examining contraceptive use, six studies that evaluated pregnancy outcomes commented on family planning as crucial to the improvement of health outcomes for women of reproductive age during TB treatment after finding high numbers of pregnancies during the study period.(19,20,41,69,75,78) One study noted 36 out of 38 pregnancies were "unplanned" among a DR-TB cohort of women, and another reported fetal exposure to two potentially teratogenic TB drugs (ethionamide and levofloxacin) during the first trimester.(19,78) One study recommended risk counseling for women initiating TB treatment.(20) All of these studies, regardless of design or location, recognized one important theme: integrated family planning services, including risk counseling and provision of effective contraceptives, are an important preventative strategy to reduce adverse maternal and fetal/newborn outcomes.

## DISCUSSION

To our knowledge, this is the first review to evaluate evidence for TB-associated pregnancy outcomes and the literature for family planning strategies in TB programs. We found that there were consistent evidence across level III and V studies of poor TB-associated pregnancy outcomes including fetal outcomes (LBW, premature birth, spontaneous or induced abortions) and maternal outcomes (anemia, pre-eclampsia, and deaths). Despite the findings of poor maternal and fetal outcomes, there is a paucity of family planning research for women with TB and DR-TB. Such a gap in literature may be the rationale for the absence of family planning recommendations in the WHO TB and updated WHO DR-TB guidelines.(86)

Integration of maternal, fetal and infant health care in the setting of TB disease is essential. Knowing that TB disproportionally affects women of reproductive age in low resource settings, (1) there is unquestionably high value in implementing family planning measures as part of TB care. This paucity of research has not gone unnoticed by other scientists. Schnippel et al. (87) wrote a call-to-action paper suggesting the need for greater integration of family planning services in DR-TB facilities in South Africa, however as to date, only one study specifically

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evaluated family planning uptake among women initiating DR-TB treatment.(60) This work provided preliminary data on family planning use and the potential benefit of a provider training and contraceptive supply on increasing uptake of Depo-Provera. However, it did not expand on other aspects of family planning services like addressing factors associated with contraceptive use, pregnancy testing, other family planning options beside the injectable Depo-Provera, or discussion of counseling of pregnancy risks during treatment, all of which could have an impact on a woman's decision to use contraception.

Given the evidence of outcomes, care and clinical management of pregnant women with TB or DR-TB described in this review, as well as limited data on integrated family planning into TB care, it is imperative for national stakeholders to recommend clear strategies to integrate family planning services into TB care. Our team developed a list of recommendations to guide the integration of family planning into TB and DR-TB services for women of reproductive age, incorporating the evidence from this systematic review, WHO family planning recommendations as well as the United States Agency for International Development and Family Health International (FHI) 360 guidelines where relevant (Table 1. Recommendations for Integrated TB/DR-TB and Family Planning Services). (88,89,90)

Due to the limited studies on family planning strategies among women with TB or DR-TB, as well as low level of evidence of included studies (level III and V), it is important for future research to conduct more rigorous studies with higher level of evidence, such as experimental or mixed methods studies to explore family planning needs and strategies at the individual, programmatic, and policy levels. Furthermore, it may be safer and beneficial for pregnant women to have the option to be considered for TB and DR-TB research in order to facilitate our understanding of how TB and DR-TB treatment may impact maternal child health outcomes and inform future practice.(87,91-93) Specifically, the WHO suggests it would be helpful to compare effectiveness of shorter DR-TB regimens in pregnant women.(94) As of now, the 2020 WHO's guidelines recommend placing pregnant women on longer DR-TB regimen.(94) This is primarily due to the exclusion of pregnant women from a large randomized controlled trial testing the efficacy of shorter DR-TB regimen (e.g. STREAM trial). (95) With a longer treatment regimen, there is a greater exposure time in which women could become pregnant, and longer time pregnant women are being exposed to DR-TB drugs. As of now, there

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is still not sufficient evidence to support a standardized DR-TB regimen for pregnant women. (91) Research guiding regimen selection in pregnant women is urgently needed in the long-term.

In addition to our recommendations for practice, program, and research, local and national level policy action is needed to ensure family planning services are integrated into TB and DR-TB care settings. Our policy-specific recommendations reflect the evidence from this review and best practices from the field of family planning. FHI 360 created guidance for family planning and HIV integration, which may be used as a blueprint and adapted to TB and DR-TB programs. (89,90) In brief, policymakers should monitor family planning data such as contraceptive prevalence rate, unmet need, and demand for family planning among women with TB and DR-TB to appropriately direct funding and resources. Allocation of resources should prioritize efforts to integrate family planning services into TB and DR-TB programs and include monitoring and evaluation plan to inform program and service delivery improvement, as well as scale-up. This could be done by measuring additional indicators specific to family planning and TB integration, including service delivery integration, contraceptive method availability, uptake, informed choice, and training and human resources. (89) Healthcare provider training and ensuring supply of contraceptive methods during TB or DR-TB treatment will also be critical. Finally, subsequent updates to WHO TB and DR-TB guidelines should provide recommendations for family planning as part of quality care for women of reproductive age undergoing treatment. Of note, such a recommendation was offered in the companion handbook to the WHO 2014 DR-TB guideline but absent from the 2020 update. (86,96)

TABLE 1. Recommendations for Integrated TB/DR-TB and Family Planning Services

| **For clinicians**[1] |
|---|
| 1. Women of reproductive age should be tested for pregnancy at initiation of TB or DR-TB treatment as part of quality TB care. (41,75,69,78) |
| 2. Women who are found to be pregnant at treatment initiation should be referred to antenatal care with appropriate follow ups and more intensive monitoring. (19,20) |
| 3. Women who are not pregnant should be informed of potential risks to maternal and fetal/infant health if they become pregnancy during treatment and counseled to delay pregnancy until after completion of treatment. (73) |
| 4. Family planning counseling should be done at baseline, with information provided to women to include options for contraceptive methods depending on the TB or DR-TB regimen they may be on (such as rifampicin inclusive regimen), side effects they may experience on contraceptive of choice, and how to switch method or discontinue method. Women should be offered a method is the most effective, such as a long-acting reversible contraceptive (LARC), that does not interfere with either TB/DR-TB or HIV treatment and is well-suited for her lifestyle and belief. (69,75,88,96)[2] |
| 5. Women who agree to be on a LARC should be educated on the correct use of method and when to follow up. (88) |

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6. Women who do not agree to be on a contraceptive should be followed up routinely throughout treatment for changes in relationship status, desire for contraceptive, pregnancy testing, and immediate linkage to antenatal care if pregnancy status changes during TB treatment. (88)

**For TB programs**

1. TB treatment site should keep stock of various contraceptive methods including male and female condoms, oral contraceptives, implants, injectables, intrauterine devices to minimize referring women to outside clinics for family planning. (60,88)
2. TB providers should be trained at least annually on family planning counseling, family planning knowledge, and proper techniques to administer contraceptive methods so they may provide these services to women who need them. (60,88)
3. Recommendations for family planning should be standardized and integrated into all TB and DR-TB treatment sites as part of high-quality TB services. (60,87)

**For researchers**

1. Pregnant women should be considered for clinical studies, particular those looking at the efficacy of shorter regimens compared to longer regimens, which expose women to DR-TB drugs for a longer period of time. (97,92,93)
2. Research, particularly pragmatic and intervention research, should address family planning needs in TB care and the implementation of integrated services. (93)
3. Given a lack of high level of evidence of studies, future research on pregnancy outcomes and family planning among women with TB or DR-TB should prioritize more rigorous study designs where possible, for example, level 1 experimental designs, level 2 mixed methods design, or level 3 prospective cohort studies with larger sample size.

1. Clinicians refer to providers who have direct patient contact including doctors, nurses, and pharmacists
2. This citation refers to the companion handbook to the WHO 2014 DR-TB guidelines, which were updated in 2020. However, in the updated 2020 guidelines family planning was not mentioned.

## Strength and limitations of the studies

There were several limitations to our systematic review. This review was limited by the inclusion of only peer-reviewed, English journal articles published in the 11 years. This likely excludes important findings in other languages and important programmatic data at a country level. Most studies in this review were of lower level of evidence, with the majority being case reports, case series, and retrospective cohort studies. The inclusion of lower level of evidence such as case reports allow for a comprehensive view of available clinical data across the globe. Aside from five cohort studies which took place in the US and China, most had small sample sizes and primarily reported exploratory and descriptive data. This may reduce the ability to generalize results beyond these studies to places that had higher burden of TB, DR-TB, and/or HIV coinfection with less access to health resources. Only one study evaluated family planning strategies as part of a quality improvement project. It would be important for future research to consider more rigorous study designs given the low level of evidence of current studies (e.g. experimental or mixed methods studies). This is especially important to consider for studies that evaluate family planning strategies for this high-risk population of women. It is also important

For peer review only

that future research prioritize studies with larger sample size in high TB, DR-TB, and/or HIV coinfection burden settings to evaluate pregnancy outcomes and family planning strategies since the majority of the larger cohort studies took place in US, a high-income country with low burden of disease.

Another limitation of this review is the variability of study design, definitions, and methods, which made comparison of outcomes difficult. Definitions of pregnancy outcomes were often heterogenous across multiple studies; some studies looked at TB, DR-TB, HIV coinfection outcomes as we defined in our review, some looked at fetal/newborn outcomes such as fetal distress, fetal demise versus stillbirth and intrauterine growth restriction or growth retardation rather than LBW or preterm birth without establishing clear definitions. Some failed to discuss TB/DR-TB outcomes of mothers or newborns. There were studies that did not define preterm birth or LBW, and there were studies that explored different pregnancy complications as a composite outcome, all of which made it difficult to cohesively conduct a meta-analysis. We sought to include all relevant studies regardless of methodological designs or quality to further our understanding of the scientific gaps that exist around maternal-child outcomes of women with TB or DR-TB and their family planning needs in both the clinical and research settings. Despite the weakness of the study designs and inclusion of studies of all quality types, the observational evidence consistently demonstrates adverse maternal and fetal/newborn outcomes across the globe in the last decade.

## CONCLUSION

TB in pregnancy is as a major risk factor of maternal and fetal morbidity and mortality, especially in low and middle-income countries where resources are limited. Pregnancy outcomes are worsened in the setting of TB/DR-TB, with or without HIV co-infection, and lack of health resources and access to specialized care. Offering family planning services at TB/DR-TB treatment initiation and education on risk of poor pregnancy outcomes should be considered as part of routine care in women of childbearing age.

**Acknowledgments**

This manuscript was supported by The Johns Hopkins School of Nursing, Johns Hopkins School of Nursing REACH Initiative, Jonas Philanthropies, Coverdell Peace Corps Fellows Scholarship,

For peer review only

and R01 AI104488 (JF, National Institute of Allergy and Infectious Diseases).

**Contributors**: the idea for this publication and preparation of the first draft was done by YN. YN and KM conducted the systematic search and assessed quality of articles. YN, KM, JF, and NW provided significant contributions to the draft and all its revisions. All authors reviewed the draft manuscript and approved the final draft for publication.

**Funding**: This work received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

**Competing interest**: none declared.

**Data availability statement:** all data relevant to the study are included in this article, appendix and supplemental files.

**Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram: identification and selection of relevant articles.**

For peer review only

## REFERENCES


1. Tuberculosis in women [Internet]. World Health Organization. 2019. Available from: https://www.who.int/tb/publications/tb_women_factsheet.pdf?ua=1
2. Grange J, Adhikari M, Ahmed Y, Mwaba P, Dheda K, Hoelscher M, et al. Tuberculosis in association with HIV/AIDS emerges as a major nonobstetric cause of maternal mortality in Sub-Saharan Africa. Int J Gynecol Obstet. 2010;108(3):181–3.
3. Mathad JS, Gupta A. Tuberculosis in Pregnant and Postpartum Women: Epidemiology, Management, and Research Gaps. Clin Infect Dis. 2012;55(11):1532–49.
4. Rifampicin. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 15, 2020. Available from www.micromedexsolutions.com. Subscription required to view.
5. Ethambutol. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 4, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
6. Pyrazinamide. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated September 4, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
7. Isoniazid. In: In Depth Answers [database on the Internet]. Greenwood Village (CO): Truven Health Analytics; Last updated July 20, 2020 [cited October 7, 2020]. Available from www.micromedexsolutions.com. Subscription required to view.
8. World Health Organization. Global Tuberculosis Report 2019. Geneva, Switzerland; 2019.


For peer review only

9. Hancock NL, Stuart GS, Tang JH, Chibwesha CJ, Stringer JS, Chi BH. Renewing focus on family planning service quality globally. Contraception and Reproductive Medicine. 2016 Dec;1(1):1-6.

10. World Health Organization. Reproductive Health. Medical eligibility criteria for contraceptive use. Geneva, Switzerland; 2015.

11. Pakistan Chest Society. Diagnosis & Management of Drug-susceptible Tuberculosis: A National Clinical Guideline. 2017.

12. Central TB Division. National Framework for A Gender-Responsive Approach to TB in India. Nirman Bhawan, New Delhi; 2019.

13. National Tuberculosis & Leprosy Control Programme. Guidelines on the Use of the Shorter Regimen and New Drugs in the Clinical and Programmatic Management of Drug Resistant Tuberculosis and Co-Infections. 2017.

14. Aboltins C, Crown L, Curtis N, Darby J, Denholm J, Flynn M, et al. Management , Control and Prevention of Tuberculosis: Guidelines for health care providers. South Melbourne: Digital House; 2015.

15. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ. 2009;339:b2535.

16. Dearholt SL, Dang, D. Johns Hopkins nursing evidence-based practice: model and guidelines (3rd ed.). Indianapolis, IN: Sigma Theta Tau International; 2017.

17. Tufanaru C, Munn Z, Aromataris E, Campbell J, Hopp L. Chapter 3: Systematic Reviews of Effectiveness. In: JBI Manual for Evidence Synthesis. JBI; 2020.

For peer review only

18. Asuquo B, Vellore AD, Walters G, Manney S, Mignini L, Kunst H. A case–control study of the risk of adverse perinatal outcomes due to tuberculosis during pregnancy. J Obstet Gynaecol (Lahore). 2012;32(7):635–8.

19. Palacios E, Dallman R, Muñoz M, Hurtado R, Chalco K, Guerra D, et al. Drug-Resistant Tuberculosis and Pregnancy: Treatment Outcomes of 38 Cases in Lima, Peru. Clin Infect Dis. 2009;48(10):1413–9.

20. Lin H-C, Lin H-C, Chen S-F. Increased risk of low birthweight and small for gestational age infants among women with tuberculosis. BJOG An Int J Obstet Gynaecol. 2010;117(5):585–90.

21. Ali A-AA, Abdallah TM, Rayis DA, Adam I. Maternal and perinatal outcomes of pregnancies associated with tuberculosis in eastern Sudan. Int J Gynecol Obstet. 2011;114(3):286–7.

22. Chang C-W, Wu PW, Yeh C-H, Wong K-S, Wang C-J, Chang C-C. Congenital tuberculosis: case report and review of the literature. Paediatr Int Child Health. 2017; 38(216–219):1–4.

23. Inal HA, Ozturk Inal ZH. A Successful Live Twin Birth by In vitro Fertilization in an Untreated Patient with Silent Genital Tuberculosis. Electron J Gen Med. 2015; 12(2):160–3.

24. Kaushal S, Dora SK, Thakur S. Spinal tuberculosis with paraplegia in pregnancy: A case report with management of spinal TB in pregnancy. J Nepal Med Assoc. 2015; 53(198):123–5.

For peer review only

25. Loughenbury P, Pavlou G, Dunsmuir R. Surgical Treatment of Tuberculous Paraparesis in the Third Trimester: A Report of Two Cases and Review of the Literature. Gynecol Obstet Invest. 2009; 68(3):213–6.

26. Sagar T, Gupta K, Rani M, Kaur IR. Disseminated tuberculosis in a newborn infant. Journal of Family Medicine and Primary Care. 2016 Jul;5(3):695.

27. Stuart R, Lewis A, Ramsden C, Doherty R. Congenital tuberculosis after in-vitro fertilisation. Med J Aust. 2009; 191(1):41–2.

28. Yeh JJ, Lin SC, Lin WC. Congenital tuberculosis in a neonate: A case report and literature review. Frontiers in Pediatrics. 2019 Jun 21;7:255.

29. Badve SA, Ghate SD, Badve MS, Rustagi T, Macchiwala T, Parekh AN, et al. Tuberculosis of spine with neurological deficit in advanced pregnancy: a report of three cases. Spine J. 2011; 11(1):e9–16.

30. Bekker A, Schaaf HS, Draper HR, Kriel M, Hesseling AC. Tuberculosis Disease during Pregnancy and Treatment Outcomes in HIV-Infected and Uninfected Women at a Referral Hospital in Cape Town. Wilkinson KA, editor. PLoS One. 2016; 11(11):e0164249.

31. Ali RF, Siddiqi DA, Malik AA, Shah MT, Khan AJ, Hussain H, Chandir S. Integrated tuberculosis screening during antenatal visits to improve Tuberculosis diagnosis and care: Results from a pilot project in Pakistan. International Journal of Infectious Diseases. 2021 Jun 1.

32. LaCourse SM, Greene SA, Dawson-Hahn EE, Hawes SE. Risk of Adverse Infant Outcomes Associated with Maternal Tuberculosis in a Low Burden Setting: A Population-Based Retrospective Cohort Study. Infect Dis Obstet Gynecol. 2016; 2016:1–8.

For peer review only

33. Loveday M, Hughes J, Sunkari B, Master I, Hlangu S, Reddy T, et al. Maternal and Infant Outcomes Among Pregnant Women Treated for Multidrug/Rifampicin-Resistant Tuberculosis in South Africa. Clin Infect Dis. 2021; 72(7):1158–68.

34. Sade S, Wainstock T, Sheiner E, Pariente G. Perinatal Outcome and Long-Term Infectious Morbidity of Offspring Born to Women with Known Tuberculosis. Journal of Clinical Medicine. 2020 Sep;9(9):2768.

35. Salazar-Austin N, Hoffmann J, Cohn S, Mashabela F, Waja Z, Lala S, et al. Poor Obstetric and Infant Outcomes in Human Immunodeficiency Virus-Infected Pregnant Women With Tuberculosis in South Africa: The Tshepiso Study. Clin Infect Dis. 2018; 66(6):921–9.

36. Yadav V, Sharma JB, Kachhawa G, Kulshrestha V, Mahey R, Kumari R, et al. Obstetrical and perinatal outcome in pregnant women with extrapulmonary tuberculosis. Indian J Tuberc. 2019; 66(1):158–62.

37. Ahmadzia HK, Khorrami N, Carter JA, Stone J, Amdur RL. Impact of human immunodeficiency virus, malaria, and tuberculosis on adverse pregnancy outcomes in the United States. J Perinatol. 2020; 40(2):240–7.

38. Fois A, Capobianco G, Crivelli P, Meloni G, Pirina P, Cherchi P, et al. Hemoptoe by tuberculosis in near term pregnant woman: A case report. Clin Exp Obstet Gynecol. 2018; 45(1):115–7.

39. Lu P, Li F, Wang Y, Xu C, Yu J, Zhang Y, et al. Uncommon Abdominopelvic Tuberculosis in Pregnancy. J Gynecol Surg. 2013; 29(6):306–8.

40. Du J, Dong S, Jia S, Zhang Q, Hei M. Clinical characteristics and post-discharge follow-up analyses of 10 infants with congenital tuberculosis: A retrospective observational study. Pediatric Investigation. 2021 Jun;5(2):86-93.

For peer review only

41. Chopra S, Siwatch S, Aggarwal N, Sikka P, Suri V. Pregnancy outcomes in women with tuberculosis: a 10-year experience from an Indian tertiary care hospital. Trop Doct. 2017; 47(2):104–9.

42. Mokhele I, Jinga N, Berhanu R, Dlamini T, Long L, Evans D. Treatment and Pregnancy Outcomes of Pregnant Women Exposed to Second-line Anti-tuberculosis Drugs in South Africa.

43. El-Messidi A, Czuzoj-Shulman N, Spence AR, Abenhaim HA. Medical and obstetric outcomes among pregnant women with tuberculosis: a population-based study of 7.8 million births. Am J Obstet Gynecol. 2016; 215(6):797.e1-797.e6.

44. Klimko A, Brandt A, Brustan MI. A Case of Miliary Perinatal Tuberculosis in a Preterm Newborn Infant Presenting as Peritonitis. Cureus. 2020 May;12(5).

45. Lee JS, Lim CH, Kim E, Lim H, Lee Y, Choung JT, Yoo Y. Congenital miliary tuberculosis in an 18-day-old boy. Korean Journal of Pediatrics. 2016 Nov;59(Suppl 1):S64.

46. Peng W, Yang J, Liu E. Analysis of 170 cases of congenital TB reported in the literature between 1946 and 2009. Pediatric Pulmonology. 2011 Dec;46(12):1215-24.

47. Rajput D, Gupta A, Roshan R, Kumar A. Stillbirth as the primary manifestation of disseminated tuberculosis in a young immunocompetent mother with multiple perforations of the ileum. BMJ Case Reports CP. 2021 Feb 1;14(2):e239386.

48. Li Q, Song Y, Chen H, Xie L, Gao M, Ma L, Huang Y. Retrospective Analysis of 28 Cases of tuberculosis in pregnant Women in china. Scientific Reports. 2019 Oct 25;9(1):1-8.

For peer review only

49. Toro PL, Schneider KL, Carter RJ, Abrams EJ, El-Sadr WM, Howard AA. Maternal and Infant Outcomes With Concurrent Treatment of Tuberculosis and HIV Infection in Pregnant Women. JAIDS J Acquir Immune Defic Syndr. 2011; 56(2):e63–6.

50. Islam B, Islam N, Mehkri H, AlQaasimi M. Abdominal tuberculosis and spontaneous miscarriage. BMJ Case Rep. 2017; 2017:220022.

51. Jacquemyn Y, Van Casteren C, Luijks M, Colpaert C. Disseminated tuberculosis in pregnancy unknown to doctors in Western Europe case presentation:'part of the routine study in infertility'. Case Reports. 2012 Sep 13;2012:bcr2012006227.

52. Ahuja V, Gombar S, Kumar N, Goyal N, Gupta K. Pregnancy with bilateral tubercular pleural effusion: challenges. Trop Doct. 2014; 44(2):116–8.

53. Gai X, Chi H, Cao W, Zeng L, Chen L, Zhang W, Song D, Wang Y, Liu P, Li R, Sun Y. Acute miliary tuberculosis in pregnancy after in vitro fertilization and embryo transfer: a report of seven cases. BMC Infectious Diseases. 2021 Dec;21(1):1-9.

54. Rathod AK, Singh V, Patil P, Singh H. Is decompressive surgery the only treatment option? A case series of patients with spinal tuberculosis in advanced pregnancy. Eur Spine J. 2017; 26(12):3170–7.

55. Ye R, Wang C, Zhao L, Wu X, Gao Y, Liu H. Characteristics of miliary tuberculosis in pregnant women after in vitro fertilisation and embryo transfer. The International Journal of Tuberculosis and Lung Disease. 2019 Feb 1;23(2):136-9.

56. Barbosa OA, Teles FM, Maia ACC, Pessoa GMF, de Alencar IM, de Sousa ET, et al. Disseminated hematogenous tuberculosis in puerperium—case report. Oxford Med Case Reports. 2019; 2019(11):479–81.

For peer review only

57. Raznatovska OM, Fedorec AV, Shalmina MO, Grekova TA. Clinical features of pregnancy in multidrug-resistant tuberculosis and type 1 diabetes mellitus comorbidities (a case report). Медичні перспективи. 2019;24(3).

58. Shi YF, Shi XH, Zhang Y, Chen JX, Lai WX, Luo JM, Ba JH, Wang YH, Chen JN, Wu BQ. Disseminated Tuberculosis Associated Hemophagocytic Lymphohistiocytosis in a Pregnant Woman With Evans syndrome: A Case Report and Literature Review. Frontiers in Immunology. 2021 Jun 10;12:2302.

59. Zhu Z, Zhang M, Li Y. Anti-tuberculosis drug-induced acute liver failure requiring transplantation in the second trimester of pregnancy: a case report. BMC Pegnancy and Childbirth. 2021 Dec;21(1):1-6.

60. Cornish EF, Hudson J, Sayers R, Loveday M. Improving access to contraception through integration of family planning services into a multidrug-resistant tuberculosis treatment programme. BMJ Sex Reprod Heal. 2020; 46(2):152–5.

61. Bishara H, Goldstein N, Hakim M, Vinitsky O, Shechter-Amram D, Weiler-Ravell D. Tuberculosis during Pregnancy in Northern Israel, 2002-2012: Epidemiology and Clinical Practices. Isr Med Assoc J. 2015; 17(6):346–50.

62. Walt MV, Masuku S, Botha S, Nkwenika T, Keddy KH. Retrospective record review of pregnant women treated for rifampicin-resistant tuberculosis in South Africa. PloS one. 2020 Sep 24;15(9):e0239018.

63. Alaoui FZF, Rachad M, Chaara H, Bouguern H, Melhouf MA. Peritoneal tuberculosis in pregnancy: a case report. Pan Afr Med J. 2012;12:65.

For peer review only

64. Brouwer A, Degrieck N, Rasschaert M, Lockefeer F, Huizing M, Tjalma W. Tuberculous mastitis presenting as a lump: a mimicking disease in a pregnant woman case report and review of literature. Acta Clinica Belgica. 2014 Oct 1;69(5):389-94.

65. Erenel H, Alpay V, Alakbarova U, Mut A, Atilgan H, Demiroz AS, Ocal P. Intraperitoneal tuberculosis abscess in pregnancy presenting as an adnexal mass: a case report. Journal of Obstetrics and Gynaecology. 2021 Feb 17;41(2):305-7.

66. John M, Chhikara A, John DM, Khawar N, Brown B, Narula P. Diagnosis of tuberculosis in an asymptomatic child, sibling, and symptomatic pregnant mother in new York City by tuberculin skin testing and the importance of screening high-risk urban populations for tuberculosis. The American Journal of Case Reports. 2018;19:1004.

67. Lahbabi M, Brini J, Massaoudi K. Tuberculous peritonitis in pregnancy: a case report. Journal of Medical Case Reports. 2014 Dec;8(1):1-4.

68. Masukume G, Sengurayi E, Moyo P, Feliu J, Gandanhamo D, Ndebele W, et al. Massive hemoptysis and complete unilateral lung collapse in pregnancy due to pulmonary tuberculosis with good maternal and fetal outcome: a case report. BMC Res Notes. 2013; 6(1):335.

69. Shital P, Mirza M. Laryngeal & Lower lung field tuberculosis in pregnancy: A case report. Electron J Gen Med. 2018; 15(2):1–5.

70. Fernandez D, Salami I, Davis J, Mbah F, Kazeem A, Ash A, et al. HIV-TB Coinfection among 57 Million Pregnant Women, Obstetric Complications, Alcohol Use, Drug Abuse, and Depression. J Pregnancy. 2018; 2018:1–8.

For peer review only

71. Dennis EM, Hao Y, Tamambang M, Roshan TN, Gatlin KJ, Bghigh H, et al. Tuberculosis during pregnancy in the United States: Racial/ethnic disparities in pregnancy complications and in-hospital death. Jhaveri R, editor. PLoS One. 2018; 13(3):e0194836.

72. Adhikari EH, Duryea EL, Rac MWF, Sheffield JS. Genitourinary Tuberculosis: A Rare Cause of Obstructive Uropathy in Pregnancy. Case Rep Obstet Gynecol. 2014; 2014(985682):1–4.

73. Lin S, Guan W, LaZhou C, Shi Y. Left lung hypoplasia with a right tuberculous pleural effusion after childbirth: A case report. Medicine. 2018 May;97(21).

74. Muin DA, Wagner K, Burian R, Ghaem Maghami N, Lapaire O. Brainstem Tuberculoma in Pregnancy. Case Rep Obstet Gynecol. 2015; 2015:1–6.

75. Raouf S, Sharma S, Sunanda G V., Jafettasj J, Papaioannou S. Disseminated extra pulmonary tuberculosis in an immune competent pregnant woman. J Obstet Gynaecol (Lahore). 2009; 29(2):148–50.

76. Modi P, Khanna R, Reddy N, Patankar A, Patel S, Nair G, Gopal S, Uppe A. COVID-19 and tuberculosis co-infection in pregnancy–A case series and review. Journal of Mother and Child. 2021 Oct 29.

77. Tabarsi P, Moradi A, Baghaei P, Marjani M, Shamaei M, Mansouri N, et al. Standardised second-line treatment of multidrug-resistant tuberculosis during pregnancy [Short communication]. Int J Tuberc Lung Dis. 2011; 15(4):547–50.

78. Laniado-Laborín R, Carrera-López K, Hernández-Pérez A. Unexpected Pregnancy during Treatment of Multidrug-resistant Tuberculosis. Turkish Thorac J. 2018; 19(4):226–7.

For peer review only

79. Rendell NL, Batjargal N, Jadambaa N, Dobler CC. Risk of tuberculosis during pregnancy in Mongolia, a high incidence setting with low HIV prevalence. Int J Tuberc Lung Dis. 2016; 20(12):1615–20.

80. Sharma S, Ahad A, Gupta N, Sharma V. Primary gingival tuberculosis in pregnancy: A rare combination. J Indian Soc Periodontol. 2018; 22(2):178.

81. Suresh S, Sharath BN, Anita S, Lalitha R, Prasad TJ, Rewari BB. TB-HIV co-infection among pregnant women in Karnataka, South India: A case series. J Infect Public Health. 2016; 9(4):465–70.

82. Baluku JB, Bongomin F. Treatment outcomes of pregnant women with drug-resistant tuberculosis in Uganda: A retrospective review of 18 cases. International Journal of Infectious Diseases. 2021 Apr 1;105:230-3.

83. Acquah R, Mohr-Holland E, Daniels J, Furin J, Loveday M, Mudaly V, Reuter A. Outcomes of Children Born to Pregnant Women With Drug-resistant Tuberculosis Treated With Novel Drugs in Khayelitsha, South Africa: A Report of Five Patients. The Pediatric Infectious Disease Journal. 2021 May;40(5):e191.

84. van de Water BJ, Brooks MB, Huang CC, Trevisi L, Lecca L, Contreras C, Galea J, Calderon R, Yataco R, Murray M, Becerra MC. Tuberculosis clinical presentation and treatment outcomes in pregnancy: a prospective cohort study. BMC Infectious Diseases. 2020 Dec;20(1):1-8.

85. Rickman HM, Cohn S, Lala SG, Waja Z, Salazar-Austin N, Hoffmann J, et al. Subclinical tuberculosis and adverse infant outcomes in pregnant women with HIV. Int J Tuberc Lung Dis. 2020; 24(7):681–5.

For peer review only

86. World Health Organization. WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-resistant tuberculosis treatment. Geneva, Switzerland; 2020.

87. Schnippel K, Ndjeka N, Conradie F, Berhanu R, Claasen Z, Banoo S, et al. A call to action: Addressing the reproductive health needs of women with drug-resistant tuberculosis. South African Med J. 2016; 106(4):333.

88. World Health Organization. Quality of care in the provision of sexual and reproductive health services. Geneva, Switzerland; 2011.

89. Freyder M, Craig L, Kaji A. Monitoring the Integration of Family Planning and HIV Services: A Manual to Support the Use of Indicators to Measure Progress toward PEPFAR's 90–90-90 Targets and Protect Women's Reproductive Rights. Chapel Hill, North Carolina; 2016.

90. FHI 360, World Health Organization, Uniter States Agency for International Development. Strategic Considerations for Strengthening the Linkages between Family Planning and HIV/AIDS Policies, Programs, and Services. Geneva, Switzerland; 2009.

91. Nahid P, Mase SR, Migliori GB, Sotgiu G, Bothamley GH, Brozek JL, et al. Treatment of Drug-Resistant Tuberculosis. An Official ATS/CDC/ERS/IDSA Clinical Practice Guideline. Am J Respir Crit Care Med. 2019; 200(10):e93–142.

92. Gupta A, Mathad JS, Abdel-Rahman SM, Albano JD, Botgros R, Brown V, et al. Toward Earlier Inclusion of Pregnant and Postpartum Women in Tuberculosis Drug Trials: Consensus Statements From an International Expert Panel. Clin Infect Dis. 2016; 62(6):761–9.

For peer review only

93. McKenna L, Frick M, Lee C, Namutamba D, Smit L, Theunissen M, et al. A Community Perspective on the Inclusion of Pregnant Women in Tuberculosis Drug Trials. Clin Infect Dis. 2017; 65(8):1383–7.

94. World Health Organization. WHO consolidated guidelines on drug-resistant tuberculosis treatment. Geneva, Switzerland; 2019.

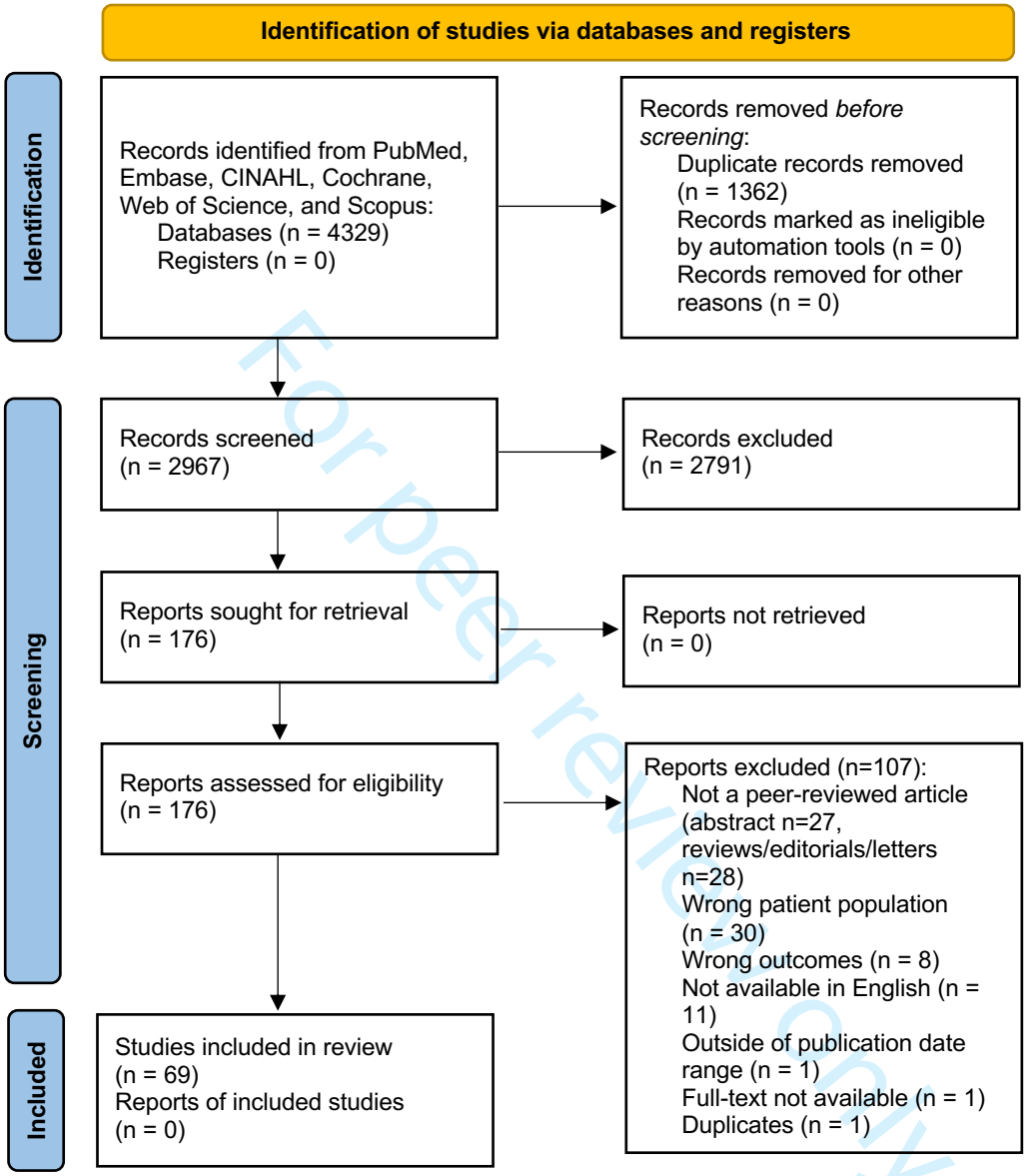
95. Nunn AJ, Rusen ID, Van Deun A, Torrea G, Phillips PP, Chiang CY, Squire SB, Madan J, Meredith SK. Evaluation of a standardized treatment regimen of anti-tuberculosis drugs for patients with multi-drug-resistant tuberculosis (STREAM): study protocol for a randomized controlled trial. Trials. 2014 Dec;15(1):1-0.

96. World Health Organization. Companion handbook to the WHO guidelines for the programmatic management of drug-resistant tuberculosis. Geneva, Switzerland: World Health Organization; 2014.

97. Sun Q, Zhang H, Zhang Y, Peng Z, Lu J, Ma X. Increased Risk of Stillbirth among Women whose Partner Has Tuberculosis. BioMed research international. 2021 Sep 15;2021.

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Figure 1. PRISMA flow diagram: identification and selection of relevant articles




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Supplemental Table 1. Critical Appraisal of the included studies

| Design | Authors | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Case Reports (Level of evidence: 5)** | | Demographic characterstics | Patient's history | Clinical presentation | Diagnostics and assessment | Intervention or treatment procedures | Post-intervention clinical condition | Adverse events identified? | Provision of takeaway lessions | | | Number of criteria met (quality) n/8 |
| Case Report | Adhikari (2014)[72] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Ahuja (2014)[52] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Alaoui (2012)[63] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Barbosa (2019)[56] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Brouwer (2014)[64] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Chang (2018)[22] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Erenel (2019)[65] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Fois (2017)[38] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Inal (2015)[23] | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 7 |
| Case Report | Islam (2017)[50] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Jacquemyn (2021)[51] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | John (2018)[66] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Kaushal (2015)[24] | x | ✓ | ✓ | x | ✓ | x | ✓ | ✓ | | | 5 |
| Case Report | Kimko (2020)[44] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Lahbabi (2014)[67] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Laniado-Laborin (2017)[78] | x | ✓ | ✓ | x | ✓ | ✓ | x | ✓ | | | 5 |
| Case Report | Lee (2015)[45] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Lin (2018)[73] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Lu (2013)[39] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Loughenbury (2009)[25] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Masukume (2013)[68] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Muin (2015)[74] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Rajput (2021)[47] | ✓ | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | | | 6 |
| Case Report | Raznatovska (2021)[57] | ✓ | x | ✓ | ✓ | x | x | ✓ | ✓ | | | 5 |
| Case Report | Sagar (2016)[26] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Sharma (2018)[80] | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 7 |
| Case Report | Shi (2021)[58] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Shital (2018)[69] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Stuart (2009)[27] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| Case Report | Yeh (2019)[28] | ✓ | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | | | 7 |
| Case Report | Zhu (2021)[59] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | 8 |
| **Case Series (Level of evidence: 5)** | | Clear criteria for inclusion? | Condition measured in a standard, reliable way for all participants? | Valid methods for identification of condition included? | Consecutive inclusion of participants? | Complete inclusion of participants? | Clear reporting of demographics? | Clear reporting of clinical information? | Outcomes or follow up results clearly reported? | Clear reporting of presenting sites/clinics demographic information? | Statistical analysis appropriate? | Number of criteria met (quality) n/10 |
| Case Series | Acquah (2021)[83] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Case Series | Badve (2011)[29] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | x | x | 6 |
| Case Series | Bishara (2015)[61] | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Case Series | Du (2021)[40] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Li (2019)[48] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Modi (2021)[76] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Palacios (2009)[19] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Case Series | Peng (2011)[46] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 10 |
| Case Series | Rathod (2017)[54] | ✓ | ✓ | x | x | x | ✓ | ✓ | ✓ | x | x | 5 |
| Case Series | Suresh (2015)[81] | ✓ | ✓ | x | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | 8 |
| Case Series | Tabarsi (2011)[77] | ✓ | ✓ | x | x | x | ✓ | ✓ | ✓ | ✓ | ✓ | 7 |
| Case Series | Toro (2011)[49] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | 9 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Case Series | Ye (2019)[55] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | | 9 |
| **Case-control (Level of evidence: 3)** | | Groups comparable? | Cases and controls appropriately matched? | Same criteria for identification of cases and controls? | Exposure measured in a standard, reliable, and valid way? | Exposure measured in the same way for cases and controls? | Confounding factors stated? | Strategies to deal with confounding factors stated? | Outcomes assessed in a standard, reliable, and valid way for cases and controls? | Exposure period of interest long enough to be meaningful? | Appropriate statistical analysis? | | Number of criteria met (quality) n/10 |
| Case-control | Asuquo (2012)[18] | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | ✓ | x | x | | 7 |
| Case-control | Chopra (2017)[41] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | ✓ | | 9 |
| Case-control | Lin (2010)[20] | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | 9 |
| **Cohort Study (Level of evidence: 3)** | | Groups similar and recruited from the same population? | Exposures measured similarly to exposed and unexposed groups? | Exposure measured in a valid and reliable way? | Confounding factors identified? | Strategies to deal with confounding factors stated? | Groups free of outcome(s) at start of study? | Outcomes measured in valid and reliable way? | Follow up time sufficiently long enough for outcome(s) to occur? | Follow up complete? Reasons for loss to follow up described? | Were strategies to address incomplete follow up used? | Appropriate statistical analysis? | Number of criteria met (quality) n/11 |
| Retrospective Cohort Study | Ahmadzia (2020)[37] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | Ali (2011)[21] | ✓ | | | x | x | ✓ | ✓ | ✓ | | | x | 4 |
| Retrospective Cohort Study | Ali (2021)[31] | | | ✓ | | | ✓ | ✓ | ✓ | ✓ | | ✓ | 6 |
| Retrospective Cohort Study | Baluku (2021)[82] | | | ✓ | x | x | ✓ | ✓ | ✓ | | | ✓ | 5 |
| Prospective Cohort Study | Bekker (2016)[30] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | x | ✓ | 9 |
| Retrospective Cohort Study | Dennis (2018)[71] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | x | x | ✓ | 9 |
| Retrospective Cohort Study | El-Messidi (2016)[43] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | Fernandez (2018)[70] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | LaCourse (2016)[32] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Retrospective Cohort Study | Loveday (2020)[33] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | 10 |
| Retrospective Cohort Study | Mokhele (2021)[42] | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | | ✓ | 9 |
| Retrospective Cohort Study | Rendell (2016)[79] | x | x | ✓ | x | x | | ✓ | x | | | ✓ | 3 |
| Prospective Cohort Study | Salazar-Austin (2007)[35] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | | ✓ | 8 |
| Retrospective Cohort Study | Sade (2020)[34] | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Retrospective Cohort Study | Sun (2021)[97] | ✓ | ✓ | ✓ | ✓ | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 9 |
| Retrospective Cohort Study | van der Walt (2020)[62] | | | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 6 |
| Retrospective Cohort Study | Yadav (2019)[36] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | | ✓ | 8 |
| **Other (Level of evidence: 3)** | | Groups similar and recruited from the same population? | Exposures measured similarly to exposed and unexposed groups? | Exposure measured in a valid and reliable way? | Confounding factors identified? | Strategies to deal with confounding factors stated? | Groups free of outcome(s) at start of study? | Outcomes measured in valid and reliable way? | Follow up time sufficiently long enough for outcome(s) to occur? | Follow up complete? Reasons for loss to follow up described? | Were strategies to address incomplete follow up used? | Appropriate statistical analysis? | Number of criteria met (quality) n/11 |
| Secondary Data Analysis | Rickman (2020)[85] | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | ✓ | 9 |
| Secondary Data Analysis | van de Water (2020)[84] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | x | ✓ | 8 |
| **Project Improvement (Level of evidence: 5)** | | Cause and effect? | Participants in each comparison group similar? | Participants received similar treatment? | Control group? | Multiple measurements of outcome pre and post intervention? | Follow up complete? | Outcomes in comparison groups measured the same way? | Outcomes measured in a reliable way? | Appropriate statistical analysis used? | | | Number of criteria met (quality) n/9 |
| Project Improvement | Cornish (2020)[60] | ✓ | ✓ | ✓ | x | x | ✓ | ✓ | ✓ | ✓ | | | 7 |

***Blank cells denote not applicable

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Supplemental Table 2. Summary of Pregnancy Outcomes and Other Relevant Findings

| Design | First Author (Year), Reference | Setting | Sample | PTB, EPTB, congenital | HIV Co-infection | Maternal outcomes | Fetal/Newborn outcomes | Other relevant findings |
|---|---|---|---|---|---|---|---|---|
| **Case Report** | | | | | | | | |
| Case Report | Adhikari (2014)[72] | US | 1 (30 yo) | DR EPTB (genitourinary) | no | cured of TB | induction of labor due to severe pre-eclampsia at term | postpartum complicated by pyelonephritis; fibrosis and chronic inflammation of ureteral segments |
| Case Report | Ahuja (2014)[52] | India | 1 (27 yo) | PTB | no | discharged home to continue ATT | fetal demise | delayed TB diagnosis; status post ventilator with resolved pulmonary embolism; resolved anemia |
| Case Report | Alaoui (2012)[63] | Morocco | 1 (37 yo 23 weeks gestation) | EPTB (peritoneal) | no | cured of TB | normal birth weight, delivered at term: vaginal delivery 37 weeks | NA |
| Case Report | Barbosa (2019)[56] | Brazil | 1 (22 yo 22 weeks gestation) | PTB, EPTB (disseminated) | no | death | induced abortion due to suspicion of chorioamnionitis | NA |
| Case Report | Brouwer (2014)[64] | Belgium | 1 (33 yo 4 weeks gestation) | EPTB (left breast) | no | completion of TB treatment | delivered at term | misdiagnosis of TB and delayed treatment x13 weeks from presentation |
| Case Report | Chang (2017)[22] | Taiwan | 1 (14 day old infant, delivered at 35 weeks) | congenital | no | not discussed | LBW; completed TB treatment, gradual resolution of miliary TB | NA |
| Case Report | Erenel (2021)[65] | Turkey | 1 (24 yo 15 weeks gestation) | EPTB (intraperitoneal) | no | Completed treatment | delivered at term, normal weight, negative for TB; BCG vaccine given; INH prophylaxis given x6 months | NA |
| Case Report | Fois (2017)[38] | Italy | 1 (28 yo 36 weeks gestation) | PTB | no | cured of TB | preterm cesarean delivery at 36 weeks, outcome unknown | NA |
| Case Report | Inal (2015)[23] | Turkey | 1 (34 yo, twins at 27 and 30 weeks gestation) | EPTB (genital) | no | cured of TB | preterm cesarian delivery for both twins at 30 week gestation; LBW | NA |
| Case Report | Islam (2017)[50] | Saudi Arabia | 1 (23 yo 23 weeks gestation) | EPTB (abdominal) | no | cured of TB | spontaneous abortion | dilatation and curretage caused uterine perforation and sepsis; resolved sepsis, bilateral pleural effusion; TB treatment was interrupted during transfer of care due to miscommunication |
| Case Report | Jacquemyn (2012)[51] | Belgium | 1 (28 yo 13 weeks gestation) | PTB, EPTB (disseminated) | no | cured of TB | spontaneous abortion | acid-fast bacilli on examination of placenta, but not fetus |
| Case Report | John (2018)[66] | US | 1 (37 yo 19 weeks gestation) | PTB | no | recovered fully | healthy infant at birth, treated with isoniazid prophylaxis, negative T-spot test at age 3-month and isoniazid was discontinued | mother had delayed diagnosis of TB; cough, fever, night sweats, 10-lb weight loss in the last 3 months - T-spot test + (30mm induration) and referred to pulmonologist - placed on azithromycin for presumptive PNA; admitted to inpatient TB treatment when 2-yo daughter was screened + for TB |
| Case Report | Kaushal (2015)[24] | India | 1 (27 yo 35 weeks gestation) | EPTB (spinal) | no | continued TB treatment after discharge; doing well at 3 month post-discharge | fetal distress, emergency cesarian section at 37 weeks; ATT prophylaxis x3 months; no congenital TB at 3 month post-discharge | improvement in motor power over weeks |
| Case Report | Klimko (2020)[44] | Romania | 1 (27 days old infant delivered at 33 weeks) | congenital | no | mom diagnosed with latent TB and isolated from infant; TB treatment outcome not discussed | improved TB; premature birth, LBW (2100g) | presents with hepatosplenomegaly, respiratory distress, hypotonic, irritable; misdiagnosis of late-onset culture-negative neonatal sepsis - patient's condition worsened prompting TB screening of mother, paracenthesis performed + for TB and started on treatment |

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|---|---|---|---|---|---|---|---|---|
| Case Report | Lahbabi (2014)[67] | Morocco | 1 (31 yo 22 weeks gestation) | EPTB (peritoneal) | no | cured of TB | delivered at term, normal weight, no TB | NA |
| Case Report | Laniado-Laborin (2017)[78] | Mexico | 1 (15 yo 8 weeks gestation) | DR PTB | no | cured of TB; normal vaginal delivery | NA | fetal exposure to levofloxacin and ethionamide; provider offered induced abortion |
| Case Report | Lee (2015)[45] | South Korea | 1 (18 days old infant) | congenital | no | No symptoms at 8-month follow up | improvement of lung infiltration and no symptoms after 8 months | delayed diagnosis of TB in mother; baby's initial presentation of respiratory distress and fever, initial misdiagnosis of PNA, then started on TB treatment on day #2 of hospitalization |
| Case Report | Lin (2018)[73] | China | 1 (31 yo, gestation age unknown) | PTB | no | cured of TB | spontaneous vaginal delivery; TB outcome on neonate not provided | 1 month delay of TB diagnosis after presenting with symptoms |
| Case Report | Lu (2013)[39] | China | 1 (38 yo 28 weeks gestation) | TB (abdominopelvic) | no | cured of TB | preterm cesarean delivery at 29 weeks, outcome unknown | abdominal hysterectomy, bilateral salpingectomy, and omentectomy |
| Case Report | Loughenbury (2009)[25] | UK | 2 (33 weeks and 33 weeks) | EPTB (spinal) | no | cured of TB | preterm cesarian delivery for both at 33 and 34 weeks; LBW reported (1980g and 2100g) | moms given steroids to aid fetal lung maturation |
| Case Report | Masukume (2013)[68] | Zimbabwe | 1 (33 yo 18 weeks gestation) | PTB | no | cured of TB; vaginal delivery at 41 weeks | normal birth weight | delayed TB diagnosis |
| Case Report | Muin (2015)[74] | Switzerland | 1 (19 yo, 28 weeks gestation) | EPTB (brainstem) | no | cured of TB; failed induction at 40 weeks, underwent cesarian section | not discussed | improvement neurological symptoms; steroids to aid fetal lung maturation; recommendation for treatment via a multidisciplinary approach inclusive of neonatal care |
| Case Report | Rajput (2021)[47] | India | 1 (21 yo female postpartum) | EPTB (intestinal) | no | died from TB due to sepsis secondary to 20 ileal perforations from TB and delayed presentation to the hospital; asymptomatic TB during pregnancy with symptom manifestation postpartum | full term stillbirth | delayed TB diagnosis, was not started on TB treatment prior to death |
| Case Report | Raouf (2009)[75] | UK | 1 (23 yo) | EPTB (brain and eye) | no | cured of TB; complications resolved with scarring of eye | full term induction vaginal birth; normal birth weight | Patient stopped ATT by herself 3 weeks before delivery because she wanted to breastfeed |
| Case Report | Raznatovska (2019)[57] | Ukraine | 1 (38 yo became pregnant during TB treatment) | DR PTB | no | treatment failure and extensive resistant TB by month 7 of treatment; recommended induced abortion based on clinical indications of: non-conversion of culture after 3 months of DR-TB treatment, nephropathy and intoxication syndrome secondary to diabetes, adenomyosis and bloody vaginal discharge | induced abortion due to advanced TB disease | Initial diagnosis found resistance to rifampicin, unclear what treatment patient was on; on month 1, sputum found resistance to further isoniazid and pyrazinamide, unclear what therapy was placed on her; on month 7, was further found to be resistant to ofloxacin, kanamycin, and capreomycin, unclear what therapy was placed on her at this time |
| Case Report | Sagar (2016)[26] | India | 1 (20 day old baby) | Congenital | no | unknown TB outcome; diagnosed with PTB at 8 month pregnancy and started on TB treatment 8 days prior to baby's presentation of fever, decreased appetite, and rapid breathing | preterm birth at 24 week gestation; LBW 1800g at 20 days old; X-ray after 12 month treatment normal | treated x12 months on isoniazid, rifampicin, pyrazinamide, and amikacin |
| Case Report | Sharma (2018)[80] | India | 1 (25 yo old 4th month of pregnancy) | EPTB (gingiva) | no | cured of TB; normal vaginal delivery | NA | NA |
| Case Report | Shi (2021)[58] | China | 1 (26 yo 15 week gestation) | PTB, EPTB (disseminated) | no | improved with combined therapeutic therapy; follow up at 2 month doing well on isoniazid and rifampicin; Patient with history of Evans syndrome on daily corticosteroid. On day 3 hospitalization, condition deteriorated; final diagnoses included disseminated TB-associated hemophagocytic lymphohistiocytosis, severe PNA, respiratory failure, severe acute respiratory distress syndrome | fetal death day 5 of hospitalization, caseating tuberculous granulomas found in placenta | was initiated on isoniazid, rifampicin, ethambutol, pyrazinamide, and moxifloxacin, then on "four drug…regimen" with methylprednisolone and immunoglobin and plasma exchange with improvement and d/c day 19 |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Case Report | Shital (2018)[69] | India | 1 (25 yo 24 weeks gestation) | PTB, EPTB (laryngeal) | no | cure of TB; normal vaginal delivery | normal birth weight | NA |
| Case Report | Stuart (2009)[27] | Australia | 1 (newborn delivered at 28 weeks) | Congenital | no | TB outcome not reported; no other maternal outcome reported | premature birth, emergency cesarian section, LBW; ventilator use due to respiratory distress; neonatal jaundice, right-sided Erb palsy | neutropenia due to ethambutol in newborn; all newborn's complications resolved and newborn was discharged from hospital |
| Case Report | Yeh (2019)[28] | Taiwan | 1 (8 day old neonate) | Congenital | no | initiated on TB treatment but died 3 days later | delivered at 37 weeks with LBW (2380g); presented with fever x2 days; diagnosed with TB day 20, initiated on TB treatment at day 24; no TB outcome reported | delayed TB diagnosis; 33 yo mother with dry cough 1 week postpartum, admitted to ICU for altered mental status 24 day postpartum, tested for TB due to suspected TB in neonate |
| Case Report | Zhu (2021)[59] | China | 1 (26 yo 11 weeks gestation) | EPTB (plural space) | no | TB treatment completion without relapse or liver damage at month 8 follow up; underwent liver transplant at 17 week gestation; uterine curettage day 27 due to incomplete abortion | induced termination of pregnancy by induction at 21 week gestation (30 day postop liver transplant of mother) due to ventricle widening in the fetus | TB treatment changed 3 times: isoniazid, rifampicin, and pyrazinamide at initiation; then amikacin, levofloxacin, and meropenem after liver transplant; then linezolid, levofloxacin, and pyridoxine at postop day 20 |
| **Case Series** | | | | | | | | |
| Case Series | Acquah (2021)[83] | South Africa | 5 (18-29 years old) | DR PTB | yes (n=3) | 3 cured, 2 lost to follow up | delivered at term, no congenital abnormalities, none developed TB or HIV | all women were exposed to regimens containing bedaquiline, delamanid, and linezolid |
| Case Series | Badve (2011)[29] | India | 3 (third trimester pregnancies, median age=30) | EPTB (spinal) | no | 1 woman with residual spastic paraparesis, otherwise complete neurological recovery in the other 2 women; unclear TB outcomes | 1 spontaneous abortion; urgent 2 cesarian section with 1 LBW (2400g); 1 newborn suffers neonatal sepsis in premature care facility; 1 newborn normal birth weight | kyphosis, spinal cord compression |
| Case Series | Bishara (2015)[61] | Israel | 6 (35-41 weeks gestation, median age=22) | both (n=4 PTB, n=2 EPTB) | no | all 6 cured of TB | 1 intrauterine growth restriction; 2 out of 6 PNA post-delivery (moms had EPTB), 2 newborns treated for latent TB | TB diagnostic and treatment delay in all 6 women |
| Case Series | Du (2021)[40] | China | 10 (mean gestational age 36.6 SD 2.2 weeks) | congenital | no | none treated for TB during pregnancy though 1 woman was diagnosed during pregnancy | 5 were preterm infants (<37 weeks); during treatment: 4 failed to thrive, 3 had neurological sequelae including seizures, 1 had seizure requiring shunt and physical therapy; at 2 month follow up: 1 baby lost to follow up, five fully recovered, 3 babies had hepatotoxicity secondary to TB treatments; at 6 months of age, 3/9 babies had normal growth and development | NA |
| Case Series | Gai (2021)[53] | China | 7 (ages 28-34) | PTB, EPTB (miliary TB, with 2 TB meningitis cases) | not reported | achieved pregnancy after in-vitro fertilization; 4 spontaneous abortion, 3 induced abortions; all recovered after TB treatment | NA | clinical pregnancy and live birth rates significantly lower in infertile patients with untreated prior PTB than in those without |
| Case Series | Li (2019)[48] | China | 28 pregnant women with TB (IQR 27-33) | PTB, EPTB | no | 10/28 were initially misdiagnosed as other respiratory illnesses; 27 cured of TB, 1 death | 4 spontaneous abortions, 9 induced abortions; 4 neonates were healthy; 1 small for gestational age, 1 stillbirth, 2 congenital TB, 1 died at 4 months old | 35.7% of cases of misdiagnosis potentially due to atypical presentation leading to treatment delays and adverse pregnancy outcomes |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Case Series | Modi (2021)[76] | Inida | 2 pregnant women with TB-COVID-19 coinfection (third trimester pregnancy) | PTB | no | patient 1 treated with quadruple therapy for TB and remdesivir and oxygen in ICU x10 days and was discharged; patient 2 underwent c-section at 38 weeks and managed in ICU with O2 and remdesivir and quadruple therapy for TB | term delivery with normal weight baby for patient 2; no discussion regarding patient 1's fetus | NA |
| Case Series | Palacios (2009)[19] | Peru | 38 (mean age=24.4 SD=5.8) | DR PTB (n=37), EPTB (n=1) | yes (n=3) | 23 (n=38, 61%) cured of TB; 4 clinical TB deterioration, 8 deaths; 2 episodes of vaginal bleed, 1 placenta previa, 1 premature membrane rupture | 5 spontaneous abortions; 1 stillborn; of 32 infants born alive: three with LBW <2500g, one fetal distress; two latent TB treated, 1 active TB treated at mo 9 | treatment was suspended with confirmed pregnancy (n=14); second-line drugs may be continued despite teratogenic risk; most had advanced disease at baseline (n=36) |
| Case Series | Peng (2011)[46] | China | 6 neonates (5-42 days old) | congenital | NA | 3 mothers had miliary TB, 1 had genital TB, 3 unknown TB; outcomes of mothers not discussed/unknown | 4 full term, 2 premature babies; all 6 babies demonstrated cough and hepatosplenomegaly, 4 with rales, 3 fever, respiratory distress, and poor feeding; two with abdominal distention, 1 jaundice, 1 with ascites; 4 improved with TB treatment, 2 died without TB treatment initiation | study also conducted a literature review of congenital TB - 95% of cases had mothers with active TB during pregnancy; laboratory results may cause misdiagnosis of other bacterial infections vs TB |
| Case Series | Rathod (2017)[54] | India | 5 (third trimester pregnancies, median age=24) | EPTB (spinal) | no | 3 normal vaginal deliveries | 1 spontaneous abortion; 1 fetal demise due to adverse reaction to anesthesia | 3 with resolution of TB, no sequelae; 1with mild back pain; 1 with mild neck pain; 1 patient with recovery of paraplagia; 1 with destruction of spinal vertebra and kyphosis; 1 with 15 degrees kyphosis |
| Case Series | Suresh (2015)[81] | India | 17 (median age=24) | both | yes | 5 deaths, 3 died before delivery, 4 elected abortions | 1 death | high maternal mortality in low-resource high HIV burden countries |
| Case Series | Tabarsi (2011)[77] | Iran | 5 (median age=22) | DR PTB | NA | cured of TB; normal vaginal delivery | normal weight | offer of induced abortion; treatment was suspended with confirmed pregnancy and resumed at patient's decision |
| Case Series | Toro (2011)[49] | sub-Saharan Africa | 33 (median age=29 IQR 27-31) | Not reported | yes | 32 survived, 25 with live births, 1 induced abortion, 5 spontaneous abortions/stillbirths, 1 infant outcome unknown | rate of spontaneous abortion/stillbirths among women with TB-HIV coinfection more than twice that compared to women receiving ART alone during pregnancy (15.2% versus 5.9%); 1 stillbirth was HIV+ | analyzed women with TB-HIV coinfection; median gestational age 33 IQR 27-37, median CD4 97 IQR 48-196 |
| Case Series | Ye (2019)[55] | China | 6 (mean age 32.17± 3.87 years) | EPTB | no | cured of TB | 3 spontaneous abortions, 3 elected abortions | delayed TB diagnosis (10-32 days from onset of symptoms to diagnosis) |
| **Case-control** | | | | | | | | |
| Case-control | Asuquo (2012)[18] | UK | 24 (pregnant TB) + 72 (pregnant control, no TB) | PTB (n=13), EPTB (n=10), both (n=1) | yes (n=1) | cured of TB not reported | LBW (p=0.03) explained by pre-term delivery (OR 179.9, p=<0.001) | complications include hepatitis, cholestasis, and vomiting |
| Case-control | Chopra (2017)[41] | India | 50 women matched with TB control | both | NA | anemia (23% vs. 4% control, p-value not reported); 2/50 died due to TB disease progression, 2/50 lost to follow up | 3x more likely to have fetal growth retardation (OR 3.2, 95% CI 1.3-7.9), preterm labor x5 more likely (OR 5.9 95% CI 2.5-13.9) | NA |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Case-control | Lin (2010)[20] | Taiwan | 761 women matched with 3805 on age and year of delivery | not reported | no | NA | SGA (aOR 1.22, 95% CI 1.00-1.49); LBW (aOR 1.35, 95% CI 1.01-1.81) | NA |
| **Cohort Study** | | | | | | | | |
| Retrospective Cohort Study | Ahmadzia (2020)[37] | US | 1450 TB cases out of 24,149,664 deliveries | PTB, EPTB | NA | TB group more likely to experience pre-term labor (90, 13% versus 162435, 6.8%, p=0.004), to have UTIs ( 7.0% versus 3.3%, p=0.02), and undergo cesarian delivery (30.9% versus 21.5%, p=0.007) compared to those without TB | fetal distress more likely in fetuses with mothers infected with TB compared to control group (5% versus 2.1%, p=0.02) | NA |
| Retrospective Cohort Study | Ali (2011)[21] | Sudan | 42 (mean ae 30.7, SD 3.7) out of 15252 pregnant women | 35 PTB, 7 EPTB | yes (n=5) | anemia (83.3% versus 61.9%, p=0.02) compared to those without TB, 2 deaths in women with TB-HIV coinfection | LBW for newborns to women with tuberculosis (28.5% vs 14.3%, p=0.03); higher numbers of preterm births (28.5% versus 19.0%, p=0.04) compared to those without TB | NA |
| Retrospective Cohort Study | Ali (2021)[31] | Pakistan | 27 pregnant women with TB out of 2,896 screened | 26 PTB, 1 DR PTB | NA | Compared to those without TB, women with TB had higher prevalence of reported weight loss (59.2% vs 23.0%, P <0.01); 25 of women with TB successfully linked for treatment; 24 completed treatment, 1 lost to follow up | 26 live births, 1 intrauterine death at 32 weeks, 8 had LBW (<2500g), 1 had very LBW (<1700g); 16 gastric samples collected and no vertical transmission was found | NA |
| Retrospective Cohort Study | Baluku (2021)[82] | Uganda | 18 (mean age 27.5, SD 5.2 years) | DR PTB | yes (n=8) | Treatment success among 15 (83.3%) patients, 2 (11.1%) were lost to follow up and 1 (5.6%) died; elevated liver enzymes (81.8%, n = 11) and hearing loss (33.3%, n = 15) were the most frequently encountered drug adverse events | NA | 6 were pregnant at initiation of TB treatment |
| Prospective Cohort Study | Bekker (2016)[30] | SA | 74 comparingmaternal TB-HIV and TB | both (14% DR PTB) | yes (n=53, 72% sample) | 5 deaths in HIV-coinfected attributed to TB; women delivering LBW infants were 3.83 times more likely to have unfavourable TB treatment outcome such as death, treatment failure, loss to follow up (95% CI 1.40–10.53, p = 0.009), compared to women delivering infants weighing> 2500 grams. | 6 infants died, 4 stillbirths, all to HIV positive women; 49 premature, 44 LBW, 2 congenital TB; 57 infants given isoniazid prophylaxis, 39 with outcomes at 6 months: 24/39 (62%) completed treatment, 13/39 (33%) did not, and 2/39 (5%) died; of 55 HIV exposed infants, HIV PCR test performed on 45, 3 were + and initiated on ART | median CD4 count 155 IQR 11-565, 29/53 (55%) on ART at time delivery, 10 (19%) no ART, Women with HIV-coinfection had more severe EPTB (15% versus 0%); only 64% had documented ART use |
| Retrospective Cohort Study | Dennis (2018)[71] | US | 4053 TB positive among 57393459 pregnancy and delivery hospitalizations | not reported | yes | rate of pregnancy complications (severe pre-eclampsia, eclampsia, placenta previa/accreta/abruption, antepartum/ postpartum hemorrhage, sepsis, anemia) 80% higher among TB infected mothers compared to non-TB infected mothers (Composite rate 293.2 TB positive versus 166.5 TB negative) | NA | non-Hispanic black mothers have an aOR 2.77 (95% C.I. 2.35, 3.25) of non-TB related death compared to white mothers; TB infected mothers have similar risk level across all ethnicities (white, black, Hispanic); over 1/3 of all TB cases were diagnosed among mothers of Hispanic origin |
| Retrospective Cohort Study | El-Messidi (2016)[43] | US | 2064 | not reported | NA | increased aOR of chorioamniotitis (OR=1.35, 95% C.I., 1.04-1.74), postpartum anemia (OR=1.51; 95% C.I., 1.22-1.87), PNA (OR 8.42, 95% C.I., 5.77-12.29), ARDS (OR, 2.85; 95% CI, 1.35-6.10), mechanical vent (OR 3.33, 95% C.I. 1.66-6.68), maternal mortality increased among pregnant women with TB (OR=6.27, 95% C.I. 2.01-19.58) | congenital anomalies (aOR=1.8, 95% C.I 1.24-2.62) | Medicaid the most common payer for TB-related hospitalizations; more pregnant women with TB from lower income quartiles |

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|---|---|---|---|---|---|---|---|---|
| Retrospective Cohort Study | Fernandez (2018)[70] | US | 4053 TB positive cases among 57393459 pregnancy and delivery hospitalizations | not reported | yes (n=110) | rate of postpartum hemorrhage highest in TB only group (45.5 per 1,000 hospitalizations) compared to TB and HIV negative group (25.8 per 1000), similar to anemia (216.6 compared to 102.7 per 1000 hospitalizations); compared to HIV & TB negative mothers, mothers with HIV or TB have significantly higher odds of pregnancy complications (HIV, aOR=1.4, 95% C.I. 1.32-1.47; TB, aOR=1.91, 95% C.I. 1.64-2.23) | NA | Rate of TB among HIV-positive mothers was 21 times higher compared to rate of TB in HIV-negative mothers; low socioeconomic status and having public or no health insurance were biggest risk factors for HIV or TB monoinfection and HIV-TB coinfection' pregnancy complications highest among HIV-coinfected mothers (331.28 per 1,000 hospitalizations), followed by those with TB mono-infection (292.1 per 1,000 hospitalizations) |
| Retrospective Cohort Study | LaCourse (2016)[32] | US | 134 matched by year delivery and TB status | not reported | NA | NA | LBW (aRR=3.74, 95% C.I. 1.40-10.00); SGA (uRR=1.95, 95% C.I. 1.11-3.41) | NA |
| Retrospective Cohort Study | Loveday (2020)[33] | SA | 108 | DR PTB | yes (n=88, 81%) | completed treatment or cured of TB (67%), treatment failure (3%), death (3%) | fetal and neonatal deaths (9%), treated for DR PTB in the first year of life (2%); LBW for those exposed to bedaquilline (45% vs 24%, p=0.034) | mean age 28, SD ±6.13; baseline cd4 median 353 IQR 165-511<br>low baseline hemoglobin a predictor of poor TB treatment outcome (uHR 0.67, p= 0.006); women living with HIV had a higher risk of a poor pregnancy outcomes (uHR 3.35; p=0.030); bedaquilline and levofloxacin are predictors of LBW |
| Retrospective Cohort Study | Mokhele (2021)[42] | SA | 35 | DR PTB | yes (n=24) | Overall 48.6% had adverse events; 82.4% had 1 and 17.6 two or more; most prevalent was nephrotoxicity (35.3%), anemia (23..5%), hypokalemia (23.5%), ototoxicity and rash (11.8%); HIV-negative patients had fewer adverse events (18.2% vs 62.5%, p<0.05); of 28 with TB outcomes, 17 completed treatment, 8 lost to follow up, 3 died; for 20 women with pregnancy outcomes, 65% of these had an adverse pregnancy outcomes | of 20 known pregnancy outcomes, there were 4 term births, 11 preterm births, 1 miscarriage, 1 neonatal death, and 3 pregnancy terminations; 10/11 preterm births were from mothers with HIV coinfection and 1 neonatal death of mother with HIV coinfection as well | median age 30.1 IQR 28-35, median CD4 145 IQR 51-301<br>Patients with HIV coinfection had more severe immunosuppression (median baseline CD4 145) with more adverse outcomes; authors propose that DR PTB & HIV coinfection during pregnancy are cause of poor birth outcomes, recommending increased TB prevention strategies and TB treatment in antenatal care & HIV programs |
| Retrospective Cohort Study | Rendell (2016)[79] | Mongolia | 107 | PTB and EPTB | no | 10/85 elected abortion after TB diagnosis | NA | 2/82 TB treatment failure, 4/82 lost to follow-up; higher incidence of TB in pregnancy compared to general population (IRR 1.31, CI 1.08-1.59) |
| Retrospective Cohort Study | Sade (2020)[34] | Israel | 46 neonates and mothers with TB compared to 243636 mothers without TB | PTB, latent TB | yes (n=2 mothers) | 3 cases of placental abruption, 1 resulting in perinatal death; placental abruption is significantly associated with mothers with TB compared to those without (23.1% vs. 2.9%, p<0.001); TB was an independent risk factor for placental abruption and very LBW when controlling for maternal age (10.76, 95% CI 3.37–34.36) and gestational age (11.27, 95% CI 3.44-36.96) | higher rates of very LBW (<1500g) newborns in women with latent TB when compared to those with active TB during pregnancy, latent TB, and those without TB (0.0%, 5.6%, and 0.6%, p < 0.001); no significant difference in long-term infectious morbidity between children of mothers with and without TB | NA |

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|---|---|---|---|---|---|---|---|---|
| Prospective Cohort Study | Salazar-Austin (2007)[35] | SA | 80 (pregnant women with TB) matched with 155 controls (no TB) | PTB and DR PTB | yes | anemia more in cases than control (27.5% versus 11.0%, p=0.001); pre-eclampsia 8-fold more common among case patients than controls (5.2% vs. 0.7%, p=0.03); higher rates of maternal hospitalization (25% vs. 11%, p=0.005) | infants born to HIV/TB moms twice as likely to have LBW (20.8% vs. 10.7%, p=0.04), prolonged hospitalizations at birth (51% vs. 16%, p=<0.001) than born to HIV only; infants born to HIV/TB moms had 10- fold increased infant mortality risk (p=<0.001) | median age 29 IQR 26-31, median gestation 30 weeks (26-34), CD4 among cases 242 IQR 137-379) compared to controls 369, IQR 263-476, p<0.001), less cases were virally suppressed than controls (30.8% versus 47.7%, p<0.001); over 90% of both cases and controls were on ART at screening |
| Retrospective Cohort Study | Sun (2021)[97] | China | 6389 cases (pregnant women or partner with TB) matched with 31945 controls (couple without TB) | PTB | unknown | no significant differences in PTB (OR: 0.87, 95% CI 0.66–1.12) between women with and without TB; women whose partner had TB had a higher risk of stillbirth (OR 2.13, 95% CI 1.10–3.86) than women whose partner did not have TB | no significant differences in stillbirth (OR 1.48, 95% CI 0.61–3.09) and LBW (OR 1.04, 95% CI 0.79–1.35) between women with and without TB | also stratified analysis by TB treatment and found similar results of pregnancy outcomes |
| Prospective Cohort Study | van der Walt (2020)[62] | SA | 26 | DR PTB | yes (n=20) | 65.4% successful treatment outcome; no statistical significance between HIV coinfection and pregnancy outcome of mother or fetus; 57.7% had at least one adverse drug event (GI disturbance, neuropsychiatric symptom, rash, thrombocytopenia, arthalgia, dizziness, hearing loss, anemia) | Birth outcome significantly associated with the trimester in which DR TB treatment was initated (60% in 1st trimester vs 90% in 2nd vs 100% in 3rd, p=0.036); 2 abortions/miscarriages, 1 stillbirth, unclear of abortions were induced or spontaneous | mean age 29 years, SD ±5.1; missing information of type of delivery, complications of pregnancy, and birth weight |
| Retrospective Cohort Study | Yadav (2019)[36] | India | 30 | EPTB | no | significant increased incidence of oligoamnios and preterm rupture of membranc/preterm delivery compared to non-TB patients (13.3% versus 1.6%, p=0.001); death secondary to shock and end-organ failure from miliary TB. 7 women underwent emergent cesarian delivery for fetal distress | 1 stillborn due to severe miliary TB in mother; LBW in newborns of moms with TB compared to non-TB moms (2324.26±379.5 grams versus 2712.3±635.7, p=0.001); more incidence of SGA in moms with EPTB compared to non-TB moms. (33.3% versus 7.8%, p=0.001) | NA |
| **Other** | | | | | | | | |
| Service Improvement Project | Cornish (2020)[60] | SA | 56 (ages 13-50) pre-intervention, 58 post-intervention | DR PTB | yes (n=34 pre-intervention, n=23 post-intervention) | 3 pregnancies during treatment before improvement project; after improvement project: 0 pregnancy | 2 termination of pregnancy; 1 uncomplicated c-section at full term before improvement project | training intervention included family planning training of DRTB nurses and stocking the DRTB unit with long-acting injectable contraceptive; proportion of women on injectable contraceptives by the time of DRTB treatment initiation increased more than 3-fold, from 23.9% to 77.4% (p<0.0001) |
| Secondary Data Analysis | Rickman (2020)[85] | US | 235 pregnant women (n=73 with TB, n=162 without TB of which 7 later identified to have TB | not reported | yes | 43% of prevalent cases experienced complications (hypertension, antepartum hemorrhage, rupture of membranes, unplanned cesarian section, sepsis, maternal death, spontaneous abortion, stillbirth, prematurity, LBW, SGA, perinatal death, fetal abnormalities) compared to 35% of control (p=1.0) | 43% infants experienced complications including TB, neonatal sepsis, and infant hospitalization among TB prevalent cases compared to 11% of control (p=0.045) | median age 29 IQR 25-31, CD4 361 IQR 303-510; ART at baseline 2/7 (29%), 1 undetectable VL; in a cohort of women living with HIV, 7 subclinical cases were identified and discussed in this study; subclinical maternal TB may result in elevated risk of adverse infant outcomes |
| Secondary Data Analysis | van de Water (2020)[84] | Peru | 36 pregnant women with TB compared with 1298 non-pregnant with TB | PTB and DR PTB | yes (n=1 in pregnant and n=28 in non-pregnant women | 97.1% pregnant women were HIV-negative; 28 had reported successful TB outcome (cured, completed treatment, treatment ended early by clinical team), 1 was treatment failure; 8 pregnant women had DR PTB - 7 completed treatment/cured, 1 lost to follow up | neonate outcome not reported | median age of pregnant women 24.5 (IQR 21-30.5) |

Abbreviations: MTCT-Plus, maternal to child transmission-Plus; PTB, pulmonary TB; EPTB, extrapulmonary TB; DR PTB, drug resistant pulmonary TB; DR EPTB, drug resistant extrapulmonary TB; LBW, low birth weight <2500g;

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SD, standard deviation; SGA, small for gestational age; PNA, pneumonia; ARDS, acute respiratory distress syndrome; NA, not applicable; aOR, adjusted odds ratio; aRR, adjusted risk ratio; uHR, unadjusted hazard ratio; NA, not applicable

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Appendix. Detailed Search of Systematic Review.

Pubmed (1160 citations)
("Tuberculosis"[Mesh] OR tuberculos* [tiab] OR "tb" [tiab])
AND
("Pregnancy Outcome"[Mesh] OR ((pregnan* [tiab] OR obstetric [tiab] OR maternal [tiab] OR fetal [tiab]) AND (outcome* [tiab] OR morbidit* [tiab] OR mortalit* [tiab] OR "family planning services" [MESH] "OR family planning" [tiab] OR "contraception" [tiab] OR "contracept*" [tiab])))

Embase (1370 citations)
'tuberculosis'/exp OR tuberculos*:ti,ab OR 'tb':ti,ab
AND
'pregnancy outcome'/exp OR (((pregnan* OR obstetric* OR maternal* OR fetal*) NEAR/5 (outcome* OR morbidity OR mortality)):ti,ab) OR 'family planning'/exp OR ((family planning services):ti,ab OR (contraception OR contracept*):ti,ab)

CINAHL (504 citations)
(MH "Tuberculosis+") OR ( tuberculos* OR "tb" )
AND
(MH "Pregnancy Outcomes") OR ( ((pregnan* OR obstetric* OR maternal OR fetal) N5 (outcome* OR morbidity OR mortality)) ) OR (MH "family planning") OR ((family planning services) OR (contraception OR contracept*))

WEB OF SCIENCE (678 citations)
TOPIC: (tuberculos* OR tb)
AND
TOPIC: ((pregnan* OR obstetric* OR maternal OR fetal) NEAR/5 (outcome* OR morbidity OR mortality) OR family planning OR family planning services OR contraception OR contracept*)

COCHRANE ( citations - Reviews and Trials Only)
#1 MeSH descriptor: [Tuberculosis] explode all trees 2116
#2 tuberculos* or tb 12120
#3 #1 or #2 12125
#4 MeSH descriptor: [Pregnancy Outcome] explode all trees 3820
#5 ((pregnan* or obstetric* or maternal or birth or fetal) near/5 outcome*) OR ((fetal or maternal) near/5 morbidity) OR ((fetal or maternal) near/5 mortality) 21371
#6 #4 or #5 21739
#7 MeSH descriptor: [Family Planning Services] explode all trees 264
#8 contraception or contracept* 15515
#9 #7 or #8 15576
#10 #6 or #9 37590
#11 #3 and #10 317

SCOPUS (310 citations)

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TITLE-ABS-KEY(( tuberculos* OR tb ) AND (family planning OR family planning services OR contraception OR contracept* OR ((pregnan* W/5 outcome*) OR (obstetric* W/5 outcome*) OR (maternal W/5 outcome*) OR (maternal W/5 morbidity*) OR (maternal W/5 mortality) OR (fetal W/5 outcome*) OR (fetal W/5 morbidity) OR (fetal W/5 mortality))))

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | Title page |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | 4 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | 5 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | 5-6 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | 6 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | 6, Appendix |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | 6-7 |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | 6-7 |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | 6 |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | 6 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | 6-7 |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | Supplemental Table 2 |
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | 7 |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | 6-7 |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | 8, Supplemental Table 1<br>Supplemental Table 2 |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | 6 |
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | N/a |

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | N/a |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | 6-7 |
| Certainty assessment | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | 6-7 |
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | 7, Figure 1 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | 7 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | Supplemental Table 2 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | Supplemental Table 1 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | Supplemental Table 2, as appropriate |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | 7-8<br>Supplemental Table 1 |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | N/a |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | N/a |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | N/a |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | N/a |
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | 7-8 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | 13-14 |
| | 23b | Discuss any limitations of the evidence included in the review. | 14,16-17 |
| | 23c | Discuss any limitations of the review processes used. | 16-17 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | 14-16, Table 1 |
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | 2 |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | N/a |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | N/a |

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## PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | 17 |
| Competing interests | 26 | Declare any competing interests of review authors. | 17-18 |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | 17-18 |

*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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